

REHOSPACE REAL ESTATE PLATFORM (RREP)

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APPROVAL

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CHAPTER 1

PRODUCT OVERVIEW

1.1 Introduction

The **RehoSpace Real Estate Platform (RREP)** is a commercial, standalone, white-label, modular software platform engineered by RehoSpace Company Limited for deployment to real estate businesses.

Unlike traditional custom-developed websites, RREP is a reusable commercial software product. Every customer receives an independent installation configured according to purchased modules and branding requirements.

Each deployment operates independently while sharing a common master codebase maintained by RehoSpace.

The platform has been designed to eliminate repeated software development by providing a configurable foundation capable of supporting different types of real estate businesses through modular licensing.

1.2 Purpose

The purpose of RREP is to provide a complete digital ecosystem for real estate organizations, enabling them to manage:

- Properties
- Customers
- Marketing
- Sales
- Reservations
- Bookings
- Payments
- Communication
- Reporting
- Business operations

through one integrated platform.

1.3 Product Classification

Product Category

Enterprise Business Platform

Deployment Model

Standalone

Architecture

Modular

Branding

White-label

Ownership

Licensed Software

Maintenance

Centralized by RehoSpace

Deployment

Independent

1.4 Product Identity

Product Name

RehoSpace Real Estate Platform

Short Name

RREP

Developer

RehoSpace Company Limited

License Type

Commercial

1.5 Product Positioning

RREP is not a website.

RREP is not a single client project.

RREP is a commercial software product.

Every client implementation is a deployment and configuration of the same master platform.

This approach provides:

- Reduced development costs
- Faster implementation
- Easier maintenance
- Consistent quality
- Continuous upgrades
- Long-term sustainability

CHAPTER 2

PRODUCT VISION

Vision Statement

To become Africa's leading reusable digital platform for real estate businesses by delivering enterprise-grade technology that enables organizations to manage, market, and grow their businesses through configurable software rather than custom development.

Mission

To engineer a modern, scalable, secure, and modular software platform that empowers real estate businesses to rapidly establish a professional digital presence while reducing operational complexity through automation.

Long-term Vision

The long-term vision of RREP extends beyond real estate.

The engineering principles, architecture, and modular foundation shall enable RehoSpace to develop additional industry-specific products sharing the same platform core.

Examples include:

- Property Management Platform
- Construction ERP
- Hotel Management Platform
- Vehicle Marketplace
- Equipment Marketplace
- Facility Management System
- Smart City Property Solutions

The real estate platform serves as the flagship implementation of the broader RehoSpace Platform ecosystem.

CHAPTER 3

PRODUCT PHILOSOPHY

Every engineering decision within RREP shall adhere to the following philosophies.

3.1 Build Once, Deploy Many

The software shall be developed only once.

Every customer receives a configured deployment rather than a newly developed application.

3.2 Configuration Over Customization

Whenever technically feasible, business behavior shall be controlled through configuration instead of source code modification.

Examples include:

- Company branding
- Property types
- Property statuses
- Homepage layouts
- Contact information
- Payment methods
- Notification templates
- Menus
- Services
- Languages
- Currencies
- Regions

3.3 Modular Architecture

Every major business capability shall exist as an independent software module.

Modules may be activated or deactivated according to the customer's purchased license.

3.4 Upgrade Safety

Customer-specific configuration shall never prevent future upgrades.

Upgrading the platform shall not require rewriting business modules.

3.5 White-label Philosophy

Every deployment shall appear as the customer's own software platform.

Branding elements including logos, colors, domains, contact information, invoices, emails, and user interfaces shall be configurable without source code modification.

3.6 Enterprise Quality

The platform shall be engineered according to enterprise software principles emphasizing:

- Reliability
- Security
- Performance
- Maintainability
- Extensibility
- Scalability

rather than rapid one-off development.

CHAPTER 4

ENGINEERING PRINCIPLES

The following principles govern all future development of RREP.

Every feature developed for the platform shall satisfy these principles unless explicitly exempted by the Product Architect.

Principle 1

Everything possible shall be configurable.

Principle 2

Nothing shall be hardcoded unless technically unavoidable.

Principle 3

Every feature shall respect role-based permissions.

Principle 4

Every business module shall be independently testable.

Principle 5

Every module shall support future API integration.

Principle 6

Every module shall generate audit records where applicable.

Principle 7

Every module shall be upgrade-safe.

Principle 8

Business logic shall remain separated from presentation logic.

Principle 9

Every module shall expose events for future integrations and automation.

Principle 10

Every customer deployment shall remain independent while originating from the same master product.

CHAPTER 5

BUSINESS OBJECTIVES

5.1 Introduction

The RehoSpace Real Estate Platform (RREP) has been developed to solve common operational, marketing, sales, communication, and management challenges experienced by modern real estate businesses.

The platform aims to replace fragmented manual processes and disconnected software solutions with one integrated, configurable, enterprise-grade platform.

Every feature implemented within RREP shall contribute toward achieving one or more of the following business objectives.

5.2 Primary Business Objectives

BO-001 – Digital Transformation

Enable real estate organizations to digitize their business processes, customer engagement, and service delivery.

BO-002 – Business Visibility

Provide organizations with a modern digital presence capable of attracting customers through web search engines, social media integration, and online marketing.

BO-003 – Property Marketing

Provide comprehensive tools for advertising, promoting, organizing, and managing properties available for sale or rent.

BO-004 – Customer Relationship Management

Allow organizations to capture, manage, nurture, and convert customer inquiries into successful transactions.

BO-005 – Operational Efficiency

Reduce manual work through automation of repetitive business processes.

Examples include:

- Email notifications
 - SMS notifications
 - Invoice generation
 - Appointment reminders
 - Reservation confirmations
 - Lead assignment
 - Workflow approvals
-

BO-006 – Revenue Generation

Support multiple business revenue streams including:

- Property sales
 - Property rentals
 - Survey services
 - Consultation services
 - Property management services
 - Advertising services
 - Premium listing subscriptions (future)
-

BO-007 – Data Centralization

Provide a centralized repository for all business information including:

- Customers
 - Properties
 - Documents
 - Images
 - Videos
 - Financial records
 - Reports
 - Communications
-

BO-008 – Scalability

Allow organizations to expand operations without replacing their software platform.

The platform shall support:

- Additional users

- Additional branches
- Increased listings
- Additional services
- Additional business workflows

BO-009 – Product Reusability

Allow RehoSpace to deploy the same platform repeatedly for different customers while maintaining a single master product.

5.3 Secondary Objectives

The platform shall also support:

- Improved customer satisfaction
 - Faster decision making
 - Better reporting
 - Reduced operational costs
 - Business intelligence
 - Higher marketing efficiency
 - Better communication
 - Brand consistency
-

CHAPTER 6

TARGET CUSTOMERS

6.1 Introduction

RREP has been engineered as a reusable software platform capable of serving different organizations operating within the real estate industry.

Rather than targeting one business model, the platform supports configurable deployment according to each customer's operational requirements.

6.2 Primary Customers

The following organizations represent the primary target market.

Real Estate Agencies

Organizations involved in marketing and selling properties on behalf of owners.

Property Developers

Organizations constructing residential, commercial, or mixed-use developments.

Land Selling Companies

Businesses specializing in plot acquisition, subdivision, surveying, and resale.

Property Management Companies

Organizations managing rental properties, tenants, landlords, and maintenance.

Apartment Operators

Businesses managing apartment rentals, occupancy, bookings, and tenants.

Commercial Property Companies

Organizations dealing with offices, warehouses, shopping centers, and commercial buildings.

Land Survey Companies

Businesses providing surveying, mapping, and land verification services.

Investment Companies

Organizations investing in real estate assets.

Government Institutions (Future)

Departments managing public land information.

6.3 Secondary Customers

The platform may later be deployed for:

- Housing cooperatives
 - Universities
 - Churches
 - NGOs
 - Industrial parks
 - Smart cities
 - Facility managers
-

CHAPTER 7

SUPPORTED BUSINESS TYPES

RREP supports multiple property business models.

Examples include:

Land Sales

- Residential plots
 - Commercial plots
 - Agricultural land
 - Industrial land
-

Residential Properties

- Houses
 - Villas
 - Townhouses
 - Apartments
 - Duplexes
-

Commercial Properties

- Offices

- Shops
 - Warehouses
 - Hotels
 - Retail centers
-

Industrial Properties

- Factories
 - Industrial parks
 - Logistics centers
-

Mixed-use Developments

Properties containing multiple business purposes.

Special Services

The platform also supports service-based businesses including:

- Survey services
- Consultation
- Property valuation
- Property inspections
- Legal documentation assistance
- Property management

The list of services shall be configurable.

CHAPTER 8

PLATFORM ARCHITECTURE

8.1 Architecture Overview

The RehoSpace Real Estate Platform is designed as a **standalone modular application**.

Each customer deployment includes:

- Independent source deployment
- Independent database
- Independent hosting
- Independent domain
- Independent configuration
- Independent branding
- Licensed feature modules

Although every deployment is independent, all deployments originate from the same master codebase maintained by RehoSpace.

8.2 Architectural Characteristics

The platform shall exhibit the following characteristics:

- Modular
 - Configurable
 - Secure
 - Extensible
 - Upgrade-safe
 - API-ready
 - Responsive
 - White-label
 - Enterprise-grade
-

8.3 Module-Based Architecture

Business functionality shall be organized into independent software modules.

Examples include:

- Core Platform
- Website CMS
- Property Management
- Marketplace
- CRM
- Reservations
- Payments
- Survey Management
- Marketing
- Reports
- Communication
- Media Library
- Licensing Engine
- AI Services (Future)

Each module shall define:

- Functional requirements
 - Dependencies
 - Business rules
 - Permissions
 - Configuration
 - Notifications
 - Reports
 - Acceptance criteria
-

8.4 White-Label Architecture

The following items shall be configurable without modifying source code:

- Company name
 - Company logo
 - Favicon
 - Brand colors
 - Typography
 - Domain
 - Footer
 - Homepage
 - Contact details
 - Social media
 - Invoice branding
 - Email templates
 - SMS sender identity
 - Legal pages
-

8.5 Licensing Architecture

Every deployment shall contain the Licensing Engine.

The Licensing Engine shall determine:

- Enabled modules
- Premium features
- Edition level
- Upgrade eligibility
- License validity

Only RehoSpace shall control commercial licensing.

CHAPTER 9

PRODUCT STANDARDS

To ensure consistency across all future development, every component of RREP shall comply with the following product standards.

Standard PS-001

Business logic shall never be hardcoded where configuration is technically feasible.

Standard PS-002

Every module shall implement role-based access control.

Standard PS-003

Every configurable value shall be manageable from the administration interface unless explicitly documented otherwise.

Standard PS-004

Every business action affecting operational or financial data shall generate an audit log entry.

Standard PS-005

Every module shall expose events for future automation and integrations.

Standard PS-006

Every module shall be independently testable.

Standard PS-007

Every module shall support localization and internationalization.

Standard PS-008

Every module shall maintain backward compatibility with future upgrades wherever feasible.

Standard PS-009

Database entities shall include appropriate audit fields, including creation, update, and deletion metadata where applicable.

Standard PS-010

The user interface shall maintain consistency in navigation, layout, terminology, and interaction patterns across all modules.

CHAPTER 10

GLOBAL BUSINESS RULES

The following rules apply across the entire platform.

GBR-001

Every authenticated user shall belong to at least one role.

GBR-002

Every permission shall be assigned through roles.

GBR-003

Every business module shall respect licensing restrictions.

GBR-004

Every configurable feature shall have a sensible default value.

GBR-005

Every business transaction shall be traceable.

GBR-006

Critical operations shall require appropriate authorization.

GBR-007

Deleted business records shall be recoverable through soft deletion unless permanent deletion is explicitly required.

GBR-008

Every notification generated by the platform shall be logged.

GBR-009

Every uploaded file shall undergo validation before storage.

GBR-010

Every platform deployment shall remain isolated from other customer deployments.

CHAPTER 11

USER ROLES & PERMISSION MODEL

11.1 Introduction

The RehoSpace Real Estate Platform (RREP) implements a comprehensive **Role-Based Access Control (RBAC)** model. Every authenticated user shall be assigned one or more roles, and each role shall be associated with specific permissions. Direct assignment of permissions to users should be avoided except where explicitly supported for exceptional cases.

The RBAC model ensures that users can only access the modules, menus, records, and operations that are authorized for their role.

Objectives

The User Roles & Permission Model aims to:

- Protect sensitive business information.
 - Enforce separation of duties.
 - Simplify user administration.
 - Support organizational hierarchies.
 - Enable module-level access control.
 - Support future expansion without redesigning the security model.
-

11.2 Role Hierarchy

The platform recognizes the following standard user roles.

Public Visitor

↓

Registered Customer

↓

Property Owner

↓

Sales Agent

↓

Senior Agent / Broker

↓

Department Manager

↓
System Administrator

↓
Super Administrator

Organizations may create additional custom roles based on operational requirements.

11.3 Public Visitor

Purpose

A Public Visitor is any individual accessing the platform without authentication.

Responsibilities

- Browse public website content.
- Search publicly available properties.
- Submit inquiries.
- Request consultations.
- Register an account.
- Subscribe to newsletters.

Permissions

The Public Visitor may:

- View homepage.
- Browse listings.
- Search properties.
- Filter listings.
- View company information.
- Contact the organization.
- Submit inquiry forms.
- View blogs.
- View gallery.

Restrictions

The Public Visitor shall not:

- Access administrative pages.
- View unpublished listings.
- Make administrative changes.
- Access confidential information.

11.4 Registered Customer

Purpose

A Registered Customer is an authenticated platform user interested in purchasing, renting, or investing in properties.

Capabilities

Customers may:

- Update profile.
- Save favorite properties.
- Compare listings.
- Book appointments.
- Reserve properties.
- Submit applications.
- View reservation history.
- Receive notifications.
- Download invoices.
- Track requests.

Restrictions

Customers shall not:

- Publish properties.
- Approve listings.
- Access administrative reports.

11.5 Property Owner

Purpose

Represents individuals or organizations owning properties managed through the platform.

Capabilities

Property owners may:

- Submit property listings.
- Upload documents.
- Upload images.

- Track inquiries.
- Monitor listing performance.
- View reservation requests.
- Respond to administrative requests.

Depending on the purchased modules, owners may also manage rental information and tenancy details.

11.6 Sales Agent

Purpose

Sales Agents represent the organization in customer engagement and property sales.

Responsibilities

- Manage customer inquiries.
- Follow up leads.
- Schedule appointments.
- Demonstrate properties.
- Update lead status.
- Assist reservations.
- Record customer interactions.

Permissions

Agents may:

- View assigned leads.
- Update lead progress.
- Manage appointments.
- Record notes.
- View approved properties.
- Generate quotations.

Agents may only access records assigned to them unless additional permissions are granted.

11.7 Senior Agent / Broker

Senior Agents possess all Sales Agent permissions together with supervisory capabilities.

Additional permissions include:

- Review team performance.
 - Assign leads.
 - Approve selected activities.
 - Monitor sales pipeline.
 - Generate departmental reports.
-

11.8 Department Manager

Managers supervise one or more operational departments.

Examples include:

- Sales Manager
- Marketing Manager
- Property Manager
- Survey Manager
- Customer Service Manager

Managers may:

- Approve workflows.
 - Monitor departmental KPIs.
 - Generate reports.
 - Configure departmental settings.
 - Manage subordinate staff.
-

11.9 System Administrator

The Administrator manages the organization's deployment of RREP.

Responsibilities include:

- User management.
- Role management.
- Configuration.
- Content management.
- Property approval.
- Marketing configuration.
- Report generation.
- System monitoring.

Administrators cannot modify platform licensing.

11.10 Super Administrator

The Super Administrator represents the highest authority within an individual deployment.

Responsibilities include:

- Full platform administration.
- Licensing visibility.
- Security configuration.
- Backup management.
- API configuration.
- Payment gateway configuration.
- Integration management.
- Organization branding.
- Module configuration.

Only authorized RehoSpace personnel may possess platform-wide master privileges outside customer deployments.

11.11 Custom Roles

Organizations may define unlimited custom roles.

Examples:

- Receptionist
- Accountant
- Cashier
- Survey Officer
- Marketing Officer
- Customer Care Officer
- Legal Officer
- Property Inspector
- Photographer
- Content Editor

Each custom role shall receive permissions through the RBAC engine.

11.12 Permission Categories

Permissions shall be grouped by module.

Example:

Property Module

View

Create

Edit

Delete

Approve

Reject

Archive

Export

CRM Module

View Leads

Create Leads

Assign Leads

Close Leads

Delete Leads

Export Leads

Payments Module

View Payments

Create Payments

Refund

Cancel

Export

Print Receipt

Reports Module

View

Generate

Export

Schedule

Delete

11.13 Permission Levels

Each module shall support permission granularity.

Examples include:

View Only

Create

Update

Delete

Approve

Reject

Import

Export

Print

Share

Assign

Archive

Restore

Configure

Administrate

This allows organizations to define highly customized access policies.

11.14 Security Principles

The RBAC implementation shall follow these principles:

UR-001

Every authenticated user shall belong to at least one role.

UR-002

Permissions shall be inherited through assigned roles.

UR-003

Permission evaluation shall occur before every protected operation.

UR-004

Unauthorized operations shall be denied and logged.

UR-005

Administrative actions shall generate audit records.

UR-006

Changes to roles and permissions shall require administrative authorization.

CHAPTER 12

MODULE CLASSIFICATION & PRODUCT CATALOGUE

12.1 Introduction

The RehoSpace Real Estate Platform is organized into independent software modules. Each module provides a specific set of business capabilities and may be licensed individually or bundled into commercial editions.

Modules are classified into five categories to simplify architecture, licensing, deployment, and future expansion.

12.2 Foundation Modules

Foundation Modules provide the essential services required by all deployments. These modules are mandatory and cannot be disabled.

Included modules:

- Core Platform
 - Authentication
 - RBAC
 - System Configuration
 - Branding
 - Security
 - Audit & Activity Logs
 - Notification Framework
 - Media Management
-

12.3 Business Modules

Business Modules implement the core operational functions of a real estate organization.

These include:

- Property Management
 - Property Categories
 - Property Types
 - Marketplace
 - Customer Relationship Management (CRM)
 - Leads Management
 - Property Owners
 - Agents Management
 - Survey Management
 - Reservation Management
 - Appointment Booking
 - Document Management
-

12.4 Commerce Modules

Commerce Modules facilitate revenue generation and financial transactions.

Modules include:

- Payments
 - Invoicing
 - Receipts
 - Billing
 - Commission Management
 - Discount Management
 - Financial Reports
-

12.5 Growth Modules

Growth Modules help organizations expand market reach and improve customer engagement.

Modules include:

- Website CMS
 - Blog
 - News & Announcements
 - Testimonials
 - Gallery
 - SEO Management
 - Social Media Integration
 - Email Campaigns
 - SMS Campaigns
 - WhatsApp Campaigns
 - Newsletter Management
 - Landing Pages
 - Banner Management
-

12.6 Premium Modules

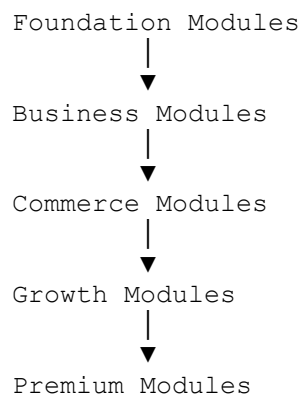
Premium Modules provide advanced enterprise capabilities and may require additional licensing.

Planned Premium Modules include:

- Artificial Intelligence Assistant
 - GIS & Interactive Mapping
 - Property Valuation
 - Auction Management
 - Tenant Management
 - Facility Management
 - Mobile Applications
 - Advanced Analytics
 - Business Intelligence Dashboard
 - Electronic Document Signing
 - Government Systems Integration
 - ERP Integrations
 - API Marketplace
 - Workflow Automation Engine
-

12.7 Module Dependencies

Modules are interconnected through well-defined dependencies.



A module may depend on one or more lower-level modules but should avoid unnecessary coupling to maintain modularity and upgrade safety.

CHAPTER 13

FOUNDATION MODULES

13.1 FOUNDATION MODULE OVERVIEW

Purpose

Foundation Modules provide the core infrastructure upon which every other module in the platform operates. They establish authentication, authorization, configuration, branding, security, logging, notifications, and media management capabilities.

These modules are mandatory for every deployment and cannot be removed or disabled.

Objectives

The Foundation Modules shall:

- Provide secure access control.
- Centralize platform configuration.
- Maintain platform security.
- Standardize administration.
- Support reusable architecture.

- Enable future scalability.
- Ensure maintainability.
- Provide consistent services to all business modules.

Included Modules

The Foundation Layer consists of:

Code	Module
FM-001	Core Platform
FM-002	Authentication
FM-003	Role & Permission (RBAC)
FM-004	System Configuration
FM-005	Branding & White-label
FM-006	Security
FM-007	Audit & Activity Logs
FM-008	Notification Framework
FM-009	Media & File Management

FOUNDATION MODULE 1

FM-001 — CORE PLATFORM MODULE

1. Purpose

The Core Platform Module is the backbone of the RehoSpace Real Estate Platform. It provides the common services, infrastructure, navigation, dashboard, and framework required by all other modules.

No other module shall operate independently of the Core Platform.

2. Business Value

The Core Platform provides:

- Consistent user experience.
 - Centralized administration.
 - Shared services.
 - Common navigation.
 - Standardized architecture.
 - Easier maintenance.
 - Faster development of future modules.
-

3. Objectives

The Core Platform shall:

- Provide the main application shell.
 - Load licensed modules.
 - Display dashboards.
 - Manage navigation.
 - Provide search capabilities.
 - Display notifications.
 - Manage application menus.
 - Support future module expansion.
-

4. Scope

Included

- Dashboard
- Sidebar
- Top Navigation
- Breadcrumbs
- Global Search
- Quick Actions
- Widgets
- Notifications Center
- User Profile
- Module Loader
- Settings Framework

Excluded

Business-specific functionality including:

- Property Management
- CRM
- Payments
- Marketing

These belong to separate modules.

5. Functional Requirements

FR-CORE-001

The system shall provide a responsive administration dashboard.

FR-CORE-002

The dashboard shall display widgets based on user permissions.

FR-CORE-003

The system shall dynamically generate menus according to:

- Licensed modules
 - User role
 - Permissions
-

FR-CORE-004

The platform shall support unlimited dashboard widgets.

FR-CORE-005

The platform shall support customizable dashboards.

FR-CORE-006

Users shall be able to pin frequently used menu items.

FR-CORE-007

The platform shall support global search.

Global search shall locate:

- Properties
- Customers
- Owners
- Reservations
- Invoices
- Agents
- Documents

subject to user permissions.

FR-CORE-008

Users shall receive real-time notifications for authorized events.

FR-CORE-009

The dashboard shall display organization-wide announcements.

FR-CORE-010

The platform shall display the current software version.

FR-CORE-011

The system shall display currently licensed modules.

FR-CORE-012

Navigation shall automatically update whenever modules are enabled or disabled.

6. Dashboard Components

The dashboard may contain configurable widgets including:

- Today's Appointments
- Pending Reservations
- Pending Property Approvals
- New Customers
- Sales Summary
- Revenue Summary
- Recent Activities
- Calendar
- Notifications
- Quick Links
- Recent Properties
- Recent Inquiries
- System Health
- License Status

Organizations shall configure dashboard layouts.

7. Global Search

The search engine shall support searching across:

Properties

Customers

Property Owners

Agents

Reservations

Bookings

Invoices

Payments

Documents

Blogs

Media

Reports

Search results shall respect permissions.

8. Quick Actions

Examples include:

Add Property

Register Customer

Create Reservation

Schedule Appointment

Upload Media

Generate Invoice

View Reports

These actions shall be configurable.

9. Business Rules

BR-CORE-001

The Core Platform module shall never be disabled.

BR-CORE-002

Every module shall register itself with the Core Platform.

BR-CORE-003

Navigation shall only display licensed modules.

BR-CORE-004

Navigation shall only display authorized menu items.

BR-CORE-005

Widgets shall respect permissions.

BR-CORE-006

The dashboard shall never expose confidential information to unauthorized users.

10. Configuration

Administrators shall configure:

Dashboard Layout

Default Landing Page

Widgets

Sidebar Position

Theme

Homepage

Shortcuts

Announcements

Search Options

Navigation Style

11. Notifications

Examples:

System Updates

License Expiration

New Reservations

Pending Approvals

Failed Jobs

Backup Completion

Security Alerts

12. Reports

Dashboard Usage

Login Statistics

Most Used Modules

Most Active Users

System Activity

Navigation Analytics

13. Acceptance Criteria

The Core Platform shall be accepted when:

- ✓ Dashboard loads successfully.
- ✓ Widgets load according to permissions.
- ✓ Navigation reflects licensed modules.
- ✓ Search returns authorized records.
- ✓ Notifications function correctly.
- ✓ Dashboard configuration persists.

FOUNDATION MODULE 2

FM-002 — AUTHENTICATION MODULE

1. Purpose

The Authentication Module manages user identity verification and secure access to the platform.

Only authenticated users may access protected resources.

2. Objectives

The Authentication Module shall:

- Verify identities.
 - Protect confidential data.
 - Secure user sessions.
 - Prevent unauthorized access.
 - Support future authentication providers.
-

3. Scope

Included:

Login

Logout

Forgot Password

Reset Password

Email Verification

Remember Me

Session Management

Account Locking

Password Policies

Future Multi-Factor Authentication

Future Single Sign-On

4. Functional Requirements

FR-AUTH-001

The platform shall authenticate registered users.

FR-AUTH-002

Passwords shall be securely hashed.

FR-AUTH-003

Users shall reset forgotten passwords.

FR-AUTH-004

The platform shall verify email addresses.

FR-AUTH-005

Failed login attempts shall be logged.

FR-AUTH-006

Accounts shall lock after configurable failed login attempts.

FR-AUTH-007

Users shall terminate active sessions by logging out.

FR-AUTH-008

The platform shall support configurable password complexity requirements.

FR-AUTH-009

Inactive sessions shall automatically expire after a configurable timeout.

FR-AUTH-010

The authentication framework shall be extensible to support future identity providers such as Google, Microsoft, LDAP, and OAuth-compliant services.

5. Password Policy

The system shall allow administrators to configure:

- Minimum password length.
 - Uppercase requirement.
 - Lowercase requirement.
 - Numeric requirement.
 - Special character requirement.
 - Password expiry period.
 - Password reuse restrictions.
-

6. Business Rules

BR-AUTH-001

Every protected operation shall require an authenticated session.

BR-AUTH-002

Plain-text passwords shall never be stored.

BR-AUTH-003

Authentication events shall be recorded in the audit log.

BR-AUTH-004

Only active user accounts shall be permitted to authenticate.

BR-AUTH-005

Session identifiers shall be securely generated and invalidated on logout.

7. Notifications

The Authentication Module shall support notifications for:

- Successful password changes.
 - Password reset requests.
 - New device login (future).
 - Account lockout.
 - Suspicious login attempts.
 - Administrator-forced password reset.
-

8. Reports

The system shall provide authentication reports including:

- Successful login history.
 - Failed login attempts.
 - Locked accounts.
 - Password reset activity.
 - Active user sessions.
-

9. Acceptance Criteria

The Authentication Module shall be accepted when:

- Registered users can log in using valid credentials.
- Invalid credentials are rejected.
- Password reset workflows function correctly.
- Email verification operates as configured.

- Session timeout is enforced.
- Audit logs capture authentication events.
- Configured password policies are consistently applied.

FOUNDATION MODULE 3

FM-003 – ROLE & PERMISSION MANAGEMENT (RBAC)

1. Purpose

The Role & Permission Management Module provides centralized authorization throughout the platform. It determines what each user is allowed to see and perform.

Authorization shall be based primarily on **Roles**, with Permissions assigned to Roles. Users inherit permissions through their assigned roles. The platform may support direct user-specific permissions only for exceptional administrative cases.

2. Objectives

The RBAC Module shall:

- Protect business data.
 - Enforce least-privilege access.
 - Support unlimited roles.
 - Support unlimited permissions.
 - Enable module-level security.
 - Enable field-level and record-level restrictions where required.
 - Provide auditability of permission changes.
-

3. Scope

Included:

- Role Management
- Permission Management
- Permission Groups
- User Role Assignment
- Multiple Roles per User
- Role Inheritance (future)

- Temporary Roles (future)
 - Record Ownership Rules
 - Department-based Access (future)
-

4. Functional Requirements

FR-RBAC-001

The system shall support unlimited user roles.

FR-RBAC-002

The system shall support unlimited permissions.

FR-RBAC-003

Permissions shall be grouped by module.

FR-RBAC-004

Users may belong to multiple roles.

FR-RBAC-005

Permissions shall be evaluated before every protected operation.

FR-RBAC-006

Unauthorized operations shall return an access denied response.

FR-RBAC-007

Permission changes shall immediately take effect.

FR-RBAC-008

Role assignments shall be recorded in the audit log.

FR-RBAC-009

The system shall allow administrators to clone existing roles.

FR-RBAC-010

The platform shall support permission import/export.

5. Permission Categories

Each module shall expose standardized permissions:

- View
- Create
- Update
- Delete
- Approve
- Reject
- Archive
- Restore
- Export
- Import
- Print
- Configure
- Manage

6. Business Rules

BR-RBAC-001

Every authenticated user shall belong to at least one role.

BR-RBAC-002

No protected feature shall bypass permission validation.

BR-RBAC-003

Permission modifications require administrative authorization.

BR-RBAC-004

Deleting a role assigned to active users shall be prohibited unless reassignment is completed.

7. Reports

- User Role Report
- Permission Matrix
- Role Assignment History
- Permission Change Log

8. Acceptance Criteria

- ✓ Roles can be created, edited and archived.
 - ✓ Permissions are enforced consistently.
 - ✓ Unauthorized actions are blocked.
 - ✓ Audit records are generated.
-

FOUNDATION MODULE 4

FM-004 – SYSTEM CONFIGURATION

1. Purpose

The System Configuration Module centralizes all configurable platform settings, eliminating the need to modify source code for customer-specific behavior.

This module is one of the most important architectural components of RREP.

2. Objectives

Allow administrators to configure the platform without developer intervention.

3. Configuration Categories

Organization

- Company Name
- Registration Number
- Tax Number
- Logo
- Favicon
- Motto

- About Company
- Office Locations

Contact Information

- Telephone Numbers
- Mobile Numbers
- WhatsApp
- Email
- Website
- Postal Address

Regional Settings

- Country
- Region
- City
- Time Zone
- Currency
- Date Format
- Time Format
- Measurement Units
- Language

Business Settings

- Property Types
- Property Statuses
- Amenities
- Services
- Booking Rules
- Reservation Expiry
- Working Days
- Working Hours

Notification Settings

- SMTP
 - SMS Gateway
 - WhatsApp Gateway
 - Push Notifications
-

Payment Settings

- Payment Gateways
 - Tax Rates
 - Invoice Prefixes
 - Receipt Prefixes
-

Maps

- Google Maps API
 - Default Coordinates
 - Default Zoom
-

SEO

- Meta Title
 - Meta Description
 - Meta Keywords
 - Sitemap
 - Robots
-

Functional Requirements

FR-CONF-001

Administrators shall configure all business settings through the administration interface.

FR-CONF-002

Configuration changes shall take effect without requiring application recompilation.

FR-CONF-003

Configuration history shall be auditable.

Business Rules

Configuration changes affecting security shall require elevated permissions.

Acceptance Criteria

All supported configuration items can be modified and saved successfully.

FOUNDATION MODULE 5

FM-005 – BRANDING & WHITE-LABEL ENGINE

Purpose

Provide complete branding customization for each customer deployment.

Configurable Branding

- Company Name
 - Logo
 - Favicon
 - Theme Colors
 - Fonts
 - Email Templates
 - Invoice Templates
 - Login Screen
 - Splash Screen
 - Homepage Hero
 - Footer
 - Social Media Links
 - Copyright
-

Functional Requirements

FR-BRAND-001

Branding changes shall not require source code modification.

FR-BRAND-002

Branding shall be isolated per deployment.

FR-BRAND-003

Branding assets shall support version replacement.

Business Rules

Customer branding shall not alter platform ownership metadata where licensing requires attribution.

Acceptance Criteria

A deployed platform can be fully rebranded through the administration interface.

FOUNDATION MODULE 6

FM-006 – SECURITY

Purpose

Protect the platform, users, and business data against unauthorized access and common security threats.

Security Features

- Password Hashing
 - CSRF Protection
 - XSS Prevention
 - SQL Injection Protection
 - File Validation
 - Session Management
 - Encryption
 - HTTPS Enforcement
 - Rate Limiting
 - Backup Management
-

Functional Requirements

FR-SEC-001

Sensitive information shall be encrypted where appropriate.

FR-SEC-002

Uploaded files shall undergo validation.

FR-SEC-003

Administrative sessions shall expire after configurable inactivity.

FR-SEC-004

Security-related events shall be logged.

Business Rules

The platform shall follow secure-by-default principles.

Reports

- Failed Login Attempts
 - Security Events
 - Backup History
 - Session Activity
-

FOUNDATION MODULE 7

FM-007 – AUDIT & ACTIVITY LOGS

Purpose

Maintain a complete historical record of important platform activities.

Activities Logged

- Login
 - Logout
 - User Creation
 - Property Approval
 - Payment Recording
 - Configuration Changes
 - Role Changes
 - Reservation Approval
 - Booking Updates
 - Document Uploads
-

Functional Requirements

FR-AUD-001

Critical business actions shall generate immutable audit records.

FR-AUD-002

Audit records shall include:

- User
 - Date
 - Time
 - IP Address
 - Module
 - Action
 - Previous Value (where applicable)
 - New Value (where applicable)
-

Acceptance Criteria

All configured business events generate searchable audit records.

FOUNDATION MODULE 8

FM-008 – NOTIFICATION FRAMEWORK

Purpose

Provide a centralized event-driven notification system.

Notification Channels

- In-App
 - Email
 - SMS
 - WhatsApp
 - Push Notifications (Future)
-

Notification Types

- Informational
 - Warning
 - Error
 - Success
 - Reminder
 - Escalation
-

Business Events

Examples:

- Reservation Created
 - Booking Confirmed
 - Property Approved
 - Invoice Generated
 - Payment Received
 - Survey Scheduled
 - New Inquiry
 - License Expiry
-

Functional Requirements

FR-NOT-001

Business modules shall publish notification events rather than sending notifications directly.

FR-NOT-002

Notification templates shall be configurable.

Acceptance Criteria

Configured notifications are delivered through enabled channels.

FOUNDATION MODULE 9

FM-009 – MEDIA & FILE MANAGEMENT

Purpose

Centralize storage, organization, and lifecycle management of all digital assets used throughout the platform.

Supported Assets

- Images
 - Videos
 - PDF Documents
 - Office Documents
 - Floor Plans
 - Survey Reports
 - Contracts
 - Brochures
-

Features

- Folder Management
- File Categorization

- Image Optimization
 - Version Replacement
 - Secure Downloads
 - Access Permissions
 - Storage Quotas
 - Bulk Upload
 - Bulk Download
-

Functional Requirements

FR-MEDIA-001

The platform shall support configurable storage providers.

FR-MEDIA-002

Media files shall be reusable across multiple modules.

FR-MEDIA-003

Large file uploads shall support resumable upload in future versions.

Business Rules

Only supported file types may be uploaded.

Maximum upload size shall be configurable.

Deleted files shall follow configurable retention policies.

Acceptance Criteria

Users with appropriate permissions can upload, organize, retrieve, preview, and securely download media assets.

FOUNDATION MODULE SUMMARY

The Foundation Layer provides the essential services required by every deployment of RREP. It establishes the platform's security, configuration, branding, notifications, media

management, auditing, and authorization capabilities. No Business Module may bypass or duplicate these services.

CHAPTER 14

BUSINESS MODULES

14.1 BUSINESS MODULE OVERVIEW

Purpose

Business Modules provide the operational capabilities required by real estate organizations. They represent the core functions of the platform and implement the workflows that support day-to-day business activities.

Unlike Foundation Modules, Business Modules may be licensed individually or bundled into different product editions.

Objectives

Business Modules shall enable organizations to:

- Manage property portfolios.
 - Publish and market listings.
 - Capture and manage customer inquiries.
 - Coordinate reservations and appointments.
 - Manage agents and property owners.
 - Maintain business records.
 - Generate operational reports.
 - Support configurable business processes.
-

Included Modules

Code	Module
BM-001	Property Management
BM-002	Marketplace
BM-003	Customer Relationship Management (CRM)
BM-004	Property Owners

Code	Module
BM-005	Agents Management
BM-006	Reservation Management
BM-007	Appointment & Booking
BM-008	Survey Management
BM-009	Document Management

BUSINESS MODULE 1

BM-001 — PROPERTY MANAGEMENT MODULE

1. Purpose

The Property Management Module (PMM) is the central business engine of RREP.

It manages the complete lifecycle of every property within the platform—from initial registration through publication, customer engagement, reservation, sale or rental, archival, and historical reporting.

Every property displayed on the website or marketplace originates from this module.

2. Business Value

The Property Management Module enables organizations to:

- Centralize property information.
 - Maintain accurate listings.
 - Improve property marketing.
 - Reduce duplicate records.
 - Standardize property data.
 - Increase operational efficiency.
 - Accelerate property sales and rentals.
 - Improve customer confidence through consistent information.
-

3. Objectives

The module shall:

- Register properties.
 - Categorize properties.
 - Manage property media.
 - Track ownership.
 - Track availability.
 - Publish listings.
 - Archive completed listings.
 - Maintain historical records.
-

4. Scope

Included

- Property Registration
- Property Editing
- Property Approval
- Property Publishing
- Property Categorization
- Property Search
- Property Availability
- Property Media
- Property Documents
- Property Pricing
- Property Status
- Property History
- Property Archiving

Excluded

Customer relationship management.

Payments.

Marketing campaigns.

Reservations.

These are handled by separate modules.

5. Property Lifecycle

Every property shall move through the following lifecycle.

Draft

↓

Pending Review

↓

Approved

↓

Published

↓

Available

↓

Reserved

↓

Sold / Rented

↓

Archived

Organizations may configure additional workflow states.

6. Property Types

The platform shall support unlimited configurable property types.

Examples include:

Residential Land

Commercial Land

Agricultural Land

Industrial Land

Apartment

House

Villa

Townhouse

Duplex

Studio Apartment

Office

Shop

Warehouse

Factory

Hotel

Hostel

Farm

Mixed Use Property

Industrial Park

Future property types shall be added without modifying source code.

7. Property Categories

Categories shall classify properties according to organizational requirements.

Examples:

Residential

Commercial

Industrial

Agricultural

Investment

Luxury

Affordable Housing

Rental

Vacation Property

Categories shall be configurable.

8. Property Statuses

Default statuses include:

Available

Reserved

Under Negotiation

Sold

Rented

Unavailable

Suspended

Archived

Organizations may define additional statuses.

9. Property Information

Each property shall maintain comprehensive information.

Basic Information

Property Reference Number

Property Name

Title

Short Description

Detailed Description

Property Type

Category

Status

Listing Type

Featured Status

Priority

Ownership Information

Owner

Agent

Broker

Organization

Ownership Documents

Ownership History

Location Information

Country

Region

District

City

Ward

Street

Physical Address

Postal Address

GPS Coordinates

Google Map Location

Nearby Landmarks

Directions

Pricing Information

Sale Price

Rental Price

Negotiable Status

Currency

Tax

Discount

Commission

Promotion Price

Price Visibility

Property Measurements

Area

Length

Width

Height

Number of Floors

Number of Rooms

Bedrooms

Bathrooms

Parking Capacity

Building Age

Year Built

Land Size

Measurement Units shall be configurable.

Amenities

Examples:

Electricity

Water

Internet

Swimming Pool

Garden

Security

Fence

Garage

Air Conditioning

Balcony

Elevator

CCTV

Gym

Conference Room

Solar Power

Generator

Borehole

Children Playground

Unlimited amenities shall be supported.

10. Property Media

Each property shall support:

Unlimited Images

Property Videos

Virtual Tours

Floor Plans

Site Plans

Survey Maps

Brochures

PDF Documents

Legal Documents

Certificates

Videos shall support:

Upload

Embedded YouTube

Embedded Vimeo

Future cloud providers.

11. Property Documents

Supported document types include:

Title Deed

Survey Plan

Building Permit

Tax Certificate

Ownership Certificate

Environmental Approval

Inspection Report

Valuation Report

Lease Agreement

Sale Agreement

Organizations may define additional document types.

12. Property Features

Property features shall be configurable.

Examples:

Water Front

Ocean View

Mountain View

Smart Home

Solar Powered

Gated Community

Corner Plot

Near Main Road

Fully Furnished

New Construction

Future Ready

Energy Efficient

Green Building

13. Functional Requirements

Property Registration

FR-PROP-001

The platform shall register new properties.

FR-PROP-002

Every property shall receive a unique Property Reference Number.

FR-PROP-003

Property Reference Numbers shall be configurable.

FR-PROP-004

Duplicate Property Reference Numbers shall not be permitted.

Property Editing

FR-PROP-005

Authorized users shall update property information.

FR-PROP-006

Property modifications shall generate audit records.

Property Approval

FR-PROP-007

Organizations may enable approval workflows.

FR-PROP-008

Only authorized users shall approve listings.

FR-PROP-009

Rejected properties shall include rejection reasons.

Property Publishing

FR-PROP-010

Approved properties may be published.

FR-PROP-011

Published properties shall automatically appear on the website when configured.

FR-PROP-012

Future publication dates shall be supported.

FR-PROP-013

Publication expiry dates shall be configurable.

Property Search

FR-PROP-014

The system shall support searching by:

Reference Number

Property Name

Property Type

Category

Location

Price Range

Bedrooms

Bathrooms

Area

Amenities

Status

Owner

Agent

Keywords

Property Media

FR-PROP-015

Properties shall support unlimited images.

FR-PROP-016

The system shall automatically generate image thumbnails.

FR-PROP-017

One image shall be designated as the primary cover image.

FR-PROP-018

Media ordering shall be configurable through drag-and-drop functionality.

Property Documents

FR-PROP-019

Properties shall support multiple document attachments.

FR-PROP-020

Document visibility shall be configurable by permission.

Property History

FR-PROP-021

The system shall maintain a complete history of changes made to every property.

FR-PROP-022

Historical records shall not be editable.

Property Archiving

FR-PROP-023

Completed listings shall be archived rather than permanently deleted.

FR-PROP-024

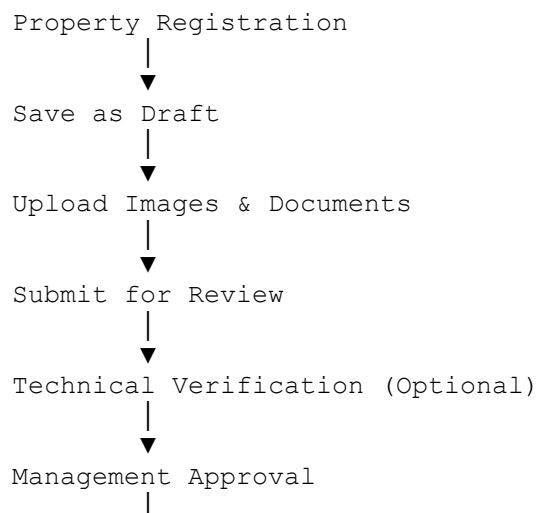
Archived properties may be restored by authorized users.

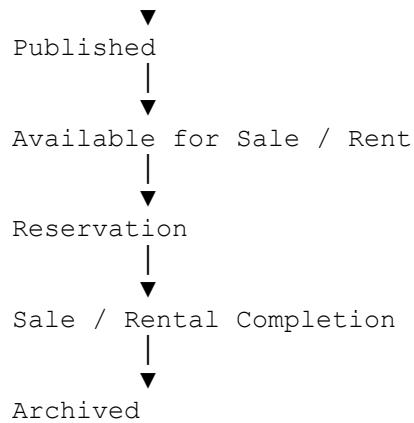
BM-001 — PROPERTY MANAGEMENT MODULE (Continued)

14. Property Workflows

The Property Management Module shall support configurable workflows. Organizations may adopt the default workflow or customize it according to their operational processes.

14.1 Default Property Registration Workflow





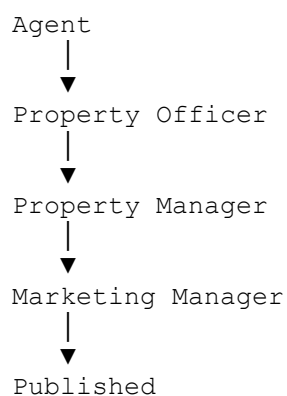
14.2 Property Editing Workflow

1. User selects a property.
 2. System verifies edit permissions.
 3. User updates information.
 4. System validates the changes.
 5. Audit record is created.
 6. If approval is enabled, the property returns to **Pending Review**.
 7. Otherwise, changes become effective immediately.
-

14.3 Property Approval Workflow

Where enabled, the platform shall support multi-step approval workflows.

Example:



Organizations may configure the number of approval stages.

15. Property Business Rules

BR-PROP-001

Every property shall have a unique Property Reference Number.

BR-PROP-002

A property shall belong to at least one Property Type.

BR-PROP-003

A property shall belong to at least one Property Category.

BR-PROP-004

Published properties shall contain at minimum:

- Title
 - Description
 - Primary Image
 - Price
 - Location
 - Status
-

BR-PROP-005

Properties missing mandatory information shall not be published.

BR-PROP-006

Only approved properties may be published.

BR-PROP-007

Archived properties shall not appear in public searches.

BR-PROP-008

Deleting a property shall require elevated permissions.

BR-PROP-009

Soft deletion shall be used by default.

BR-PROP-010

Property ownership history shall never be permanently lost.

16. Property Duplication

The platform shall allow authorized users to duplicate an existing property record.

Duplicated records shall:

- Generate a new Property Reference Number.
- Copy media (configurable).
- Copy documents (optional).
- Reset workflow status to Draft.
- Require approval before publication.

This feature is intended to reduce repetitive data entry for similar listings.

17. Property Templates

Organizations may define templates for recurring property types.

Example templates:

- Apartment
- Residential House
- Office Building
- Commercial Plot

- Farm
- Warehouse

Templates may include:

- Default amenities.
 - Default descriptions.
 - Default document requirements.
 - Default pricing structures.
 - Default SEO settings.
-

18. Bulk Import & Export

The system shall support bulk property operations.

Import

Supported formats:

- CSV
- Excel
- JSON (future)

Imported data shall undergo validation before processing.

Invalid records shall be rejected with detailed error reports.

Export

Supported formats:

- PDF
- Excel
- CSV

Exports shall respect user permissions.

19. Property Versioning

The platform shall maintain historical versions of significant property changes.

Examples:

- Price changes
- Description updates
- Status changes
- Ownership updates
- Location corrections

Users with permission shall be able to view version history.

Historical versions shall be read-only.

20. Related Properties

The platform shall automatically recommend similar properties.

Matching criteria may include:

- Property Type
- Category
- Location
- Price Range
- Bedrooms
- Amenities
- Availability

Administrators may override automatic recommendations manually.

21. Featured Properties

Administrators may designate properties as **Featured**.

Featured properties shall receive priority display on:

- Homepage
- Search Results
- Category Pages
- Landing Pages
- Marketing Campaigns

The number of featured properties displayed shall be configurable.

22. Property SEO

Each property shall support Search Engine Optimization.

Configurable SEO fields include:

- SEO Title
- Meta Description
- Meta Keywords
- Canonical URL
- Open Graph Image
- Social Sharing Image

The system shall automatically generate SEO defaults where not manually specified.

23. Notifications

The module shall publish events for:

- Property Created
- Property Updated
- Property Submitted for Approval
- Property Approved
- Property Rejected
- Property Published
- Property Archived
- Price Changed
- Availability Changed
- Media Uploaded

Supported delivery channels:

- In-App
- Email
- SMS
- WhatsApp
- Push Notifications (future)

Notification templates shall be configurable.

24. Reports

The Property Management Module shall provide the following reports:

Operational Reports

- Property Register
- Available Properties
- Sold Properties
- Rented Properties
- Archived Properties
- Pending Approvals

Financial Reports

- Properties by Value
- Revenue by Property
- Commission Summary

Marketing Reports

- Most Viewed Properties
- Most Inquired Properties
- Featured Property Performance

Administrative Reports

- Property Audit Trail
- User Activity
- Approval History
- Status Distribution

All reports shall support:

- Filtering
- Sorting
- Grouping
- Export
- Printing
- Scheduled generation (future)

25. Configuration Options

Administrators shall configure:

- Property Number Format
- Approval Workflow
- Mandatory Fields
- Listing Expiry
- Featured Property Limits
- Default Currency

- Default Measurement Units
 - Image Compression
 - Maximum Upload Size
 - Watermark Settings
 - Allowed File Types
 - Auto-Publishing Rules
 - SEO Defaults
 - Property Visibility Rules
-

26. Permissions Matrix

The module shall expose at least the following permissions:

- View Properties
 - Create Property
 - Edit Property
 - Delete Property
 - Restore Property
 - Archive Property
 - Approve Property
 - Reject Property
 - Publish Property
 - Unpublish Property
 - Upload Media
 - Remove Media
 - Manage Documents
 - Export Properties
 - Import Properties
 - Print Property Reports
 - Configure Property Settings
-

27. Acceptance Criteria

The Property Management Module shall be considered complete when:

- Properties can be created, edited, and archived.
- Property reference numbers are unique.
- Configured approval workflows operate correctly.
- Search returns accurate results.
- Media uploads function correctly.
- Documents are securely stored.
- Audit records are generated.
- Notifications are published.
- Reports generate correctly.
- Permissions are consistently enforced.

- Archived properties remain recoverable.
 - Public listings reflect only approved and published properties.
-

28. Future Enhancements

The architecture shall accommodate future features without major redesign, including:

- AI-generated property descriptions.
 - AI-assisted image enhancement.
 - AI-powered property recommendations.
 - Automated property valuation.
 - Drone image integration.
 - 3D virtual property tours.
 - GIS parcel overlays.
 - Blockchain-based ownership verification.
 - IoT-enabled smart property monitoring.
 - Market price prediction.
 - Property comparison engine.
 - Multi-language listing translation.
-

29. Module Summary

The Property Management Module is the foundation of the business layer. It serves as the central repository for all property-related information and provides the workflows required to manage properties from registration through marketing, reservation, transaction, and archival. Every other business module interacts with this module through defined interfaces and business events.

BUSINESS MODULE 2

BM-002 — MARKETPLACE MODULE

1. Purpose

The Marketplace Module is the public-facing digital marketplace of the RehoSpace Real Estate Platform. It enables visitors to discover, search, compare, evaluate, and inquire about available properties.

This module transforms the Property Management Module's approved listings into an engaging online marketplace that generates leads, reservations, and sales opportunities.

2. Business Value

The Marketplace Module enables organizations to:

- Showcase available properties professionally.
 - Increase online visibility.
 - Generate qualified sales leads.
 - Improve customer engagement.
 - Reduce response time.
 - Improve property discoverability.
 - Increase conversion rates.
 - Build trust through transparent property information.
-

3. Objectives

The Marketplace Module shall:

- Display approved properties.
 - Provide advanced property search.
 - Support filtering and sorting.
 - Display property details.
 - Support property comparison.
 - Allow users to save favorites.
 - Generate inquiries.
 - Recommend similar properties.
 - Support SEO-friendly property pages.
-

4. Scope

Included

- Public Property Listings
- Property Search
- Advanced Filters
- Property Details Page
- Property Gallery
- Virtual Tours
- Property Comparison

- Favorite Properties
- Recently Viewed Properties
- Inquiry Forms
- Share Property
- Similar Property Recommendations

Excluded

- Property editing
 - Property approval
 - Property pricing management
 - Reservation processing
 - Payment processing
-

5. Marketplace Home

The Marketplace homepage shall display configurable sections.

Examples include:

- Hero Banner
- Featured Properties
- Latest Properties
- Premium Listings
- Popular Locations
- Property Categories
- Property Types
- Recently Added Properties
- Luxury Collection
- Affordable Collection
- Testimonials
- Statistics
- Partner Logos
- Blog Highlights
- Call-to-Action Sections

Each section shall be individually configurable and may be enabled, disabled, or reordered by administrators.

6. Property Listings

Each property card shall display configurable information, including:

- Cover Image

- Property Reference Number
 - Property Title
 - Property Type
 - Category
 - Price
 - Currency
 - Location
 - Property Status
 - Bedrooms
 - Bathrooms
 - Area
 - Featured Badge
 - New Listing Badge
 - Verification Badge
 - Favorite Button
 - Compare Button
 - View Details Button
-

7. Advanced Search

The Marketplace shall provide powerful search capabilities.

Users shall be able to search using:

- Keywords
- Property Reference Number
- Property Type
- Category
- Listing Type (Sale/Rent)
- Country
- Region
- District
- City
- Price Range
- Bedrooms
- Bathrooms
- Area Range
- Amenities
- Furnished Status
- Property Status
- Year Built
- Agent
- Owner (optional)
- Featured Only
- Verified Only

Search shall support partial matching where appropriate.

8. Sorting

Search results shall support sorting by:

- Newest
- Oldest
- Lowest Price
- Highest Price
- Most Viewed
- Most Popular
- Featured First
- Alphabetical
- Area
- Bedrooms

Administrators may configure the default sorting option.

9. Property Details Page

The Property Details page shall provide comprehensive information.

Sections include:

Basic Information

- Property Title
- Reference Number
- Category
- Type
- Status
- Price
- Availability

Description

- Summary
- Detailed Description

Gallery

- Images
- Videos

- Floor Plans
- Virtual Tours

Features & Amenities

- Bedrooms
- Bathrooms
- Parking
- Utilities
- Security
- Outdoor Features
- Smart Features

Location

- Address
- Interactive Map
- Nearby Schools
- Hospitals
- Shopping Centers
- Public Transport
- Places of Worship
- Other Landmarks

Documents

Where permitted, visitors may download:

- Brochures
- Floor Plans
- Survey Maps
- Information Sheets

Contact Section

Visitors may:

- Call Agent
 - Email Agent
 - WhatsApp Agent
 - Book Appointment
 - Request Viewing
 - Submit Inquiry
-

10. Property Gallery

The gallery shall support:

- Image Slideshow
 - Fullscreen Preview
 - Zoom
 - Image Captions
 - Image Ordering
 - Lazy Loading
 - Video Playback
 - 360° Images (Future)
 - Virtual Reality Tours (Future)
-

11. Property Comparison

Visitors may compare multiple properties side-by-side.

Comparison attributes include:

- Price
- Area
- Bedrooms
- Bathrooms
- Parking
- Location
- Amenities
- Property Type
- Property Status
- Year Built
- Energy Features
- Listing Date

The maximum number of properties in a comparison shall be configurable.

12. Favorite Properties

Authenticated users may:

- Save favorites
- Remove favorites
- Organize favorites into collections (Future)
- Receive notifications when favorite properties change

Examples of tracked changes:

- Price reduction
 - Status update
 - New media
 - Reservation
 - Sale
-

13. Recently Viewed Properties

The Marketplace shall maintain a configurable history of recently viewed properties to improve user experience and facilitate quick navigation.

14. Property Sharing

Visitors shall be able to share properties via:

- WhatsApp
- Facebook
- X (Twitter)
- LinkedIn
- Email
- Copy Link

The system shall generate SEO-friendly share previews using Open Graph metadata.

15. Inquiry Management

Visitors may submit inquiries from any property page.

Inquiry form fields may include:

- Full Name
- Email Address
- Phone Number
- Preferred Contact Method
- Subject
- Message
- Preferred Visit Date
- Budget Range
- Additional Notes

Upon submission:

- A CRM lead shall be created.
 - Notifications shall be sent to assigned staff.
 - A confirmation shall be sent to the customer.
-

16. Similar Property Recommendations

The platform shall recommend similar properties using configurable criteria, such as:

- Same property type
 - Same location
 - Similar price range
 - Similar amenities
 - Same category
 - Recently viewed preferences (Future AI)
-

17. Functional Requirements

FR-MARKET-001

Only approved and published properties shall appear in the Marketplace.

FR-MARKET-002

Search results shall be permission-aware where restricted listings exist.

FR-MARKET-003

Property URLs shall be SEO-friendly.

FR-MARKET-004

The Marketplace shall support pagination and infinite scrolling (configurable).

FR-MARKET-005

Property pages shall support structured data (Schema.org) for improved search engine indexing.

FR-MARKET-006

The system shall record property views for analytics.

FR-MARKET-007

Property inquiry forms shall integrate directly with the CRM Module.

FR-MARKET-008

Favorite properties shall be synchronized with authenticated user accounts.

FR-MARKET-009

The Marketplace shall be fully responsive across desktop, tablet, and mobile devices.

FR-MARKET-010

The system shall support configurable featured listing placements.

18. Business Rules

BR-MARKET-001

Only published properties shall be visible to the public.

BR-MARKET-002

Archived, suspended, or deleted properties shall not appear in public searches.

BR-MARKET-003

Every public listing shall display at least one approved cover image.

BR-MARKET-004

Inquiry submissions shall automatically create CRM leads.

BR-MARKET-005

Property statistics (views, favorites, inquiries) shall be retained even after listing updates.

19. Reports

The Marketplace Module shall provide:

- Most Viewed Properties

- Most Favorited Properties
 - Most Shared Properties
 - Inquiry Conversion Report
 - Search Trends
 - Popular Locations
 - Property Performance Report
 - Visitor Engagement Report
 - Click-Through Rate (CTR) Report
 - SEO Performance Summary (where integrated)
-

20. Configuration

Administrators shall configure:

- Default search filters
 - Default sorting
 - Listings per page
 - Maximum comparison count
 - Featured listing rules
 - Inquiry form fields
 - Public contact visibility
 - Gallery layout
 - Similar property algorithm
 - Property URL structure
 - SEO defaults
 - Social sharing settings
-

21. Permissions Matrix

The Marketplace Module shall expose permissions including:

- View Marketplace
 - Configure Marketplace
 - Manage Featured Listings
 - Moderate Public Reviews (Future)
 - Manage Homepage Sections
 - Manage Search Configuration
 - Export Marketplace Reports
 - View Analytics
-

22. Acceptance Criteria

The Marketplace Module shall be considered complete when:

- Approved properties appear correctly on the public website.
 - Advanced search returns accurate results.
 - Property comparison functions correctly.
 - Favorite properties are stored and retrieved successfully.
 - Inquiry submissions create CRM leads.
 - Property view statistics are recorded.
 - SEO-friendly URLs are generated.
 - Marketplace pages are responsive.
 - Social sharing functions correctly.
 - Reports accurately reflect marketplace activity.
-

23. Future Enhancements

The architecture shall support future capabilities including:

- AI-powered property recommendations.
- Personalized search results.
- Voice-based property search.
- Interactive GIS map search.
- Augmented Reality (AR) property visualization.
- AI chatbot-assisted property discovery.
- Saved search alerts.
- Heat maps showing property demand.
- Real-time availability indicators.
- Property bidding and auction support.

BUSINESS MODULE 3

BM-003 — CUSTOMER RELATIONSHIP MANAGEMENT (CRM)

1. Purpose

The Customer Relationship Management (CRM) Module manages the complete lifecycle of customer interactions, from the first inquiry through qualification, assignment, follow-up, negotiation, reservation, purchase or rental, and post-sale engagement.

The CRM shall serve as the central repository for all customer information and sales activities, enabling organizations to build long-term relationships while improving sales efficiency and customer satisfaction.

2. Business Value

The CRM Module enables organizations to:

- Capture leads from multiple sources.
 - Maintain a centralized customer database.
 - Improve lead response times.
 - Track customer interactions.
 - Increase lead conversion rates.
 - Automate follow-ups.
 - Measure sales performance.
 - Improve customer retention.
 - Support data-driven sales decisions.
-

3. Objectives

The CRM Module shall:

- Capture customer inquiries.
 - Manage customer profiles.
 - Manage leads.
 - Track sales opportunities.
 - Schedule follow-ups.
 - Record communications.
 - Assign leads to agents.
 - Monitor sales pipelines.
 - Convert leads into customers.
 - Generate CRM analytics.
-

4. Scope

Included

- Lead Capture
- Lead Management
- Customer Profiles
- Opportunity Management

- Follow-up Scheduling
- Communication History
- Agent Assignment
- Sales Pipeline
- Activity Timeline
- Notes & Attachments
- Customer Segmentation
- CRM Reports

Excluded

- Property Management
 - Payment Processing
 - Marketing Campaign Creation (handled by Marketing Module)
 - Appointment Calendar (integrated with Appointment Module)
-

5. CRM Data Model

The CRM shall manage the following entities:

- Leads
 - Customers
 - Organizations (for corporate clients)
 - Contacts
 - Sales Opportunities
 - Activities
 - Follow-ups
 - Notes
 - Tasks
 - Documents
 - Communication Logs
 - Customer Preferences
 - Property Interests
-

6. Lead Sources

Every lead shall record its origin. Configurable lead sources include:

- Website Inquiry
- Property Detail Page
- Walk-in Customer
- Phone Call
- Email
- WhatsApp

- Facebook
- Instagram
- LinkedIn
- Google Ads
- Referral
- Existing Customer
- Trade Exhibition
- Marketing Campaign
- Manual Entry
- API Integration (Future)

Administrators may define additional lead sources.

7. Lead Information

Each lead shall include:

Personal Information

- Lead ID
 - Full Name
 - Organization Name (optional)
 - Email Address
 - Mobile Number
 - Alternative Number
 - Preferred Communication Method
-

Address Information

- Country
 - Region
 - District
 - City
 - Street
 - Postal Address
-

Business Information

- Interested Property
- Property Type
- Budget Range
- Preferred Location

- Purchase Purpose
 - Financing Method
 - Expected Purchase Date
-

CRM Information

- Lead Source
 - Assigned Agent
 - Lead Owner
 - Lead Score
 - Lead Status
 - Priority
 - Tags
 - Created Date
 - Last Contact Date
 - Next Follow-up Date
-

8. Lead Status Workflow

Default lead statuses shall include:

New

↓

Contacted

↓

Qualified

↓

Property Suggested

↓

Viewing Scheduled

↓

Negotiation

↓

Reservation

↓

Converted

↓

Closed Won

Alternative outcomes include:

- Closed Lost
- Dormant
- Disqualified

Organizations may customize the workflow.

9. Lead Assignment

The CRM shall support multiple assignment strategies.

Manual Assignment

Managers assign leads individually.

Round Robin Assignment

The system distributes leads evenly among available agents.

Territory Assignment

Leads are assigned based on:

- Region
 - District
 - Branch
 - Property Type
-

Load Balancing

Assignment based on:

- Current workload
 - Open opportunities
 - Agent availability
-

10. Customer Profiles

Once a lead is converted, a Customer Profile shall be created automatically.

Customer profiles shall maintain:

- Personal information
- Communication history
- Purchased properties
- Rental history
- Reservation history
- Payment history
- Favorite properties
- Documents
- Customer preferences
- Notes
- Activity timeline

The profile becomes the customer's permanent record.

11. Sales Opportunities

Each qualified lead may generate one or more sales opportunities.

Opportunity fields include:

- Opportunity Number
- Customer
- Property
- Estimated Value
- Expected Closing Date
- Probability of Success
- Assigned Agent
- Current Stage
- Expected Commission

Opportunity stages shall be configurable.

12. Activities & Follow-ups

The CRM shall manage activities such as:

- Phone Calls

- Meetings
- Property Viewings
- Emails
- WhatsApp Conversations
- SMS
- Internal Discussions
- Site Visits

Each activity shall include:

- Date
- Time
- Responsible User
- Outcome
- Next Action
- Related Customer
- Related Property

13. Communication History

Every customer interaction shall be recorded in chronological order.

Examples:

- Emails Sent
- Emails Received
- Phone Calls
- SMS Messages
- WhatsApp Messages
- Property Recommendations
- Meeting Notes
- Follow-up Outcomes

This creates a complete customer timeline.

14. Functional Requirements

Lead Management

FR-CRM-001

The system shall create leads from website inquiry forms.

FR-CRM-002

Users shall create leads manually.

FR-CRM-003

Duplicate lead detection shall be supported using configurable matching rules.

FR-CRM-004

Leads shall be assignable to individual agents or teams.

Customer Management

FR-CRM-005

Qualified leads shall be converted into customers without re-entering information.

FR-CRM-006

Customer profiles shall consolidate all interactions across modules.

Opportunity Management

FR-CRM-007

Multiple opportunities may exist for one customer.

FR-CRM-008

Opportunity stage changes shall be recorded in the audit log.

Follow-ups

FR-CRM-009

Users shall schedule follow-up activities.

FR-CRM-010

The system shall generate reminders for overdue follow-ups.

Communication

FR-CRM-011

Communication history shall be searchable.

FR-CRM-012

Attachments shall be linked to customer records.

Automation

FR-CRM-013

The system shall automatically create CRM leads from Marketplace inquiries.

FR-CRM-014

The system shall automatically notify assigned agents of new leads.

FR-CRM-015

Automatic lead scoring rules shall be configurable.

15. Business Rules

BR-CRM-001

Every lead shall have an identifiable source.

BR-CRM-002

A converted lead shall not be deleted if linked to business transactions.

BR-CRM-003

Every customer shall have a unique Customer ID.

BR-CRM-004

Lead assignment changes shall be recorded.

BR-CRM-005

Overdue follow-ups shall remain visible until completed.

BR-CRM-006

Communication history shall be immutable once recorded.

16. Notifications

The CRM Module shall publish notifications for:

- New Lead Created
- Lead Assigned
- Lead Reassigned
- Follow-up Due
- Follow-up Overdue
- Opportunity Updated
- Customer Converted
- Property Viewing Scheduled
- Reservation Initiated
- Deal Won
- Deal Lost

Notifications shall support:

- In-App
- Email
- SMS
- WhatsApp

17. Reports

The CRM Module shall provide:

Lead Reports

- Leads by Source
- Leads by Status
- Leads by Agent
- Leads by Region
- Lead Conversion Rate

Customer Reports

- Active Customers
- New Customers
- Returning Customers
- Customer Growth Trends

Opportunity Reports

- Sales Pipeline
- Forecast Revenue
- Won Opportunities
- Lost Opportunities
- Average Sales Cycle

Performance Reports

- Agent Performance
- Response Time Analysis
- Follow-up Compliance
- Conversion Funnel
- Activity Summary

All reports shall support filtering, grouping, exporting, printing, and scheduling.

18. Configuration

Administrators shall configure:

- Lead numbering format
 - Customer numbering format
 - Lead sources
 - Lead statuses
 - Opportunity stages
 - Follow-up reminder intervals
 - Lead scoring rules
 - Assignment strategy
 - Customer categories
 - Duplicate detection rules
 - Notification templates
-

19. Permissions Matrix

The CRM Module shall expose permissions including:

- View Leads
- Create Leads
- Edit Leads
- Delete Leads
- Convert Leads
- Assign Leads

- Reassign Leads
 - Manage Customers
 - View Opportunities
 - Manage Opportunities
 - View Reports
 - Export Reports
 - Configure CRM
-

20. Acceptance Criteria

The CRM Module shall be accepted when:

- Leads are captured from all configured sources.
 - Duplicate detection functions correctly.
 - Lead assignment follows configured rules.
 - Customer profiles are created upon conversion.
 - Follow-up reminders are generated.
 - Communication history is preserved.
 - CRM reports are accurate.
 - Permissions are consistently enforced.
 - Marketplace inquiries automatically generate CRM leads.
 - Audit records capture critical CRM activities.
-

21. Future Enhancements

The architecture shall support future capabilities including:

- AI lead scoring.
 - AI sales forecasting.
 - AI conversation summarization.
 - WhatsApp Business API integration.
 - Voice call recording and transcription.
 - Customer sentiment analysis.
 - Automated nurturing campaigns.
 - Predictive customer churn analysis.
 - Integration with external CRM systems.
 - AI-assisted property recommendations based on customer behavior.
-

22. Module Summary

The CRM Module is the sales engine of the RehoSpace Real Estate Platform. It centralizes all customer relationships, manages the progression from inquiry to transaction, and provides the operational tools needed to maximize conversions and maintain long-term customer engagement.

BUSINESS MODULE 4

BM-004 — PROPERTY OWNERS MANAGEMENT MODULE

1. Purpose

The Property Owners Management Module provides a centralized system for registering, verifying, managing, and maintaining relationships with individuals and organizations that own properties listed within the RehoSpace Real Estate Platform.

It separates **property ownership** from **property marketing**, enabling one owner to have multiple properties, multiple owners to jointly own a property, and ownership to change over time while preserving historical records.

The module serves as the authoritative source of ownership information across the platform.

2. Business Value

The Property Owners Management Module enables organizations to:

- Maintain an accurate owner registry.
 - Track ownership history.
 - Manage legal ownership documentation.
 - Improve communication with owners.
 - Simplify commission calculations.
 - Manage ownership transfers.
 - Support multiple ownership structures.
 - Provide transparency and accountability.
-

3. Objectives

The module shall:

- Register property owners.
 - Verify owner identity.
 - Manage owner documents.
 - Link owners to properties.
 - Support multiple owners per property.
 - Support one owner owning multiple properties.
 - Track ownership history.
 - Manage owner contracts.
 - Maintain communication history.
 - Generate owner reports.
-

4. Scope

Included

- Owner Registration
- Individual Owners
- Corporate Owners
- Owner Verification
- Ownership Management
- Ownership Transfers
- Ownership History
- Owner Documents
- Owner Contracts
- Commission Agreements
- Owner Notes
- Owner Communication History
- Owner Portal (Optional)

Excluded

- Customer Relationship Management
 - Property Marketing
 - Property Reservations
 - Financial Accounting
-

5. Owner Types

The platform shall support configurable owner types.

Examples include:

- Individual
- Company

- Government Institution
- Cooperative Society
- NGO
- Religious Institution
- Bank
- Investment Group
- Estate Administrator
- Legal Representative

Administrators may define additional owner types.

6. Owner Information

Each owner profile shall maintain comprehensive information.

Basic Information

- Owner ID
 - Owner Type
 - Full Name / Company Name
 - National ID / Registration Number
 - Tax Identification Number (TIN)
 - Date of Birth / Incorporation Date
 - Gender (for individuals)
 - Nationality
-

Contact Information

- Mobile Number
 - Alternative Number
 - Email Address
 - WhatsApp Number
 - Website (for organizations)
-

Address Information

- Country
- Region
- District
- City
- Ward
- Street

- Postal Address
 - Physical Address
-

Business Information

- Preferred Communication Method
 - Preferred Payment Method
 - Bank Details
 - Tax Category
 - Commission Agreement
 - Account Status
-

7. Identity Verification

The platform shall support owner verification.

Verification may include:

- National ID
- Passport
- Driving License
- Business Registration Certificate
- Tax Certificate
- Utility Bill
- Biometric Verification (Future)

Verification status shall include:

- Pending
 - Verified
 - Rejected
 - Expired
-

8. Ownership Model

The platform shall support flexible ownership structures.

One Owner → Many Properties

One owner may own any number of properties.

Many Owners → One Property

A property may have multiple owners.

Ownership percentages shall be configurable.

Example:

- Owner A — 60%
- Owner B — 40%

The total ownership allocation must equal 100%.

Ownership History

The system shall maintain historical ownership records.

Historical records shall include:

- Previous owner
- New owner
- Transfer date
- Transfer reason
- Supporting documents
- Authorizing officer

Historical ownership records shall never be permanently deleted.

9. Owner Documents

The system shall support secure storage of owner-related documents.

Examples include:

- National ID
- Passport
- Company Registration Certificate
- TIN Certificate
- Ownership Agreement
- Agency Agreement
- Power of Attorney
- Bank Details
- Compliance Certificates
- Other Legal Documents

Documents shall support:

- Versioning
 - Expiry dates
 - Renewal reminders
 - Secure downloads
-

10. Commission Agreements

Organizations may define commission agreements with owners.

Supported commission models include:

- Fixed Amount
- Percentage of Sale
- Percentage of Rent
- Tiered Commission
- Negotiated Commission

The module shall maintain a complete history of commission agreements.

11. Owner Contracts

The module shall manage contracts between the organization and property owners.

Contract information includes:

- Contract Number
- Contract Type
- Effective Date
- Expiry Date
- Renewal Status
- Commission Terms
- Exclusive / Non-Exclusive Agreement
- Digital Copy
- Status

Contract expiry reminders shall be configurable.

12. Functional Requirements

Owner Registration

FR-OWNER-001

The system shall register new property owners.

FR-OWNER-002

Each owner shall receive a unique Owner ID.

FR-OWNER-003

Duplicate owner detection shall be supported.

Ownership Management

FR-OWNER-004

Users shall associate one or more owners with a property.

FR-OWNER-005

Ownership percentages shall be validated.

FR-OWNER-006

Ownership changes shall be recorded in history.

Verification

FR-OWNER-007

The platform shall support configurable verification workflows.

FR-OWNER-008

Only verified owners may be associated with published properties if verification is mandatory.

Contracts

FR-OWNER-009

Multiple contracts may exist for one owner.

FR-OWNER-010

Contract expiry reminders shall be generated automatically.

Documents

FR-OWNER-011

Owner documents shall support version history.

FR-OWNER-012

Access to sensitive documents shall be permission-controlled.

13. Business Rules

BR-OWNER-001

Every owner shall have a unique Owner ID.

BR-OWNER-002

Ownership percentages for a property shall total 100%.

BR-OWNER-003

Ownership history shall never be deleted.

BR-OWNER-004

Verified ownership shall be required where mandated by organizational policy.

BR-OWNER-005

Expired contracts shall be clearly identified and reported.

14. Notifications

The module shall publish notifications for:

- Owner Registered
- Verification Submitted

- Verification Approved
- Verification Rejected
- Ownership Changed
- Contract Expiring
- Contract Renewed
- New Property Assigned
- Commission Agreement Updated

Supported channels:

- In-App
 - Email
 - SMS
 - WhatsApp
-

15. Reports

The Property Owners Management Module shall provide:

Owner Reports

- Owner Register
- Verified Owners
- Unverified Owners
- Active Owners
- Inactive Owners

Ownership Reports

- Properties by Owner
- Owners by Property
- Joint Ownership Register
- Ownership History Report

Contract Reports

- Active Contracts
- Expiring Contracts
- Expired Contracts
- Commission Agreements

Compliance Reports

- Missing Documents
- Expired Verification
- Outstanding Renewals

All reports shall support filtering, exporting, printing, and scheduling.

16. Configuration

Administrators shall configure:

- Owner numbering format
 - Owner types
 - Verification requirements
 - Supported document types
 - Commission models
 - Contract templates
 - Expiry reminder intervals
 - Ownership validation rules
 - Required profile fields
-

17. Permissions Matrix

The module shall expose permissions including:

- View Owners
 - Create Owner
 - Edit Owner
 - Delete Owner
 - Verify Owner
 - Manage Ownership
 - Manage Contracts
 - Upload Documents
 - View Sensitive Documents
 - Export Owner Reports
 - Configure Owner Module
-

18. Acceptance Criteria

The module shall be accepted when:

- Owners can be registered and updated.
- Duplicate detection functions correctly.
- Multiple ownership structures are supported.
- Ownership percentages are validated.
- Verification workflows operate correctly.

- Contracts and documents are securely managed.
 - Ownership history is preserved.
 - Reports are generated accurately.
 - Permissions are consistently enforced.
-

19. Future Enhancements

The architecture shall support future capabilities including:

- Electronic Know Your Customer (eKYC) verification.
 - Digital contract signing.
 - Owner self-service portal.
 - Mobile owner application.
 - Biometric identity verification.
 - Blockchain-based ownership registry integration.
 - Government land registry integration.
 - Automated commission settlements.
 - AI-powered compliance monitoring.
-

20. Module Summary

The Property Owners Management Module establishes a trusted registry of property owners and ownership relationships. It ensures legal accountability, supports complex ownership structures, preserves historical ownership records, and integrates tightly with the Property Management, CRM, Reservation, Payments, and Reporting modules. Its design emphasizes data integrity, compliance, and long-term maintainability, making it a foundational business capability for the RehoSpace Real Estate Platform.

BUSINESS MODULE 5

BM-005 — AGENTS MANAGEMENT MODULE

1. Purpose

The Agents Management Module provides a centralized system for managing all real estate agents, brokers, sales executives, and field officers operating within the organization.

The module manages the complete lifecycle of an agent, from recruitment and onboarding through assignment, performance tracking, commission management, leave, availability, and offboarding.

It also serves as the operational link between customers, properties, and sales processes.

2. Business Value

The Agents Management Module enables organizations to:

- Maintain a centralized registry of agents.
 - Improve lead distribution.
 - Monitor agent performance.
 - Track commissions and incentives.
 - Manage territories and branches.
 - Increase accountability.
 - Improve customer service.
 - Support data-driven management decisions.
-

3. Objectives

The module shall:

- Register agents.
 - Manage agent profiles.
 - Assign territories.
 - Assign leads.
 - Assign properties.
 - Monitor workload.
 - Track performance.
 - Calculate commissions.
 - Generate productivity reports.
 - Support organizational hierarchies.
-

4. Scope

Included

- Agent Registration
- Agent Profiles
- Branch Assignment

- Territory Management
- Property Assignment
- Lead Assignment
- Availability Management
- Commission Management
- Performance Monitoring
- Targets & KPIs
- Attendance Integration (Future)
- Leave Management (Optional)
- Agent Portal

Excluded

- Payroll Processing
 - Human Resource Management
 - Recruitment
 - Accounting
-

5. Agent Types

The platform shall support configurable agent classifications.

Examples include:

- Sales Agent
- Senior Sales Agent
- Property Consultant
- Broker
- Marketing Executive
- Customer Relations Officer
- Leasing Agent
- Field Officer
- Branch Sales Manager
- Regional Sales Manager

Organizations may define additional classifications.

6. Agent Profile

Each agent profile shall include:

Personal Information

- Agent ID
 - Employee Number
 - Full Name
 - Gender
 - Date of Birth
 - Nationality
 - National ID
 - Passport Number
 - Photograph
-

Contact Information

- Mobile Number
 - Alternative Number
 - Email
 - WhatsApp
 - Physical Address
-

Employment Information

- Employment Date
 - Employment Status
 - Branch
 - Department
 - Supervisor
 - Position
 - Employment Type
-

Professional Information

- License Number
 - Professional Certifications
 - Years of Experience
 - Areas of Specialization
 - Languages Spoken
-

Banking Information

- Bank Name
- Account Number

- Mobile Money Details
 - Preferred Payment Method
-

7. Territory Management

The platform shall allow assignment of one or more territories to an agent.

Territories may include:

- Country
- Region
- District
- City
- Ward
- Development Project
- Property Category

Territory overlap shall be configurable.

8. Property Assignment

Properties may be assigned to:

- One Agent
- Multiple Agents
- A Sales Team

Assignment history shall be maintained.

The system shall record:

- Assignment Date
 - Assigned By
 - Assignment Reason
 - Assignment Status
-

9. Lead Assignment

The module integrates with CRM to distribute leads.

Supported methods include:

- Manual Assignment
 - Round Robin
 - Territory-Based
 - Availability-Based
 - Workload-Based
 - Skill-Based (Future)
 - AI-Based Assignment (Future)
-

10. Availability Management

Agents shall maintain availability status.

Default statuses include:

- Available
- Busy
- In Meeting
- On Property Visit
- On Leave
- Offline
- Suspended

Availability may be updated manually or automatically based on calendar events.

11. Targets & KPIs

The platform shall support configurable performance targets.

Examples:

- Monthly Sales Value
- Monthly Property Listings
- Qualified Leads
- Closed Deals
- Property Viewings
- Customer Satisfaction Score
- Response Time
- Conversion Rate

Targets may be assigned per:

- Agent
- Team
- Branch

- Region
-

12. Commission Management

The module shall support multiple commission models.

Supported models include:

- Fixed Amount
- Percentage of Sale
- Percentage of Rental
- Tiered Commission
- Team Commission
- Bonus Commission

Commission calculations shall be configurable and may integrate with the Finance Module.

13. Agent Performance Dashboard

Each agent shall have a dashboard displaying:

- Assigned Leads
 - Assigned Properties
 - Pending Follow-ups
 - Upcoming Appointments
 - Monthly Sales
 - Commission Earned
 - Active Opportunities
 - Target Progress
 - Customer Rating (Future)
-

14. Functional Requirements

Agent Management

FR-AGENT-001

The system shall register new agents.

FR-AGENT-002

Each agent shall receive a unique Agent ID.

FR-AGENT-003

Duplicate detection shall prevent duplicate agent records.

Assignment

FR-AGENT-004

Users shall assign properties to agents.

FR-AGENT-005

Users shall assign leads to agents.

FR-AGENT-006

Assignment history shall be retained.

Availability

FR-AGENT-007

Agents shall update availability status.

FR-AGENT-008

Managers shall view real-time availability.

Performance

FR-AGENT-009

The system shall calculate KPI performance.

FR-AGENT-010

The system shall calculate earned commissions.

FR-AGENT-011

Managers shall configure targets.

Dashboards

FR-AGENT-012

Agents shall have personalized dashboards.

FR-AGENT-013

Managers shall have team dashboards.

15. Business Rules

BR-AGENT-001

Every agent shall have a unique Agent ID.

BR-AGENT-002

Inactive agents shall not receive new assignments.

BR-AGENT-003

Completed assignments shall remain in history.

BR-AGENT-004

Commission calculations shall follow configured commission rules.

BR-AGENT-005

Performance metrics shall update automatically as business transactions occur.

16. Notifications

The module shall publish notifications for:

- New Lead Assigned
- Property Assigned
- Target Achieved
- Target Missed
- Commission Calculated
- Commission Approved

- Appointment Assigned
- Territory Updated
- Leave Approved
- Performance Review Due

Supported channels:

- In-App
 - Email
 - SMS
 - WhatsApp
-

17. Reports

The Agents Management Module shall provide:

Agent Reports

- Agent Register
- Active Agents
- Inactive Agents
- Agent Availability

Assignment Reports

- Property Assignments
- Lead Assignments
- Territory Assignments

Performance Reports

- Sales by Agent
- Listings by Agent
- Conversion Rate
- Average Response Time
- Target Achievement
- Commission Summary
- Customer Satisfaction (Future)

Management Reports

- Branch Performance
- Team Performance
- Regional Performance
- Top Performing Agents
- Underperforming Agents

All reports shall support filtering, grouping, exporting, printing, and scheduling.

18. Configuration

Administrators shall configure:

- Agent numbering format
 - Agent types
 - Territory rules
 - Assignment strategy
 - KPI definitions
 - Commission models
 - Target periods
 - Availability statuses
 - Performance review intervals
 - Notification templates
-

19. Permissions Matrix

The module shall expose permissions including:

- View Agents
 - Create Agent
 - Edit Agent
 - Archive Agent
 - Assign Properties
 - Assign Leads
 - Manage Territories
 - Configure Commissions
 - View Performance Reports
 - Export Reports
 - Configure Agent Module
-

20. Acceptance Criteria

The Agents Management Module shall be accepted when:

- Agents can be registered and managed.
- Property and lead assignments function correctly.
- Availability status updates in real time.
- Performance dashboards display accurate information.

- Commission calculations follow configured rules.
 - KPI tracking is accurate.
 - Reports are generated successfully.
 - Permissions are enforced consistently.
 - Assignment and commission histories are preserved.
-

21. Future Enhancements

The architecture shall support future capabilities including:

- AI-based lead assignment.
 - AI performance coaching.
 - GPS-based field activity tracking.
 - Mobile field application.
 - Digital business cards.
 - Electronic contract signing.
 - Attendance and timesheet integration.
 - Customer rating and review system.
 - Gamification with badges and leaderboards.
 - Predictive sales forecasting per agent.
-

22. Module Summary

The Agents Management Module provides comprehensive management of sales personnel, ensuring effective lead distribution, property assignment, performance monitoring, and commission administration. It integrates closely with the CRM, Property Management, Reservation, Appointment, Finance, and Reporting modules to create a complete sales operations ecosystem.

BUSINESS MODULE 6

BM-006 — RESERVATION MANAGEMENT MODULE

1. Purpose

The Reservation Management Module enables prospective buyers or tenants to temporarily reserve properties while completing due diligence, financing, legal verification, or internal approval processes.

It ensures that reserved properties are protected from conflicting transactions, manages reservation validity periods, deposits, approvals, renewals, cancellations, and conversions into finalized sales or rental agreements.

The module acts as the bridge between **CRM, Property Management, Payments, and Sales Transactions**.

2. Business Value

The Reservation Management Module enables organizations to:

- Prevent double booking of properties.
 - Manage reservation lifecycles.
 - Improve transaction transparency.
 - Track reservation deposits.
 - Reduce manual coordination.
 - Increase customer confidence.
 - Automate reservation expiry.
 - Improve sales conversion.
-

3. Objectives

The module shall:

- Create reservations.
 - Hold properties for customers.
 - Manage reservation deposits.
 - Approve reservations.
 - Extend reservation periods.
 - Cancel reservations.
 - Convert reservations into completed transactions.
 - Notify all stakeholders.
 - Maintain reservation history.
-

4. Scope

Included

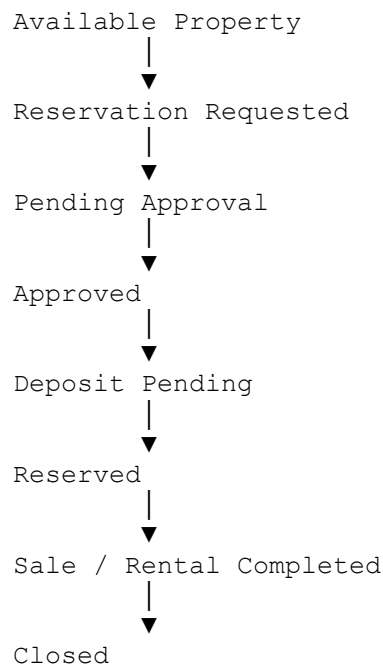
- Reservation Creation
- Reservation Approval
- Reservation Renewal
- Reservation Expiry
- Reservation Cancellation
- Reservation Deposits
- Waiting List
- Reservation History
- Reservation Documents
- Reservation Notes
- Reservation Reports

Excluded

- Final property sale registration
- Rental agreement execution
- Accounting entries
- Legal contract generation

5. Reservation Lifecycle

The default reservation workflow shall be:



Alternative paths include:

- Rejected
- Expired
- Cancelled

- Converted

Organizations may configure additional workflow stages.

6. Reservation Information

Each reservation shall maintain:

Reservation Details

- Reservation Number
 - Reservation Date
 - Reservation Type (Sale / Rent)
 - Reservation Status
 - Reservation Source
 - Reservation Priority
-

Customer Information

- Customer
 - CRM Lead
 - Organization (if applicable)
 - Assigned Agent
-

Property Information

- Property Reference
 - Property Type
 - Property Status
 - Reservation Price
 - Negotiated Price
 - Currency
-

Reservation Terms

- Reservation Start Date
- Reservation Expiry Date
- Reservation Duration
- Renewal Count

- Reservation Conditions
 - Cancellation Policy
-

Financial Information

- Deposit Required
 - Deposit Paid
 - Outstanding Deposit
 - Payment Status
 - Refund Status
-

7. Reservation Types

The platform shall support configurable reservation types, including:

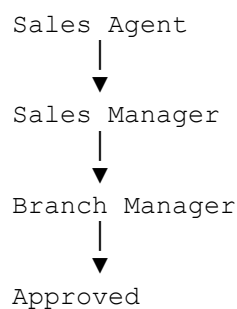
- Standard Reservation
- Priority Reservation
- Corporate Reservation
- Rental Reservation
- Off-plan Reservation
- Pre-launch Reservation
- VIP Reservation

Organizations may define additional reservation types.

8. Reservation Approval

Organizations may enable approval workflows.

Example:



Approval steps shall be configurable.

9. Reservation Deposits

Reservations may require deposits.

Supported deposit models include:

- Fixed Amount
- Percentage of Property Value
- Optional Deposit
- No Deposit

Deposit rules shall be configurable by property type or reservation type.

10. Reservation Extensions

Authorized users may extend reservation periods.

The extension process shall record:

- Previous expiry date
- New expiry date
- Extension reason
- Approved by
- Extension count

Maximum extensions shall be configurable.

11. Reservation Expiry

The system shall automatically monitor reservation expiry.

Upon expiry, configurable actions include:

- Release property automatically.
 - Notify assigned agent.
 - Notify customer.
 - Notify manager.
 - Move customer to waiting list.
 - Archive reservation.
-

12. Waiting List

If a property is already reserved, interested customers may join a waiting list.

The waiting list shall record:

- Customer
- Registration Date
- Priority
- Notes

When a reservation expires or is cancelled, the next eligible customer may be notified automatically.

13. Reservation Documents

Reservations shall support document attachments such as:

- Reservation Form
- Deposit Receipt
- Identification Documents
- Financing Letter
- Internal Approval
- Reservation Agreement

Documents shall support versioning and secure access.

14. Functional Requirements

Reservation Management

FR-RES-001

The system shall create reservations for available properties.

FR-RES-002

Each reservation shall receive a unique Reservation Number.

FR-RES-003

A property shall not have conflicting active reservations unless explicitly permitted.

Approval

FR-RES-004

The system shall support configurable reservation approval workflows.

FR-RES-005

Only approved reservations may proceed to confirmed status.

Deposits

FR-RES-006

The system shall record reservation deposits.

FR-RES-007

Deposit payment status shall synchronize with the Payments Module.

Extensions

FR-RES-008

Authorized users shall extend reservation validity periods.

Expiry

FR-RES-009

Expired reservations shall trigger configured workflow actions automatically.

Waiting List

FR-RES-010

Customers may join waiting lists for unavailable properties.

FR-RES-011

Approved reservations shall convert into sales or rental transactions without duplicate data entry.

15. Business Rules

BR-RES-001

Only available properties may be reserved.

BR-RES-002

A reservation shall not exceed the maximum configured duration unless extended by an authorized user.

BR-RES-003

Expired reservations shall no longer block property availability.

BR-RES-004

Cancelled reservations shall retain their historical records.

BR-RES-005

Reservation deposits shall follow configured refund policies.

BR-RES-006

Every reservation shall be linked to one customer and one property.

16. Notifications

The module shall publish notifications for:

- Reservation Requested
- Reservation Approved
- Reservation Rejected
- Deposit Due
- Deposit Received
- Reservation Extended

- Reservation Expiring Soon
- Reservation Expired
- Reservation Cancelled
- Reservation Converted

Supported channels:

- In-App
 - Email
 - SMS
 - WhatsApp
-

17. Reports

The Reservation Management Module shall provide:

Reservation Reports

- Active Reservations
- Pending Approvals
- Expired Reservations
- Cancelled Reservations
- Converted Reservations

Financial Reports

- Reservation Deposits
- Outstanding Deposits
- Deposit Refunds

Operational Reports

- Reservation Duration Analysis
- Waiting List Report
- Reservation Conversion Rate
- Reservation Activity Log

Reports shall support filtering, grouping, exporting, printing, and scheduled generation.

18. Configuration

Administrators shall configure:

- Reservation numbering format

- Default reservation duration
 - Maximum reservation duration
 - Maximum extensions
 - Approval workflows
 - Deposit rules
 - Refund policies
 - Waiting list behavior
 - Automatic expiry actions
 - Notification templates
-

19. Permissions Matrix

The module shall expose permissions including:

- View Reservations
 - Create Reservation
 - Edit Reservation
 - Cancel Reservation
 - Approve Reservation
 - Reject Reservation
 - Extend Reservation
 - Manage Waiting List
 - Record Deposit
 - Export Reservation Reports
 - Configure Reservation Module
-

20. Acceptance Criteria

The Reservation Management Module shall be accepted when:

- Reservations can be created for eligible properties.
 - Conflicting reservations are prevented according to business rules.
 - Approval workflows function correctly.
 - Deposits synchronize with the Payments Module.
 - Expiry automation operates as configured.
 - Waiting lists function correctly.
 - Reservations convert into transactions without data duplication.
 - Reports are accurate.
 - Notifications are delivered.
 - Permissions are enforced consistently.
-

21. Future Enhancements

The architecture shall support future capabilities including:

- Online customer self-service reservations.
 - Digital reservation agreement signing.
 - AI-based reservation conversion probability scoring.
 - Reservation queue optimization.
 - Dynamic reservation pricing.
 - Multi-property reservation packages.
 - Integration with mortgage and financing providers.
 - QR code-based reservation verification.
 - Blockchain-backed reservation records.
-

22. Module Summary

The Reservation Management Module governs the controlled allocation of properties during the pre-sale and pre-rental process. It safeguards property availability, enforces reservation policies, manages deposits and approvals, and provides a seamless transition from customer interest to finalized transactions. By integrating tightly with the CRM, Property Management, Payments, Notifications, and future Sales modules, it forms a critical component of the RehoSpace Real Estate Platform's transaction lifecycle.

BUSINESS MODULE 7

BM-007 — APPOINTMENT & PROPERTY VIEWING MANAGEMENT MODULE

1. Purpose

The Appointment & Property Viewing Management Module enables customers, agents, property owners, and administrators to efficiently schedule, coordinate, conduct, and monitor property viewings.

It manages the complete lifecycle of appointments, including requests, scheduling, confirmations, reminders, attendance, viewing outcomes, customer feedback, and follow-up actions.

This module integrates tightly with the CRM, Property Management, Agents Management, Reservation Management, Notification Framework, and Calendar services.

2. Business Value

The Appointment & Property Viewing Management Module enables organizations to:

- Eliminate manual appointment scheduling.
- Improve customer experience.
- Prevent scheduling conflicts.
- Increase property viewing attendance.
- Improve lead conversion.
- Track viewing performance.
- Reduce agent workload.
- Maintain complete viewing history.

3. Objectives

The module shall:

- Schedule appointments.
- Manage property viewing requests.
- Allocate agents.
- Prevent scheduling conflicts.
- Send reminders.
- Record attendance.
- Capture viewing outcomes.
- Collect customer feedback.
- Trigger CRM follow-up activities.

4. Scope

Included

- Appointment Requests
- Appointment Scheduling
- Calendar Management
- Property Viewing
- Agent Assignment
- Owner Coordination
- Reminder Management
- Check-in / Check-out
- Viewing Outcomes
- Customer Feedback

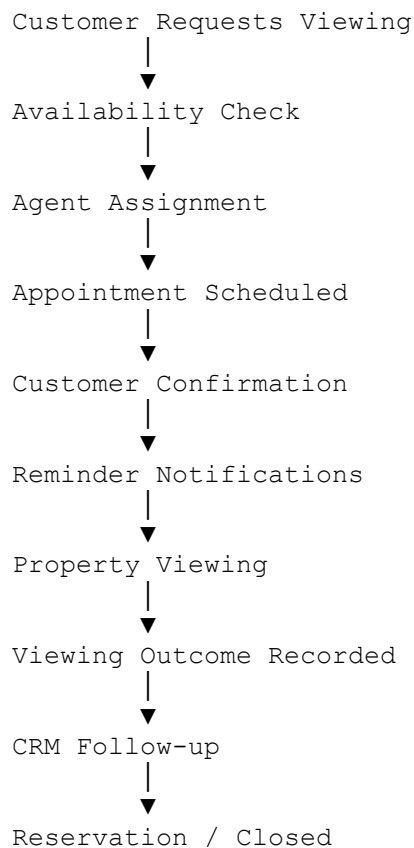
- Appointment Reports

Excluded

- Reservation approval
- Property management
- Payment processing
- Customer billing

5. Appointment Lifecycle

Default workflow:



Organizations may customize the workflow.

6. Appointment Types

Supported appointment types include:

- Property Viewing
- Virtual Property Tour
- Consultation Meeting
- Negotiation Meeting
- Property Inspection
- Site Visit
- Handover Meeting
- Owner Meeting

Administrators may configure additional appointment types.

7. Appointment Information

Each appointment shall maintain:

Basic Information

- Appointment Number
 - Appointment Type
 - Appointment Status
 - Date
 - Start Time
 - End Time
 - Duration
-

Customer Information

- Customer
 - CRM Lead
 - Contact Details
-

Property Information

- Property
 - Property Address
 - Meeting Point
-

Agent Information

- Assigned Agent
 - Backup Agent
 - Supervisor
-

Additional Information

- Special Instructions
 - Notes
 - Required Documents
 - Safety Requirements
-

8. Appointment Status

Default statuses include:

- Requested
- Pending Confirmation
- Confirmed
- Rescheduled
- In Progress
- Completed
- Cancelled
- No Show
- Expired

Organizations may define additional statuses.

9. Calendar Management

The module shall provide centralized calendar management.

Calendar views shall include:

- Daily
- Weekly
- Monthly
- Timeline
- Agenda

Appointments shall be filterable by:

- Agent

- Property
 - Branch
 - Customer
 - Appointment Type
 - Status
-

10. Availability Checking

Before scheduling an appointment, the system shall automatically verify:

- Agent availability.
- Property availability.
- Existing reservations.
- Existing appointments.
- Configured business hours.
- Public holidays (optional).

Conflicting appointments shall not be permitted unless explicitly overridden.

11. Agent Assignment

Appointments may be assigned using:

- Manual Assignment
- Round Robin
- Territory-Based Assignment
- Availability-Based Assignment
- Workload-Based Assignment

Assignment history shall be maintained.

12. Reminder Management

The platform shall automatically send reminders.

Default reminder schedule:

- 24 Hours Before
- 2 Hours Before
- 30 Minutes Before

Reminder intervals shall be configurable.

Supported channels:

- Email
 - SMS
 - WhatsApp
 - Push Notification
 - In-App Notification
-

13. Check-in & Check-out

Agents shall record:

Check-in

- Arrival Time
 - GPS Location (Optional)
 - Customer Present
 - Property Accessible
-

Check-out

- Departure Time
 - Viewing Duration
 - Notes
 - Outcome
-

14. Viewing Outcomes

The viewing outcome shall be recorded.

Examples include:

- Customer Interested
- Reservation Requested
- Follow-up Required
- Price Negotiation
- Additional Documents Requested
- Not Interested
- Customer No Show
- Property Not Accessible

Multiple outcomes may be recorded.

15. Customer Feedback

After the appointment, customers may provide feedback.

Feedback fields include:

- Overall Rating
- Agent Professionalism
- Property Condition
- Accuracy of Listing
- Satisfaction Level
- Comments

Ratings shall support configurable scales (e.g., 1–5 stars).

16. Functional Requirements

Appointment Management

FR-APT-001

The system shall create appointment requests.

FR-APT-002

Each appointment shall receive a unique Appointment Number.

FR-APT-003

Authorized users shall reschedule appointments.

Scheduling

FR-APT-004

The system shall detect scheduling conflicts.

FR-APT-005

Business hours shall be enforced.

Calendar

FR-APT-006

Users shall view appointments in multiple calendar formats.

Notifications

FR-APT-007

Automatic reminders shall be sent according to configured schedules.

Attendance

FR-APT-008

Agents shall record attendance and outcomes.

CRM Integration

FR-APT-009

Completed appointments shall automatically update CRM activities.

FR-APT-010

Interested customers may be converted into reservations without duplicate data entry.

17. Business Rules

BR-APT-001

Appointments shall not overlap for the same agent unless overrides are permitted.

BR-APT-002

Completed appointments shall become read-only except for administrative corrections.

BR-APT-003

Cancelled appointments shall remain in historical records.

BR-APT-004

Appointment reminders shall only be sent for confirmed appointments.

BR-APT-005

Every completed appointment shall record an outcome.

18. Notifications

The module shall publish notifications for:

- Appointment Requested
- Appointment Approved
- Appointment Confirmed
- Appointment Rescheduled
- Appointment Cancelled
- Reminder Sent
- Customer Checked In
- Viewing Completed
- Customer Feedback Received
- Follow-up Created

Supported channels:

- In-App
 - Email
 - SMS
 - WhatsApp
 - Push Notification
-

19. Reports

The Appointment & Property Viewing Management Module shall provide:

[Appointment Reports](#)

- Appointment Register
- Upcoming Appointments
- Completed Appointments
- Cancelled Appointments
- No-show Report

Agent Reports

- Appointments by Agent
- Average Viewing Duration
- Viewing Success Rate
- Follow-up Compliance

Customer Reports

- Viewing History
- Customer Feedback
- Customer Satisfaction

Operational Reports

- Calendar Utilization
- Peak Viewing Hours
- Property Viewing Frequency
- Appointment Conversion Rate

Reports shall support filtering, grouping, exporting, printing, and scheduled generation.

20. Configuration

Administrators shall configure:

- Appointment numbering format
 - Appointment types
 - Business hours
 - Working days
 - Reminder intervals
 - Appointment duration
 - Calendar settings
 - Conflict detection rules
 - Feedback forms
 - Notification templates
-

21. Permissions Matrix

The module shall expose permissions including:

- View Appointments
- Create Appointment
- Edit Appointment

- Cancel Appointment
 - Assign Agent
 - View Calendar
 - Record Attendance
 - Record Viewing Outcome
 - View Feedback
 - Export Reports
 - Configure Appointment Module
-

22. Acceptance Criteria

The Appointment & Property Viewing Management Module shall be accepted when:

- Appointments can be scheduled without conflicts.
 - Calendar views function correctly.
 - Automatic reminders are delivered.
 - Attendance and viewing outcomes are recorded.
 - Customer feedback is captured.
 - CRM activities are updated automatically.
 - Reservation creation can begin directly from completed viewings.
 - Reports generate accurately.
 - Permissions are consistently enforced.
-

23. Future Enhancements

The architecture shall support future capabilities including:

- Integration with Google Calendar, Microsoft Outlook, and Apple Calendar.
 - Real-time GPS tracking of field agents.
 - QR code check-in for customers.
 - AI-powered scheduling optimization.
 - Route optimization for multiple property viewings.
 - Video conferencing integration for virtual tours.
 - Electronic visitor registration.
 - Facial recognition check-in (optional).
 - Automated customer satisfaction analysis.
 - AI-generated viewing summaries.
-

24. Module Summary

The Appointment & Property Viewing Management Module provides a structured and automated process for coordinating customer interactions with properties. It reduces scheduling conflicts, enhances customer experience, improves operational efficiency, and creates a seamless transition from inquiry to reservation. By integrating with CRM, Property Management, Agents Management, and Reservation Management, it becomes a key component in maximizing sales conversion and customer satisfaction.

BUSINESS MODULE 8

BM-008 — LAND SURVEY & GEOSPATIAL MANAGEMENT MODULE

1. Purpose

The Land Survey & Geospatial Management Module provides a comprehensive system for managing land surveying activities, cadastral information, parcel boundaries, survey documentation, geographic coordinates, and spatial visualization.

The module enables organizations to capture, verify, visualize, and maintain accurate geospatial information for every property managed within the RehoSpace Real Estate Platform.

It bridges traditional property management with Geographic Information Systems (GIS), providing a powerful competitive advantage for land developers, survey firms, government agencies, and real estate companies.

2. Business Value

The module enables organizations to:

- Maintain accurate parcel boundaries.
- Reduce land ownership disputes.
- Digitize survey records.
- Improve property verification.
- Support GIS visualization.
- Improve investor confidence.
- Simplify land subdivision management.
- Prepare for government land registry integration.

3. Objectives

The module shall:

- Register land surveys.
 - Manage survey projects.
 - Store parcel coordinates.
 - Record beacon information.
 - Store survey plans.
 - Visualize parcels on maps.
 - Track survey history.
 - Manage survey approvals.
 - Maintain geospatial records.
-

4. Scope

Included

- Survey Requests
- Survey Projects
- Surveyors
- Parcel Management
- Parcel Coordinates
- GPS Data
- Beacon Management
- Survey Documents
- Survey Maps
- GIS Visualization
- Coordinate Systems
- Survey Reports
- Survey History

Excluded

- Government title registration
 - Legal ownership adjudication
 - Engineering design
 - Construction supervision
-

5. Survey Types

Supported survey types include:

- Boundary Survey
- Topographic Survey
- Engineering Survey
- Construction Survey
- Subdivision Survey
- Consolidation Survey
- As-Built Survey
- Site Survey
- Farm Survey
- Utility Survey
- GPS Survey
- Drone Survey
- Cadastral Survey

Administrators may define additional survey types.

6. Survey Information

Each survey shall include:

Basic Information

- Survey Number
 - Survey Type
 - Survey Status
 - Survey Date
 - Requested By
 - Assigned Surveyor
 - Branch
 - Priority
-

Property Information

- Property Reference
 - Parcel Number
 - Plot Number
 - Block Number
 - Village / Street
 - Ward
 - District
 - Region
-

Technical Information

- Coordinate System
 - Datum
 - Projection
 - Area
 - Perimeter
 - Elevation
 - Accuracy Level
-

Survey Outcome

- Survey Completed
 - Pending Verification
 - Approved
 - Rejected
 - Requires Resurvey
-

7. Parcel Management

Every parcel shall maintain:

- Parcel ID
- Parcel Number
- Area
- Shape
- Coordinates
- Current Status
- Linked Property
- Linked Owner(s)
- Survey History

Parcels shall remain uniquely identifiable.

8. Coordinate Management

The module shall support storage of coordinate points.

Each point shall include:

- Point Number

- Latitude
- Longitude
- Easting
- Northing
- Elevation
- Description

The system shall support configurable coordinate reference systems.

9. Beacon Management

Survey beacons shall maintain:

- Beacon Number
- GPS Coordinates
- Installation Date
- Condition
- Last Inspection Date
- Inspection Notes
- Related Parcel

Beacon status examples:

- Existing
 - Missing
 - Damaged
 - Replaced
 - Pending Inspection
-

10. GIS Mapping

The module shall visualize parcels on interactive maps.

Supported capabilities include:

- Parcel Boundary Display
- Polygon Drawing
- GPS Marker Display
- Satellite View
- Street View
- Terrain View
- Layer Management
- Distance Measurement
- Area Measurement
- Parcel Highlighting

Future integrations may support advanced GIS engines.

11. Survey Documents

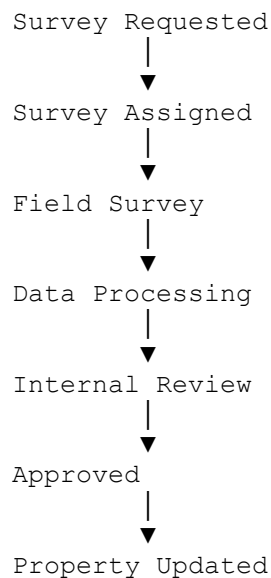
The module shall securely manage:

- Survey Plans
- Cadastral Maps
- Beacon Certificates
- Coordinate Sheets
- Survey Reports
- Field Notes
- Drone Images
- Satellite Images
- Approval Letters
- Technical Drawings

Version history shall be maintained.

12. Survey Workflow

Default workflow:



Organizations may customize workflow stages.

13. Functional Requirements

Survey Management

FR-SURV-001

The system shall register survey projects.

FR-SURV-002

Each survey shall receive a unique Survey Number.

FR-SURV-003

Survey status shall be tracked throughout its lifecycle.

Parcel Management

FR-SURV-004

The system shall register land parcels.

FR-SURV-005

Each parcel shall have a unique Parcel Number.

Coordinates

FR-SURV-006

Multiple coordinate points shall be stored for every parcel.

FR-SURV-007

Coordinate validation rules shall be configurable.

GIS

FR-SURV-008

Users shall visualize parcel boundaries on interactive maps.

FR-SURV-009

Users shall measure parcel dimensions within the map interface.

Documents

FR-SURV-010

Survey documents shall support version control.

FR-SURV-011

Large survey files shall be managed through the Media & File Management Module.

Integration

FR-SURV-012

Approved survey data shall update linked property records automatically.

14. Business Rules

BR-SURV-001

Every parcel shall have a unique identifier.

BR-SURV-002

Survey history shall be immutable.

BR-SURV-003

Approved surveys shall become the authoritative spatial record.

BR-SURV-004

Deleted survey records shall be archived rather than permanently removed.

BR-SURV-005

Only authorized surveyors or administrators may approve survey results.

15. Notifications

The module shall publish notifications for:

- Survey Requested
- Survey Assigned
- Survey Started
- Survey Completed
- Survey Approved
- Survey Rejected
- Beacon Inspection Due
- Parcel Updated
- Survey Document Uploaded

Supported channels:

- In-App
 - Email
 - SMS
 - WhatsApp
-

16. Reports

The Land Survey & Geospatial Management Module shall provide:

Survey Reports

- Survey Register
- Pending Surveys
- Completed Surveys
- Survey Status Report
- Surveyor Workload

Parcel Reports

- Parcel Register
- Parcel Area Summary
- Parcel History
- Boundary Change Report

Beacon Reports

- Beacon Register
- Missing Beacons
- Damaged Beacons
- Inspection Schedule

GIS Reports

- Spatial Coverage Report
- Area Distribution Report
- Coordinate Export Report

Reports shall support filtering, grouping, exporting, printing, and scheduled generation.

17. Configuration

Administrators shall configure:

- Survey numbering format
 - Parcel numbering format
 - Beacon numbering format
 - Coordinate systems
 - Survey types
 - Workflow stages
 - GIS layers
 - Approval workflows
 - Map providers
 - File formats
 - Notification templates
-

18. Permissions Matrix

The module shall expose permissions including:

- View Surveys
 - Create Survey
 - Edit Survey
 - Assign Survey
 - Approve Survey
 - Reject Survey
 - Manage Parcels
 - Manage Beacons
 - Upload Survey Documents
 - View GIS Maps
 - Export Survey Reports
 - Configure Survey Module
-

19. Acceptance Criteria

The module shall be accepted when:

- Survey projects can be created and managed.
 - Parcel records maintain unique identifiers.
 - Coordinates are stored accurately.
 - Interactive maps display parcel boundaries.
 - Survey documents are securely managed.
 - Approved surveys update linked property records.
 - Reports are generated correctly.
 - Notifications are delivered.
 - Permissions are consistently enforced.
-

20. Future Enhancements

The architecture shall support future capabilities including:

- Integration with national land registry systems.
 - Drone flight planning and automated image processing.
 - 3D terrain and contour visualization.
 - Real-time RTK GPS device integration.
 - Mobile field survey application with offline synchronization.
 - Automatic parcel overlap detection.
 - AI-assisted boundary validation.
 - Integration with BIM (Building Information Modeling).
 - Satellite imagery analysis for land-use monitoring.
 - Spatial analytics dashboards and heat maps.
-

21. Module Summary

The Land Survey & Geospatial Management Module transforms RREP from a conventional real estate platform into a comprehensive property and land information system. By combining cadastral data, GIS visualization, survey workflows, and parcel management, it enables organizations to maintain accurate spatial records, improve land verification processes, and support informed decision-making throughout the property lifecycle.

BUSINESS MODULE 9

BM-009 — SALES & TRANSACTION MANAGEMENT MODULE

1. Purpose

The Sales & Transaction Management Module manages the complete lifecycle of real estate transactions, from reservation conversion through negotiation, contract preparation, payment tracking, ownership transfer, transaction completion, and post-sale activities.

It serves as the authoritative record for all commercial transactions executed through the RehoSpace Real Estate Platform.

This module integrates tightly with Property Management, CRM, Reservation Management, Property Owners, Payments, Finance, and Document Management.

2. Business Value

The module enables organizations to:

- Centralize all sales and rental transactions.
 - Reduce paperwork and manual errors.
 - Improve transaction transparency.
 - Track negotiations and approvals.
 - Prevent duplicate sales.
 - Automate transaction workflows.
 - Maintain complete legal history.
 - Generate financial and operational reports.
-

3. Objectives

The module shall:

- Convert reservations into transactions.
- Manage negotiations.
- Generate transaction records.
- Track sales and rentals.
- Record transaction milestones.
- Manage ownership transfers.
- Coordinate payment milestones.

- Archive completed transactions.
-

4. Scope

Included

- Sales Transactions
- Rental Transactions
- Lease Transactions
- Transaction Workflow
- Negotiation Tracking
- Transaction Approvals
- Ownership Transfer
- Closing Process
- Transaction History
- Transaction Reports

Excluded

- General accounting
 - Payroll
 - Tax filing
 - Government title registration
-

5. Transaction Types

The platform shall support configurable transaction types.

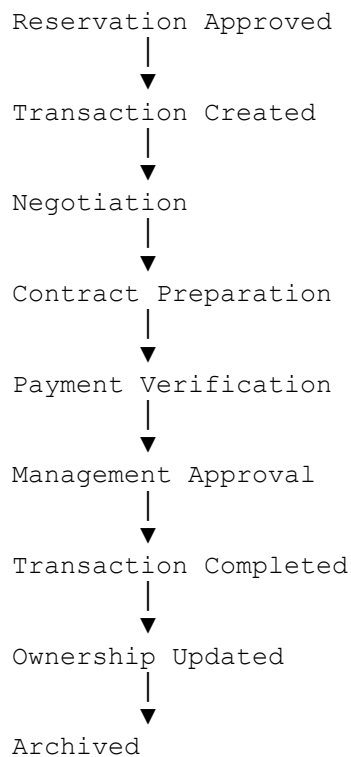
Examples include:

- Property Sale
- Property Rental
- Property Lease
- Lease Renewal
- Property Transfer
- Joint Purchase
- Installment Purchase
- Off-plan Purchase

Administrators may define additional transaction types.

6. Transaction Lifecycle

Default workflow:



Organizations may configure workflow stages.

7. Transaction Information

Each transaction shall maintain:

Basic Information

- Transaction Number
- Transaction Type
- Transaction Date
- Transaction Status
- Branch
- Responsible Agent

Customer Information

- Customer
 - Organization (if applicable)
 - Contact Information
-

Property Information

- Property Reference
 - Property Type
 - Property Address
-

Financial Information

- Listing Price
 - Agreed Price
 - Discount
 - Taxes
 - Commission
 - Deposit Paid
 - Outstanding Balance
 - Currency
-

Legal Information

- Contract Number
 - Ownership Transfer Status
 - Witnesses
 - Legal Representative
 - Closing Date
-

8. Negotiation Management

The system shall record negotiation history.

Each negotiation entry shall include:

- Date
- Negotiator
- Proposed Price
- Counter Offer
- Notes

- Outcome

All negotiation records shall remain immutable.

9. Ownership Transfer

Upon successful completion of a sale:

- Property ownership shall be updated.
- Ownership history shall be preserved.
- Previous ownership records shall remain immutable.
- Linked owner profiles shall be updated automatically.

For rental transactions, ownership shall remain unchanged while tenancy information is created.

10. Transaction Milestones

Each transaction may contain milestones such as:

- Reservation Converted
- Deposit Received
- Contract Generated
- Contract Signed
- Payment Verified
- Legal Review Completed
- Ownership Updated
- Transaction Closed

Milestones shall support configurable due dates and responsible users.

11. Functional Requirements

Transaction Creation

FR-TRX-001

The system shall create transactions from approved reservations.

FR-TRX-002

Each transaction shall receive a unique Transaction Number.

FR-TRX-003

The system shall prevent duplicate active transactions for the same property unless explicitly allowed.

Negotiation

FR-TRX-004

Users shall record negotiation history.

FR-TRX-005

Negotiation records shall be read-only after submission.

Closing

FR-TRX-006

Completed transactions shall update the property's availability and status automatically.

FR-TRX-007

Completed sales shall update ownership records.

Workflow

FR-TRX-008

Organizations shall configure approval workflows for transactions.

FR-TRX-009

Transaction status changes shall generate audit records.

Integration

FR-TRX-010

Payment milestones shall synchronize with the Payments Module.

FR-TRX-011

Documents shall be stored through the Document Management Module.

12. Business Rules

BR-TRX-001

Every transaction shall have a unique Transaction Number.

BR-TRX-002

A property marked as Sold shall not be available for new sale transactions.

BR-TRX-003

Completed transactions shall not be deleted.

BR-TRX-004

Ownership transfers shall occur only after all mandatory completion criteria are satisfied.

BR-TRX-005

Transaction history shall be immutable.

13. Notifications

The module shall publish notifications for:

- Transaction Created
- Negotiation Updated
- Contract Ready
- Payment Verified
- Approval Requested
- Transaction Approved
- Ownership Updated
- Transaction Completed
- Transaction Cancelled

Supported channels:

- In-App
- Email

- SMS
 - WhatsApp
-

14. Reports

The Sales & Transaction Management Module shall provide:

Sales Reports

- Sales Register
- Sales by Period
- Sales by Agent
- Sales by Branch
- Sales by Property Type

Rental Reports

- Rental Register
- Active Rentals
- Lease Expiry Report
- Rental Income Summary

Transaction Reports

- Transaction Pipeline
- Negotiation Summary
- Transaction Duration Analysis
- Ownership Transfer Report

Financial Reports

- Revenue by Transaction
- Commission Summary
- Outstanding Balances

Reports shall support filtering, grouping, exporting, printing, and scheduled generation.

15. Configuration

Administrators shall configure:

- Transaction numbering format
- Transaction types
- Workflow stages

- Approval workflows
 - Discount limits
 - Commission rules
 - Required milestones
 - Closing checklist templates
 - Notification templates
-

16. Permissions Matrix

The module shall expose permissions including:

- View Transactions
 - Create Transaction
 - Edit Transaction
 - Approve Transaction
 - Cancel Transaction
 - Record Negotiation
 - Update Ownership
 - View Financial Summary
 - Export Transaction Reports
 - Configure Transaction Module
-

17. Acceptance Criteria

The module shall be accepted when:

- Transactions can be created from reservations.
 - Duplicate transactions are prevented according to business rules.
 - Negotiation history is preserved.
 - Completed sales update property status and ownership automatically.
 - Transaction workflows operate correctly.
 - Payment milestones synchronize with the Payments Module.
 - Reports generate accurately.
 - Notifications are delivered.
 - Permissions are consistently enforced.
-

18. Future Enhancements

The architecture shall support future capabilities including:

- Electronic contract generation.

- Digital signatures compliant with local regulations.
 - Mortgage and bank financing integration.
 - Escrow account management.
 - Blockchain-based transaction verification.
 - AI-assisted contract review.
 - Smart contract execution.
 - Cross-border currency settlement.
 - Government property transfer integration.
-

19. Module Summary

The Sales & Transaction Management Module is the commercial execution engine of the RehoSpace Real Estate Platform. It governs the transition from customer commitment to legally completed property transactions while maintaining complete financial, operational, and ownership records. It ensures every transaction is auditable, configurable, and integrated with the broader platform ecosystem.

BUSINESS MODULE 10

BM-010 — DOCUMENT & DIGITAL RECORDS MANAGEMENT MODULE (EDMS)

1. Purpose

The Document & Digital Records Management Module provides a centralized, secure, and compliant platform for managing all electronic documents and digital records generated or received by the RehoSpace Real Estate Platform.

It serves as the single source of truth for contracts, legal documents, survey plans, identification documents, payment receipts, photographs, videos, certificates, reports, templates, and every other digital asset used throughout the platform.

The module provides enterprise-grade capabilities including document versioning, metadata management, workflow approvals, retention policies, full-text search, electronic signatures, and secure access control.

2. Business Value

The module enables organizations to:

- Centralize document storage.
 - Eliminate paper-based processes.
 - Improve legal compliance.
 - Secure sensitive documents.
 - Reduce document duplication.
 - Improve collaboration.
 - Support digital transformation.
 - Preserve complete document history.
-

3. Objectives

The module shall:

- Store documents securely.
 - Organize digital records.
 - Support document versioning.
 - Maintain document history.
 - Manage document approvals.
 - Control document access.
 - Support document templates.
 - Enable electronic signatures.
 - Provide enterprise search capabilities.
-

4. Scope

Included

- Document Repository
- Folder Management
- Metadata Management
- Version Control
- Check-in / Check-out
- Approval Workflow
- Digital Signatures
- OCR & Indexing
- Document Templates
- File Preview
- Document Sharing
- Retention Policies

- Archive Management
- Audit Trail

Excluded

- External cloud backup management
 - Third-party file hosting administration
 - Desktop document editing software
-

5. Supported Document Categories

The platform shall support configurable document categories.

Examples include:

Property Documents

- Title Deeds
 - Certificates of Occupancy
 - Survey Plans
 - Site Maps
 - Floor Plans
 - Building Permits
-

Owner Documents

- National ID
 - Passport
 - Company Registration
 - TIN Certificate
 - Power of Attorney
-

Customer Documents

- Identification
 - Loan Approval
 - Application Forms
 - Contracts
 - Correspondence
-

Transaction Documents

- Reservation Agreements

- Sales Agreements
- Lease Agreements
- Payment Receipts
- Closing Documents

Survey Documents

- Survey Reports
- Coordinate Sheets
- Beacon Certificates
- GIS Maps
- Drone Images

Financial Documents

- Invoices
- Receipts
- Credit Notes
- Statements

Administrative Documents

- Policies
- Templates
- Reports
- Internal Memos

Administrators may define additional categories.

6. Document Information

Each document shall maintain:

Basic Information

- Document ID
 - Document Number
 - Document Name
 - Category
 - Type
 - Status
-

Ownership

- Created By
 - Owned By
 - Department
 - Branch
-

Version Information

- Version Number
 - Revision Date
 - Revision Notes
 - Current Version
-

Metadata

- Related Property
 - Related Customer
 - Related Owner
 - Related Transaction
 - Related Survey
 - Tags
 - Keywords
-

Security

- Classification
 - Access Level
 - Encryption Status
 - Expiry Date
 - Retention Policy
-

7. Supported File Types

The module shall support configurable file formats including:

Documents

- PDF

- DOC
- DOCX
- XLS
- XLSX
- PPT
- PPTX
- TXT

Images

- JPG
- PNG
- TIFF
- BMP
- SVG

Media

- MP4
- MOV
- AVI

Technical Files

- DWG
- DXF
- SHP
- KML
- GeoJSON
- ZIP

Additional file formats may be configured.

8. Version Control

The system shall maintain complete version history.

Each version shall record:

- Version Number
- Modified By
- Date
- Change Summary

Previous versions shall remain accessible according to permissions.

9. Check-in / Check-out

To prevent editing conflicts:

Users may check out documents.

While checked out:

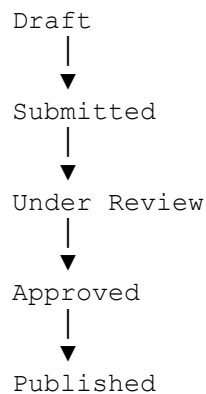
- Other users may view.
- Editing shall be restricted.
- The editor shall be identified.

Upon check-in:

- New version created.
- Audit log updated.
- Notifications generated.

10. Document Approval Workflow

Default workflow:



Alternative paths include:

- Rejected
- Returned for Revision
- Archived
- Expired

Organizations may configure workflows.

11. Electronic Signatures

The module shall support electronic document signing.

Supported capabilities:

- Single Signatory
- Multiple Signatories
- Sequential Signing
- Parallel Signing
- Signature Timestamp
- Signature Verification
- Signature Audit Trail

Future integrations with legally recognized e-signature providers shall be supported.

12. OCR & Full-Text Search

The module shall support Optical Character Recognition (OCR) for scanned documents.

Search shall include:

- Document Name
- Keywords
- Tags
- Metadata
- OCR Content
- File Contents
- Related Records

Search results shall support advanced filters.

13. Retention & Archiving

Administrators shall define retention policies.

Examples:

- Permanent
- 5 Years
- 7 Years
- 10 Years
- Custom Period

Expired documents may be:

- Archived
 - Flagged
 - Scheduled for secure deletion
 - Preserved for legal hold
-

14. Functional Requirements

Document Repository

FR-DOC-001

The system shall store digital documents centrally.

FR-DOC-002

Each document shall receive a unique Document ID.

Versioning

FR-DOC-003

Uploading a replacement shall create a new version.

FR-DOC-004

Previous versions shall remain accessible according to permissions.

Search

FR-DOC-005

Users shall search documents using metadata and full-text indexing.

Workflow

FR-DOC-006

Configurable approval workflows shall be supported.

Electronic Signatures

FR-DOC-007

Authorized users shall electronically sign supported documents.

Integration

FR-DOC-008

Every business module shall attach and retrieve documents through this module.

15. Business Rules

BR-DOC-001

Every document shall have a unique identifier.

BR-DOC-002

Approved documents shall become read-only unless a new version is created.

BR-DOC-003

Deleted documents shall follow configured retention policies.

BR-DOC-004

Every document modification shall generate an audit record.

BR-DOC-005

Sensitive documents shall only be accessible to authorized users.

16. Notifications

The module shall publish notifications for:

- Document Uploaded
- Document Updated

- Document Submitted for Approval
- Document Approved
- Document Rejected
- Document Signed
- Document Expiring
- Retention Period Ending
- Document Archived

Supported channels:

- In-App
 - Email
 - SMS
 - WhatsApp
-

17. Reports

The Document & Digital Records Management Module shall provide:

Repository Reports

- Document Register
- Documents by Category
- Documents by Department
- Storage Utilization

Workflow Reports

- Pending Approvals
- Rejected Documents
- Approval Turnaround Time

Compliance Reports

- Expired Documents
- Retention Schedule
- Missing Mandatory Documents
- Audit History

Usage Reports

- Most Accessed Documents
- Download Statistics
- User Activity
- Version History

Reports shall support filtering, exporting, printing, and scheduling.

18. Configuration

Administrators shall configure:

- Document numbering format
- Categories
- Metadata fields
- File size limits
- Supported file types
- Approval workflows
- Signature settings
- Retention policies
- Archive policies
- Notification templates

19. Permissions Matrix

The module shall expose permissions including:

- View Documents
- Upload Documents
- Edit Documents
- Delete Documents
- Approve Documents
- Reject Documents
- Check-out Documents
- Check-in Documents
- Download Documents
- Share Documents
- Sign Documents
- Export Reports
- Configure Document Module

20. Acceptance Criteria

The Document & Digital Records Management Module shall be accepted when:

- Documents can be uploaded, categorized, and retrieved.
- Version history is maintained accurately.
- Approval workflows operate as configured.
- Full-text and metadata searches return relevant results.
- Electronic signatures are recorded with audit trails.

- Retention and archiving policies are enforced.
 - Permissions consistently restrict unauthorized access.
 - All integrated modules can store and retrieve documents through this module.
-

21. Future Enhancements

The architecture shall support future capabilities including:

- AI-powered document classification and tagging.
 - Automatic document summarization.
 - OCR for handwritten documents.
 - Legal clause extraction and comparison.
 - Integration with cloud storage providers (AWS S3, Azure Blob Storage, Google Cloud Storage).
 - Watermarking and DRM (Digital Rights Management).
 - Blockchain-based document integrity verification.
 - Automated document translation.
 - AI-powered compliance checking.
-

22. Module Summary

The Document & Digital Records Management Module is the enterprise document backbone of the RehoSpace Real Estate Platform. Rather than functioning as a simple file repository, it provides a comprehensive Electronic Document Management System (EDMS) with version control, workflow automation, electronic signatures, enterprise search, compliance management, and secure access controls. Every business module relies on this shared service to ensure documents remain organized, traceable, secure, and legally defensible throughout their lifecycle.

BUSINESS MODULE 11

BM-011 — FINANCE, BILLING & PAYMENTS MANAGEMENT MODULE

1. Purpose

The Finance, Billing & Payments Management Module manages all financial transactions generated within the RehoSpace Real Estate Platform. It provides comprehensive capabilities

for billing, invoicing, payment collection, installment management, commission calculations, refunds, financial approvals, and revenue reporting.

The module ensures that every financial transaction related to properties, reservations, rentals, sales, commissions, and services is accurately recorded, tracked, and reconciled while supporting integration with external accounting systems.

2. Business Value

The module enables organizations to:

- Automate billing processes.
 - Improve payment collection.
 - Reduce financial errors.
 - Track outstanding balances.
 - Support flexible payment plans.
 - Manage agent commissions.
 - Generate financial reports.
 - Improve cash flow visibility.
 - Integrate with banks and payment gateways.
-

3. Objectives

The module shall:

- Generate invoices.
 - Record receipts.
 - Manage payment plans.
 - Track installments.
 - Calculate commissions.
 - Process refunds.
 - Reconcile payments.
 - Generate financial reports.
 - Support multiple payment methods.
 - Support multi-currency transactions.
-

4. Scope

Included

- Billing

- Invoicing
- Receipts
- Payment Collection
- Installment Plans
- Deposit Management
- Refunds
- Credit Notes
- Commission Management
- Payment Reconciliation
- Financial Approvals
- Financial Reports

Excluded

- General Ledger
 - Payroll
 - Procurement
 - Inventory Accounting
 - Tax Filing
-

5. Financial Transaction Types

The platform shall support configurable transaction categories.

Examples include:

- Property Sale Payment
- Rental Payment
- Lease Payment
- Reservation Deposit
- Security Deposit
- Survey Fees
- Service Charges
- Agent Commission
- Owner Commission
- Late Payment Penalty
- Refund
- Adjustment
- Miscellaneous Income

Administrators may define additional transaction types.

6. Invoice Management

Each invoice shall include:

Basic Information

- Invoice Number
 - Invoice Date
 - Due Date
 - Invoice Type
 - Status
-

Customer Information

- Customer
 - Organization
 - Contact Information
-

Property Information

- Property
 - Reservation
 - Transaction Reference
-

Financial Details

- Currency
 - Line Items
 - Quantity
 - Unit Price
 - Discount
 - Taxes
 - Total Amount
 - Outstanding Balance
-

Approval Information

- Prepared By
 - Approved By
 - Approval Date
-

7. Invoice Status

Supported statuses include:

- Draft
- Pending Approval
- Approved
- Issued
- Partially Paid
- Paid
- Overdue
- Cancelled
- Written Off

Administrators may configure additional statuses.

8. Payment Management

The platform shall support recording of payments.

Each payment shall maintain:

- Payment Number
 - Payment Date
 - Amount
 - Currency
 - Payment Method
 - Transaction Reference
 - Received By
 - Receipt Number
 - Payment Status
-

9. Payment Methods

Supported methods include:

- Cash
- Bank Transfer
- Mobile Money
- Credit Card
- Debit Card
- Cheque
- Online Payment Gateway
- POS Terminal

- Electronic Funds Transfer (EFT)

Organizations may configure additional payment methods.

10. Installment Management

The module shall support flexible payment schedules.

Each installment plan shall include:

- Plan Number
- Customer
- Property
- Total Amount
- Number of Installments
- Installment Frequency
- Due Dates
- Interest Rate (optional)
- Penalty Rules
- Outstanding Balance

Supported frequencies include:

- Weekly
 - Biweekly
 - Monthly
 - Quarterly
 - Semi-Annual
 - Annual
 - Custom
-

11. Commission Management

The module shall calculate and track commissions for:

- Sales Agents
- Brokers
- Property Owners
- Referral Partners

Supported commission models:

- Fixed Amount
- Percentage

- Tiered
- Performance-Based
- Shared Commission

Commission payments shall support approval workflows before disbursement.

12. Refund Management

Authorized users shall process refunds.

Refund information shall include:

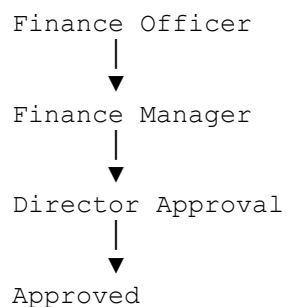
- Refund Number
- Original Payment
- Refund Amount
- Refund Reason
- Approval Status
- Payment Method
- Refund Date

Refund policies shall be configurable.

13. Financial Approvals

Organizations may configure approval workflows.

Example:



Approval rules shall be configurable by transaction amount, type, or branch.

14. Functional Requirements

Billing

FR-FIN-001

The system shall generate invoices automatically from approved transactions.

FR-FIN-002

Each invoice shall receive a unique Invoice Number.

Payments

FR-FIN-003

The system shall record payments against invoices.

FR-FIN-004

Partial payments shall be supported.

FR-FIN-005

Outstanding balances shall be updated automatically.

Installments

FR-FIN-006

Users shall create installment payment plans.

FR-FIN-007

The system shall generate installment schedules automatically.

Commissions

FR-FIN-008

The system shall calculate commissions according to configured rules.

FR-FIN-009

Commission adjustments shall be recorded with full audit history.

Refunds

FR-FIN-010

Authorized users shall process refunds.

Integration

FR-FIN-011

Payments shall synchronize with Reservations and Sales modules.

FR-FIN-012

Financial data shall be exportable to external accounting systems.

15. Business Rules

BR-FIN-001

Every invoice shall have a unique Invoice Number.

BR-FIN-002

Payments shall not exceed the outstanding balance unless overpayments are explicitly allowed.

BR-FIN-003

Completed financial transactions shall never be permanently deleted.

BR-FIN-004

Commission calculations shall follow approved commission rules.

BR-FIN-005

Refunds shall require appropriate authorization.

16. Notifications

The module shall publish notifications for:

- Invoice Generated
- Invoice Approved
- Invoice Due
- Invoice Overdue
- Payment Received
- Partial Payment Recorded
- Installment Due
- Installment Overdue
- Commission Approved
- Commission Paid
- Refund Approved
- Refund Completed

Supported delivery channels:

- In-App
- Email
- SMS
- WhatsApp

17. Reports

The Finance, Billing & Payments Management Module shall provide:

Billing Reports

- Invoice Register
- Outstanding Invoices
- Overdue Invoices
- Billing by Branch
- Billing by Property

Payment Reports

- Payment Register
- Daily Collections
- Monthly Revenue
- Payment Method Analysis
- Outstanding Balances

Commission Reports

- Commission Register
- Commission by Agent
- Commission by Property

- Pending Commission Payments

Financial Performance Reports

- Revenue Trends
- Cash Flow Summary
- Installment Collection Performance
- Refund Analysis
- Penalty Collection Report

Reports shall support filtering, grouping, dashboards, exporting, printing, and scheduled generation.

18. Configuration

Administrators shall configure:

- Invoice numbering format
 - Receipt numbering format
 - Payment numbering format
 - Supported currencies
 - Exchange rates
 - Payment methods
 - Installment frequencies
 - Penalty rules
 - Commission rules
 - Tax rules
 - Approval workflows
 - Financial notification templates
-

19. Permissions Matrix

The module shall expose permissions including:

- View Invoices
- Create Invoice
- Edit Invoice
- Cancel Invoice
- Record Payment
- Process Refund
- Manage Installments
- Manage Commissions
- Approve Financial Transactions
- Export Financial Reports

- Configure Finance Module
-

20. Acceptance Criteria

The module shall be accepted when:

- Invoices are generated accurately.
 - Payments update outstanding balances automatically.
 - Partial and installment payments function correctly.
 - Commission calculations follow configured rules.
 - Refund workflows operate correctly.
 - Financial approval workflows are enforced.
 - Reports generate accurate financial summaries.
 - External accounting exports are available.
 - Notifications are delivered.
 - Permissions are consistently enforced.
-

21. Future Enhancements

The architecture shall support future capabilities including:

- Direct integration with banks for payment confirmation.
 - Integration with mobile money providers (e.g., M-Pesa, Airtel Money, Tigo Pesa, HaloPesa).
 - Integration with international payment gateways (Stripe, Flutterwave, PayPal, Pesapal, etc.).
 - Automatic bank reconciliation.
 - AI-powered revenue forecasting.
 - AI-based fraud detection.
 - Subscription billing for managed properties.
 - Multi-company financial management.
 - IFRS-compliant financial reporting.
 - Integration with ERP/accounting systems such as QuickBooks, Sage, SAP, and Odoo.
-

22. Module Summary

The Finance, Billing & Payments Management Module is the financial operations engine of the RehoSpace Real Estate Platform. It manages billing, collections, installment plans, commissions, refunds, and financial approvals while maintaining complete auditability and integration with transaction workflows. Designed for flexibility and extensibility, it supports organizations ranging from small real estate agencies to large multi-branch enterprises and provides a solid foundation for integration with external accounting ecosystems.

BUSINESS MODULE 12

BM-012 — MARKETING, CAMPAIGNS & MARKETPLACE PROMOTION MANAGEMENT MODULE

1. Purpose

The Marketing, Campaigns & Marketplace Promotion Management Module enables organizations to plan, execute, automate, monitor, and analyze marketing activities aimed at promoting properties, attracting leads, increasing brand awareness, and improving sales conversions.

The module integrates with the Marketplace, CRM, Property Management, and Analytics modules to provide a complete marketing ecosystem for real estate businesses.

2. Business Value

The module enables organizations to:

- Promote properties effectively.
 - Generate qualified leads.
 - Increase property visibility.
 - Improve marketing ROI.
 - Automate customer engagement.
 - Manage promotional campaigns.
 - Publish to multiple channels.
 - Measure marketing performance.
 - Strengthen brand awareness.
-

3. Objectives

The module shall:

- Create marketing campaigns.
- Promote property listings.
- Publish to multiple marketing channels.

- Manage featured listings.
 - Automate customer communications.
 - Track campaign performance.
 - Measure lead sources.
 - Manage referral programs.
 - Generate marketing reports.
-

4. Scope

Included

- Marketing Campaigns
- Featured Listings
- Property Promotions
- Social Media Publishing
- Email Marketing
- SMS Marketing
- WhatsApp Campaigns
- Promotional Banners
- Referral Programs
- Discount Campaigns
- Lead Attribution
- Marketing Analytics

Excluded

- Third-party advertising account billing
 - Graphic design tools
 - External email server administration
-

5. Campaign Types

The platform shall support configurable campaign types.

Examples include:

- Property Launch Campaign
- Featured Property Campaign
- New Development Campaign
- Rental Promotion
- Open House Campaign
- Holiday Promotion
- Investment Opportunity Campaign
- Price Reduction Campaign

- Referral Campaign
- Brand Awareness Campaign

Administrators may define additional campaign types.

6. Campaign Information

Each campaign shall include:

Basic Information

- Campaign ID
 - Campaign Name
 - Campaign Type
 - Campaign Status
 - Campaign Objective
 - Priority
-

Schedule

- Start Date
 - End Date
 - Campaign Duration
 - Publishing Schedule
-

Financial Information

- Budget
 - Actual Cost
 - Expected Revenue
 - Return on Investment (ROI)
-

Ownership

- Campaign Manager
 - Branch
 - Department
-

Target Audience

- Customer Segment
 - Region
 - Property Type
 - Budget Range
 - Customer Interests
-

7. Campaign Status

Supported statuses include:

- Draft
 - Scheduled
 - Active
 - Paused
 - Completed
 - Cancelled
 - Archived
-

8. Marketplace Promotion

The module shall support enhanced property visibility through:

- Featured Listings
- Premium Listings
- Sponsored Listings
- Homepage Highlights
- Category Highlights
- Regional Highlights
- Urgent Sale Badges
- Newly Added Badges
- Price Reduced Badges

Promotion duration shall be configurable.

9. Marketing Channels

The platform shall support publishing to multiple channels.

Supported channels include:

Internal

- Marketplace Homepage
 - Property Categories
 - Featured Sections
 - Customer Portal
 - Mobile Application
-

External

- Email
- SMS
- WhatsApp
- Facebook
- Instagram
- LinkedIn
- X (formerly Twitter)
- YouTube
- TikTok

Future integrations may support automatic publishing through official APIs.

10. Email Marketing

The module shall support:

- Email Templates
 - Personalization
 - Mailing Lists
 - Scheduled Delivery
 - Open Tracking
 - Click Tracking
 - Bounce Tracking
 - Unsubscribe Management
-

11. SMS & WhatsApp Marketing

Supported capabilities include:

- Bulk Messaging
- Personalized Messages
- Scheduled Delivery
- Delivery Status

- Read Receipts (where supported)
- Campaign Tracking

Integration shall support approved messaging providers.

12. Referral Program Management

The platform shall support configurable referral programs.

Examples include:

- Customer Referrals
- Agent Referrals
- Partner Referrals
- Affiliate Referrals

Each referral shall record:

- Referrer
 - Referred Customer
 - Property
 - Referral Date
 - Reward Status
-

13. Promotional Pricing

Organizations may define promotional offers such as:

- Percentage Discounts
- Fixed Amount Discounts
- Limited-Time Offers
- Bundle Offers
- Early Bird Discounts
- Seasonal Promotions

Promotion eligibility rules shall be configurable.

14. Lead Attribution

The system shall identify the origin of each lead.

Supported attribution sources include:

- Website
- Google Search
- Social Media
- Email Campaign
- SMS Campaign
- WhatsApp Campaign
- Referral
- QR Code
- Offline Event
- Manual Entry

The attribution model shall remain linked to CRM records.

15. Functional Requirements

Campaign Management

FR-MKT-001

The system shall create and manage marketing campaigns.

FR-MKT-002

Each campaign shall receive a unique Campaign ID.

Promotions

FR-MKT-003

Authorized users shall promote selected property listings.

FR-MKT-004

Promotions shall automatically expire based on configured dates.

Multi-channel Marketing

FR-MKT-005

Campaigns shall support publishing across multiple configured channels.

FR-MKT-006

Delivery status shall be recorded where supported by integrated providers.

Lead Tracking

FR-MKT-007

Campaign-generated leads shall automatically create CRM records.

FR-MKT-008

Lead attribution shall remain associated with the originating campaign.

Referral Management

FR-MKT-009

Referral rewards shall follow configurable business rules.

Analytics

FR-MKT-010

Campaign performance metrics shall update automatically.

16. Business Rules

BR-MKT-001

Every campaign shall have a unique Campaign ID.

BR-MKT-002

Expired promotions shall automatically stop displaying.

BR-MKT-003

Lead attribution shall remain immutable after creation unless corrected by authorized users.

BR-MKT-004

Referral rewards shall only be granted after configured qualifying events.

BR-MKT-005

Campaign history shall never be permanently deleted.

17. Notifications

The module shall publish notifications for:

- Campaign Created
- Campaign Approved
- Campaign Activated
- Campaign Completed
- Promotion Expired
- Lead Generated
- Referral Qualified
- Referral Reward Issued
- Marketing Budget Exceeded

Supported channels:

- In-App
 - Email
 - SMS
 - WhatsApp
-

18. Reports

The Marketing, Campaigns & Marketplace Promotion Management Module shall provide:

Campaign Reports

- Campaign Register
- Active Campaigns
- Campaign Performance
- Campaign ROI
- Budget Utilization

Promotion Reports

- Featured Listing Performance
- Property Promotion Performance
- Promotion Expiry Report

Lead Reports

- Leads by Campaign
- Leads by Marketing Channel
- Conversion Rate by Campaign
- Cost per Lead

Referral Reports

- Referral Register
- Referral Conversion Rate
- Referral Rewards Summary

Marketing Analytics

- Website Traffic Sources
- Click-Through Rate (CTR)
- Email Open Rate
- Social Engagement
- Lead-to-Sale Conversion Funnel

Reports shall support dashboards, filtering, exporting, printing, and scheduled delivery.

19. Configuration

Administrators shall configure:

- Campaign numbering format
 - Campaign types
 - Promotion types
 - Marketing channels
 - Referral rules
 - Reward structures
 - Discount policies
 - Publishing schedules
 - Approval workflows
 - Notification templates
-

20. Permissions Matrix

The module shall expose permissions including:

- View Campaigns
- Create Campaign

- Edit Campaign
 - Delete Campaign
 - Publish Campaign
 - Manage Promotions
 - Manage Referral Programs
 - View Marketing Analytics
 - Export Marketing Reports
 - Configure Marketing Module
-

21. Acceptance Criteria

The module shall be accepted when:

- Campaigns can be created, scheduled, and executed.
 - Featured property promotions display correctly.
 - Multi-channel publishing functions through configured integrations.
 - Campaign-generated leads appear automatically in the CRM.
 - Referral programs calculate rewards correctly.
 - Marketing analytics accurately reflect campaign performance.
 - Reports generate successfully.
 - Permissions are consistently enforced.
-

22. Future Enhancements

The architecture shall support future capabilities including:

- AI-generated marketing content.
 - AI-powered audience segmentation.
 - Automated A/B testing.
 - Predictive lead scoring for campaigns.
 - Dynamic advertisement optimization.
 - Integration with Google Ads and Meta Ads APIs.
 - SEO recommendations for property listings.
 - Marketing automation workflows.
 - Customer journey visualization.
 - AI-driven campaign budget optimization.
-

23. Module Summary

The Marketing, Campaigns & Marketplace Promotion Management Module equips real estate organizations with a comprehensive marketing platform that extends beyond simple

property advertising. By combining campaign management, multi-channel publishing, referral programs, promotional pricing, and marketing analytics, it enables businesses to attract qualified prospects, improve conversion rates, and maximize return on marketing investment while maintaining seamless integration with the Marketplace and CRM modules.

BUSINESS MODULE 13

BM-013 — SELF-SERVICE PORTAL MANAGEMENT MODULE

1. Purpose

The Self-Service Portal Management Module provides secure, role-based web and mobile access for external users to interact with the RehoSpace Real Estate Platform without requiring intervention from internal staff.

The module enables customers, property owners, agents, partners, and future stakeholder groups to securely access information, perform authorized transactions, communicate with the organization, and monitor activities relevant to their roles.

The module serves as the primary digital interaction point between the organization and external stakeholders.

2. Business Value

The module enables organizations to:

- Reduce manual customer service.
 - Improve customer satisfaction.
 - Increase transparency.
 - Empower property owners.
 - Improve agent productivity.
 - Enable self-service operations.
 - Reduce operational costs.
 - Increase customer engagement.
-

3. Objectives

The module shall:

- Provide secure portal access.
 - Support multiple user roles.
 - Enable profile management.
 - Provide personalized dashboards.
 - Support secure messaging.
 - Enable document access.
 - Allow online service requests.
 - Support notifications.
 - Track user activities.
-

4. Scope

Included

- Customer Portal
- Property Owner Portal
- Agent Portal
- Partner Portal
- User Dashboard
- Profile Management
- Document Access
- Messaging
- Notifications
- Activity History
- Online Requests
- Account Settings

Excluded

- Internal administration
 - System configuration
 - User provisioning by HR
-

5. Portal Roles

The platform shall support configurable external user roles.

Examples include:

- Customer
- Property Owner
- Sales Agent

- Broker
- Business Partner
- Legal Representative
- Surveyor (External)
- Property Investor

Organizations may define additional portal roles.

6. Authentication & Security

The portal shall support secure authentication using:

- Username & Password
- Email & Password
- Mobile Number & OTP
- Two-Factor Authentication (2FA)
- Single Sign-On (Future)
- Biometric Authentication (Mobile)

Security features shall include:

- Password policies
 - Session timeout
 - Device management
 - Login history
 - Account lockout
 - CAPTCHA
 - IP monitoring
-

7. Customer Portal

Customers shall be able to:

- View profile.
- Update personal information.
- Browse saved properties.
- Track reservations.
- View transaction history.
- Download invoices.
- Download receipts.
- View contracts.
- Schedule appointments.
- Submit inquiries.
- Upload requested documents.
- Track service requests.

- Receive notifications.
 - Send secure messages.
-

8. Property Owner Portal

Property owners shall be able to:

- View owned properties.
 - View listing performance.
 - View reservations.
 - View active transactions.
 - View rental income.
 - View commission statements.
 - Upload ownership documents.
 - Download reports.
 - Communicate with assigned agents.
 - Submit property update requests.
 - Monitor contract status.
-

9. Agent Portal

Agents shall be able to:

- View assigned leads.
 - View assigned properties.
 - Manage appointments.
 - Track commissions.
 - View performance dashboard.
 - Update availability.
 - Upload documents.
 - Record viewing outcomes.
 - Submit expenses (optional).
 - Communicate with customers.
 - View notifications.
-

10. Partner Portal

Business partners may:

- View assigned projects.
- Submit referrals.

- Track referral commissions.
 - Upload documents.
 - View partnership agreements.
 - Communicate with organization representatives.
-

11. Dashboard Management

Every portal user shall have a personalized dashboard.

Dashboard widgets shall be configurable.

Examples:

Customer

- Saved Properties
- Upcoming Appointments
- Reservations
- Outstanding Payments
- Notifications

Property Owner

- Property Portfolio
- Active Listings
- Rental Income
- Occupancy
- Recent Transactions

Agent

- Assigned Leads
 - Today's Appointments
 - Monthly Sales
 - Commission Summary
 - Pending Tasks
-

12. Secure Messaging

Portal users shall communicate with authorized internal users.

Supported features include:

- Conversation Threads

- Attachments
- Read Receipts
- Message Search
- Conversation History

Future support:

- Voice Messages
 - Video Messages
-

13. Notifications

Portal users shall receive:

- Reservation Updates
- Appointment Reminders
- Payment Reminders
- New Messages
- Document Requests
- Contract Updates
- Marketing Offers (Optional)
- System Announcements

Delivery channels:

- In-App
 - Email
 - SMS
 - WhatsApp
 - Push Notifications
-

14. Online Service Requests

Portal users shall submit requests including:

- Property Inquiry
- Viewing Request
- Reservation Request
- Maintenance Request (Rental)
- Document Request
- Ownership Update Request
- Complaint
- General Support

Requests shall integrate with CRM or Service Desk modules.

15. Functional Requirements

Portal Access

FR-PORTAL-001

The system shall provide secure authentication for external users.

FR-PORTAL-002

Each portal user shall have a unique account.

Profile Management

FR-PORTAL-003

Users shall update permitted profile information.

Dashboard

FR-PORTAL-004

Dashboard content shall vary according to user role.

Documents

FR-PORTAL-005

Authorized users shall download permitted documents.

FR-PORTAL-006

Users shall upload requested documents securely.

Messaging

FR-PORTAL-007

Secure messaging shall maintain complete conversation history.

Requests

FR-PORTAL-008

Users shall submit online service requests.

Notifications

FR-PORTAL-009

Notifications shall synchronize with the Notification Framework.

16. Business Rules

BR-PORTAL-001

Users shall access only data authorized for their role.

BR-PORTAL-002

Sensitive documents shall require appropriate permissions.

BR-PORTAL-003

Every portal action shall generate an audit record where applicable.

BR-PORTAL-004

Inactive accounts may be suspended according to organizational policy.

BR-PORTAL-005

Portal access shall be secured using configured authentication policies.

17. Reports

The Self-Service Portal Management Module shall provide:

Usage Reports

- Portal Login Activity
- Active Users
- User Registration Trends
- Session Duration

Customer Activity Reports

- Property Views
- Reservation Requests
- Appointment Requests
- Downloads

Owner Activity Reports

- Portfolio Views
- Income Reports Access
- Document Uploads

Agent Activity Reports

- Portal Usage
- Lead Updates
- Appointment Management
- Commission Dashboard Access

18. Configuration

Administrators shall configure:

- Portal branding
- Authentication methods
- Password policies
- Dashboard widgets
- Notification preferences
- Messaging settings
- Portal menus
- File upload limits
- Session timeout
- Security policies

19. Permissions Matrix

The module shall expose permissions including:

- Manage Portal Users
 - Activate Account
 - Suspend Account
 - Reset Password
 - Manage Dashboards
 - Configure Messaging
 - Configure Notifications
 - Configure Portal Module
 - View Portal Reports
-

20. Acceptance Criteria

The module shall be accepted when:

- External users authenticate securely.
 - Role-based dashboards display correctly.
 - Users access only authorized information.
 - Messaging functions correctly.
 - Notifications are delivered.
 - Documents upload and download securely.
 - Service requests integrate with other modules.
 - Reports generate accurately.
 - Permissions are enforced consistently.
-

21. Future Enhancements

The architecture shall support future capabilities including:

- Native Android and iOS mobile applications.
 - Offline access with synchronization.
 - AI-powered virtual assistant for customers and owners.
 - Voice-based property search.
 - Video consultation and virtual meetings.
 - Digital identity verification.
 - Live chat with agents.
 - Personalized property recommendations.
 - Tenant community features.
 - Multi-language user interfaces.
-

22. Module Summary

The Self-Service Portal Management Module extends the RehoSpace Real Estate Platform beyond internal operations by providing secure, role-based access for customers, property owners, agents, and business partners. It empowers external stakeholders to perform routine activities independently, improves communication, reduces operational overhead, and enhances user satisfaction while maintaining enterprise-grade security and governance.

BUSINESS MODULE 14

BM-014 — COMMUNICATION, NOTIFICATIONS & WORKFLOW AUTOMATION MODULE

1. Purpose

The Communication, Notifications & Workflow Automation Module provides a centralized platform for managing all communications, business workflows, approvals, reminders, escalations, and event-driven automation across the RehoSpace Real Estate Platform.

The module enables organizations to automate repetitive business processes, ensure timely communication with internal and external stakeholders, and enforce standardized approval workflows while maintaining complete auditability.

It acts as the event orchestration engine that connects every business module within the platform.

2. Business Value

The module enables organizations to:

- Centralize business communications.
 - Eliminate manual follow-ups.
 - Automate repetitive tasks.
 - Improve customer engagement.
 - Standardize approval processes.
 - Improve SLA compliance.
 - Reduce operational delays.
 - Increase process transparency.
 - Improve organizational efficiency.
-

3. Objectives

The module shall:

- Deliver notifications.
 - Manage communication channels.
 - Automate business workflows.
 - Manage approval processes.
 - Schedule reminders.
 - Escalate overdue activities.
 - Manage templates.
 - Publish business events.
 - Integrate with external communication providers.
-

4. Scope

Included

- Notification Management
- Email Management
- SMS Management
- WhatsApp Messaging
- Push Notifications
- In-App Notifications
- Workflow Automation
- Approval Workflows
- Reminder Management
- Escalation Rules
- Event Publishing
- Message Templates
- Webhooks
- SLA Monitoring

Excluded

- Email server administration
 - Telecommunication provider management
 - Social media marketing campaigns (covered by BM-012)
-

5. Communication Channels

The platform shall support configurable communication channels.

Supported channels include:

Digital Messaging

- Email
- SMS
- WhatsApp
- Push Notification
- In-App Notification

Future Channels

- Microsoft Teams
- Slack
- Telegram
- Signal
- Voice Calls
- IVR
- Web Chat

Organizations may enable or disable channels.

6. Notification Types

Supported notification categories include:

- Informational
- Reminder
- Warning
- Alert
- Approval Request
- Confirmation
- System Announcement
- Marketing (Optional)
- Emergency Notification

Administrators may configure additional notification types.

7. Notification Information

Each notification shall maintain:

Basic Information

- Notification ID
 - Notification Type
 - Subject
 - Message
 - Priority
 - Status
-

Delivery Information

- Recipient
 - Channel
 - Scheduled Time
 - Delivery Time
 - Delivery Status
 - Retry Count
-

Tracking Information

- Opened Date
 - Read Date
 - Click Status
 - Response Status
-

8. Workflow Automation

The module shall support configurable business workflows.

Examples include:

- Reservation Approval
- Property Approval
- Survey Approval
- Invoice Approval
- Refund Approval
- Document Approval
- Property Publication
- Customer Registration Verification

Organizations may define additional workflows.

9. Workflow Components

Each workflow shall include:

- Trigger Event
 - Conditions
 - Actions
 - Approvers
 - Deadlines
 - Escalation Rules
 - Completion Criteria
-

10. Event-Driven Automation

Every business module shall publish business events.

Examples:

Property Module

- Property Created
- Property Approved
- Property Sold

CRM Module

- Lead Created
- Lead Converted

Reservations

- Reservation Submitted
- Reservation Approved

Sales

- Transaction Completed

Finance

- Payment Received
- Invoice Overdue

Survey

- Survey Approved

Document Management

- Document Signed

Portal

- User Registered

The workflow engine shall subscribe to these events and execute configured actions automatically.

11. Approval Management

Approval workflows shall support:

- Single Approver
- Multiple Approvers
- Sequential Approval
- Parallel Approval
- Conditional Approval
- Delegated Approval

Approval history shall remain immutable.

12. Reminder Management

The module shall generate automatic reminders.

Examples:

- Appointment Reminder
- Payment Reminder
- Contract Expiry Reminder
- Lease Expiry Reminder
- Survey Due Reminder
- Document Expiry Reminder
- Follow-up Reminder

Reminder intervals shall be configurable.

13. Escalation Management

If deadlines are exceeded:

The system may automatically:

- Notify supervisors.
- Escalate to managers.
- Create follow-up tasks.
- Increase priority.
- Trigger workflow actions.

Escalation rules shall be configurable.

14. SLA Monitoring

Organizations may configure Service Level Agreements (SLAs).

Examples:

- Lead response within 30 minutes.
- Reservation approval within 24 hours.
- Survey completion within 5 working days.
- Complaint resolution within 3 days.
- Refund approval within 48 hours.

The module shall monitor compliance and trigger escalations for SLA breaches.

15. Message Templates

The system shall maintain reusable templates.

Supported template types include:

- Email Templates
- SMS Templates
- WhatsApp Templates
- Push Notification Templates
- Letter Templates

Templates shall support:

- Variables
- Localization
- Rich Formatting
- Branding

- Attachments
-

16. Webhooks & Integrations

The module shall publish configurable webhooks.

Examples:

- Property Published
- Payment Received
- Customer Registered
- Transaction Completed
- Survey Approved
- Reservation Confirmed

External systems may subscribe to these events.

17. Functional Requirements

Notifications

FR-COM-001

The system shall generate notifications based on business events.

FR-COM-002

Each notification shall receive a unique Notification ID.

FR-COM-003

Notifications shall support multiple delivery channels.

Workflow Automation

FR-COM-004

Organizations shall configure workflow rules.

FR-COM-005

Workflow execution shall support conditional branching.

Approvals

FR-COM-006

Approval workflows shall support sequential and parallel approvals.

Escalations

FR-COM-007

The system shall escalate overdue workflow tasks according to configured rules.

Templates

FR-COM-008

Administrators shall manage reusable communication templates.

Integrations

FR-COM-009

Business events shall be published for internal and external subscribers.

18. Business Rules

BR-COM-001

Every notification shall have a unique Notification ID.

BR-COM-002

Workflow history shall never be deleted.

BR-COM-003

Approval actions shall generate audit records.

BR-COM-004

Failed notifications shall follow retry policies.

BR-COM-005

Workflow definitions shall be version-controlled.

19. Reports

The Communication, Notifications & Workflow Automation Module shall provide:

Notification Reports

- Notification Register
- Delivery Success Rate
- Failed Deliveries
- Read Rate
- Response Rate

Workflow Reports

- Workflow Execution Summary
- Pending Approvals
- Average Approval Time
- Workflow Bottlenecks
- Escalation Summary

SLA Reports

- SLA Compliance
- SLA Violations
- Average Resolution Time
- Escalated Cases

Integration Reports

- Webhook Delivery Status
- Event Processing Log
- Integration Errors

Reports shall support dashboards, filtering, exporting, printing, and scheduled delivery.

20. Configuration

Administrators shall configure:

- Notification channels
 - Message priorities
 - Retry policies
 - Workflow definitions
 - Approval hierarchies
 - Reminder intervals
 - Escalation rules
 - SLA policies
 - Event subscriptions
 - Webhooks
 - Communication templates
 - Localization settings
-

21. Permissions Matrix

The module shall expose permissions including:

- View Notifications
 - Send Notifications
 - Manage Templates
 - Configure Workflows
 - Approve Workflow Tasks
 - Configure SLA Policies
 - Manage Webhooks
 - View Communication Reports
 - Export Reports
 - Configure Communication Module
-

22. Acceptance Criteria

The module shall be accepted when:

- Notifications are delivered through configured channels.
 - Business events trigger automated workflows.
 - Approval workflows execute correctly.
 - Reminder schedules function accurately.
 - Escalation rules activate for overdue tasks.
 - SLA monitoring identifies and reports violations.
 - Webhooks publish events successfully.
 - Reports provide accurate operational insights.
 - Permissions are consistently enforced.
-

23. Future Enhancements

The architecture shall support future capabilities including:

- AI-generated communication content.
 - Intelligent workflow recommendations.
 - Natural language workflow designer.
 - Chatbot integration for customer support.
 - Voice notifications and IVR integration.
 - Sentiment analysis for customer messages.
 - Predictive SLA breach detection.
 - Event streaming with Apache Kafka or similar platforms.
 - Low-code/no-code workflow builder for administrators.
-

24. Module Summary

The Communication, Notifications & Workflow Automation Module is the orchestration layer of the RehoSpace Real Estate Platform. It centralizes all communications, automates business processes, manages approval workflows, monitors service levels, and coordinates event-driven interactions between modules and external systems. By separating workflow logic and communication from business modules, it enhances scalability, maintainability, and reusability while providing organizations with a flexible platform for automating complex operational processes.

BUSINESS MODULE 15

BM-015 — REPORTING, BUSINESS INTELLIGENCE & ANALYTICS MODULE

1. Purpose

The Reporting, Business Intelligence & Analytics Module provides comprehensive reporting, interactive dashboards, key performance indicators (KPIs), data visualization, forecasting, and analytical capabilities across all functional modules of the RehoSpace Real Estate Platform.

The module transforms operational data into meaningful business intelligence, enabling executives, managers, agents, finance teams, and other stakeholders to make informed, data-driven decisions.

It consolidates data from all platform modules into a unified reporting and analytics environment.

2. Business Value

The module enables organizations to:

- Monitor business performance.
 - Improve strategic decision-making.
 - Track organizational KPIs.
 - Identify operational bottlenecks.
 - Measure sales performance.
 - Analyze customer behavior.
 - Improve forecasting accuracy.
 - Support executive reporting.
 - Enable data-driven management.
-

3. Objectives

The module shall:

- Generate operational reports.
 - Provide executive dashboards.
 - Monitor KPIs.
 - Support custom report creation.
 - Visualize business data.
 - Schedule report generation.
 - Export reports.
 - Support predictive analytics.
 - Enable drill-down analysis.
-

4. Scope

Included

- Operational Reports
- Executive Dashboards

- KPI Monitoring
- Interactive Charts
- Geographic Analytics
- Custom Report Builder
- Report Scheduling
- Data Export
- Predictive Analytics
- Business Forecasting
- Data Warehouse Integration

Excluded

- External BI platform administration
 - Machine Learning model development
 - Database administration
-

5. Reporting Categories

The platform shall provide reports for all business modules.

Examples include:

Executive Reports

- Executive Dashboard
 - Organization Performance
 - Revenue Summary
 - Growth Trends
-

Property Reports

- Property Register
 - Property Availability
 - Property Performance
 - Occupancy Analysis
-

Sales Reports

- Sales Pipeline
 - Sales Performance
 - Agent Performance
 - Sales Conversion
-

CRM Reports

- Lead Generation
 - Lead Conversion
 - Customer Acquisition
 - Customer Retention
-

Finance Reports

- Revenue Analysis
 - Outstanding Payments
 - Cash Collection
 - Commission Summary
-

Survey Reports

- Survey Activities
 - Parcel Statistics
 - Surveyor Performance
-

Marketing Reports

- Campaign Performance
 - Lead Sources
 - Referral Performance
 - Marketing ROI
-

System Reports

- User Activity
 - Audit Logs
 - Login Statistics
 - Security Events
-

6. Dashboard Management

The module shall support configurable dashboards.

Dashboard categories include:

Executive Dashboard

- Revenue
- Sales
- Active Customers
- Occupancy Rate
- Pending Transactions
- Financial Summary

Branch Dashboard

- Branch Performance
- Staff Performance
- Property Inventory
- Monthly Revenue

Agent Dashboard

- Leads
- Appointments
- Sales
- Commissions
- Monthly Targets

Finance Dashboard

- Outstanding Invoices
- Payments Received
- Cash Flow
- Revenue Trends

Marketing Dashboard

- Active Campaigns
- Lead Sources
- Campaign ROI
- Website Traffic

7. Key Performance Indicators (KPIs)

The platform shall support configurable KPIs.

Examples include:

Sales KPIs

- Total Sales
- Sales Growth
- Average Property Value
- Conversion Rate

Customer KPIs

- New Customers
- Customer Satisfaction
- Customer Retention
- Referral Rate

Finance KPIs

- Monthly Revenue
- Outstanding Debt
- Collection Rate
- Average Payment Time

Operations KPIs

- Reservation Approval Time
- Appointment Attendance Rate
- Transaction Completion Time
- Survey Completion Rate

Marketing KPIs

- Cost per Lead
- Lead Conversion Rate
- Campaign ROI
- Website Conversion Rate

8. Interactive Analytics

The module shall support:

- Drill-down Analysis
- Drill-through Navigation
- Dynamic Filtering
- Cross-filtering
- Sorting
- Grouping
- Pivot Tables
- Comparative Analysis

9. Geographic Analytics

The platform shall provide map-based analytics.

Examples include:

- Property Distribution
- Sales by Region
- Customer Distribution
- Survey Coverage
- Market Heat Maps
- Investment Opportunities

Future GIS integration shall extend these capabilities.

10. Custom Report Builder

Authorized users shall create custom reports.

Supported capabilities include:

- Drag-and-drop report designer
 - Field selection
 - Grouping
 - Aggregation
 - Calculated fields
 - Conditional formatting
 - Parameterized reports
 - Saved report templates
-

11. Report Scheduling

Users shall schedule reports.

Supported frequencies include:

- Daily
- Weekly
- Monthly
- Quarterly
- Annual
- Custom Schedule

Delivery options:

- Email
- PDF
- Excel

- CSV
 - Secure Portal Download
-

12. Data Visualization

Supported visualizations include:

- Bar Charts
- Line Charts
- Pie Charts
- Donut Charts
- Area Charts
- Scatter Charts
- Heat Maps
- Tree Maps
- KPI Cards
- Gauges
- Geographic Maps

Visualization types shall be configurable.

13. Predictive Analytics

The architecture shall support predictive models.

Examples include:

- Sales Forecasting
- Revenue Forecasting
- Property Demand Prediction
- Customer Churn Prediction
- Occupancy Forecasting
- Marketing Performance Forecasting

Predictive models may leverage historical platform data.

14. Functional Requirements

Reporting

FR-BI-001

The system shall generate standard operational reports.

FR-BI-002

Users shall export reports in PDF, Excel, CSV, and printable formats.

Dashboards

FR-BI-003

Role-based dashboards shall display relevant KPIs.

FR-BI-004

Dashboard widgets shall be configurable.

Analytics

FR-BI-005

Users shall perform interactive drill-down analysis.

FR-BI-006

Geographic analytics shall support map-based visualization.

Custom Reports

FR-BI-007

Authorized users shall design custom reports without software development.

Scheduling

FR-BI-008

Reports shall support automatic scheduling and delivery.

Forecasting

FR-BI-009

The architecture shall support future predictive analytics engines.

15. Business Rules

BR-BI-001

Reports shall respect role-based security permissions.

BR-BI-002

Historical reports shall remain reproducible.

BR-BI-003

Dashboard data shall refresh according to configurable schedules.

BR-BI-004

Custom reports shall be version-controlled.

BR-BI-005

Only authorized users may publish organization-wide dashboards.

16. Notifications

The module shall publish notifications for:

- Scheduled Report Generated
- Report Delivery Failed
- Dashboard Shared
- KPI Threshold Exceeded
- Forecast Completed
- Data Refresh Completed
- Analytics Job Failed

Supported channels:

- In-App
 - Email
 - SMS (optional)
 - WhatsApp (optional)
-

17. Reports About Reports

The module shall provide administrative reports including:

Report Usage

- Most Viewed Reports
- Dashboard Usage
- Export Statistics
- Scheduled Reports

Analytics Performance

- Query Performance
 - Dashboard Load Time
 - Data Refresh Status
 - Data Source Health
-

18. Configuration

Administrators shall configure:

- Dashboard layouts
 - KPI definitions
 - Report templates
 - Visualization types
 - Refresh intervals
 - Data retention
 - Export formats
 - Scheduling policies
 - Forecasting parameters
 - Access permissions
-

19. Permissions Matrix

The module shall expose permissions including:

- View Reports
- Create Reports
- Edit Reports
- Delete Reports
- Schedule Reports
- Export Reports

- Publish Dashboards
 - Configure KPIs
 - Manage Analytics
 - Configure BI Module
-

20. Acceptance Criteria

The module shall be accepted when:

- Standard reports generate accurately.
 - Dashboards display real-time or scheduled data correctly.
 - KPIs calculate according to configured definitions.
 - Users can build custom reports.
 - Interactive filtering and drill-down function correctly.
 - Scheduled reports are delivered successfully.
 - Geographic analytics display valid spatial information.
 - Exported reports maintain formatting and accuracy.
 - Role-based access is enforced consistently.
-

21. Future Enhancements

The architecture shall support future capabilities including:

- AI-generated executive insights.
 - Natural language queries (e.g., "Show me monthly sales in Dar es Salaam").
 - Automated anomaly detection.
 - Predictive investment opportunity scoring.
 - Market price trend analysis using external datasets.
 - Embedded data warehouse and OLAP cubes.
 - Real-time streaming analytics.
 - Executive mobile analytics dashboards.
 - AI-generated narrative summaries for reports.
-

22. Module Summary

The Reporting, Business Intelligence & Analytics Module is the strategic intelligence layer of the RehoSpace Real Estate Platform. It consolidates operational data into reports, dashboards, KPIs, and predictive analytics, enabling organizations to monitor performance, identify trends, optimize operations, and make informed decisions. Its extensible architecture supports both current reporting needs and future AI-driven business intelligence capabilities.

BUSINESS MODULE 16

BM-016 — INTEGRATION, APIs & EXTERNAL SERVICES MANAGEMENT MODULE

1. Purpose

The Integration, APIs & External Services Management Module provides the infrastructure for securely integrating the RehoSpace Real Estate Platform (RREP) with internal services, third-party systems, external applications, government platforms, payment providers, GIS services, authentication providers, and partner ecosystems.

The module exposes standardized APIs and manages inbound and outbound integrations while ensuring security, reliability, scalability, and auditability.

2. Business Value

The module enables organizations to:

- Integrate with external systems.
 - Reduce duplicate data entry.
 - Automate cross-platform processes.
 - Support mobile applications.
 - Enable partner integrations.
 - Facilitate digital ecosystem growth.
 - Improve interoperability.
 - Future-proof the platform.
-

3. Objectives

The module shall:

- Expose secure APIs.
- Consume external APIs.
- Manage webhooks.
- Synchronize data.

- Integrate payment gateways.
 - Integrate GIS providers.
 - Support authentication providers.
 - Monitor integrations.
 - Maintain API documentation.
-

4. Scope

Included

- REST APIs
- GraphQL APIs (Future)
- Webhooks
- API Authentication
- API Versioning
- Third-party Integrations
- Payment Gateway Integration
- GIS Integration
- Government System Integration
- Identity Provider Integration
- API Monitoring
- API Rate Limiting
- Integration Logs

Excluded

- External application development
 - Third-party infrastructure administration
-

5. Integration Categories

Supported categories include:

Financial

- Banks
 - Mobile Money
 - Payment Gateways
 - Accounting Systems
-

Government

- Land Registry
 - Tax Authority
 - Business Registration
 - National Identity Verification
-

Mapping & GIS

- Google Maps
 - OpenStreetMap
 - Mapbox
 - ArcGIS
-

Communication

- SMS Providers
 - Email Providers
 - WhatsApp Business API
 - Push Notification Providers
-

Authentication

- OAuth2
 - OpenID Connect
 - LDAP
 - Active Directory
 - SAML
-

Enterprise

- ERP Systems
 - CRM Systems
 - Document Management Systems
 - BI Platforms
-

6. API Management

The platform shall expose APIs for:

- Properties
- Customers

- Owners
- Agents
- Reservations
- Appointments
- Transactions
- Payments
- Documents
- Surveys
- Notifications
- Reports

APIs shall follow consistent naming conventions and versioning strategies.

7. API Security

Supported security mechanisms include:

- API Keys
 - OAuth 2.0
 - JWT Tokens
 - Client Credentials
 - IP Whitelisting
 - Rate Limiting
 - Request Signing
 - HTTPS/TLS Encryption
-

8. Webhook Management

The platform shall publish webhooks for configurable business events.

Examples include:

- Property Published
- Reservation Confirmed
- Appointment Completed
- Invoice Generated
- Payment Received
- Transaction Closed
- Survey Approved
- Document Signed
- Customer Registered

Each webhook shall support:

- Retry Policies

- Authentication
 - Delivery Tracking
 - Event Filtering
 - Signature Verification
-

9. Data Synchronization

The module shall support:

- Real-time synchronization
- Scheduled synchronization
- One-way synchronization
- Two-way synchronization
- Incremental synchronization
- Full synchronization

Conflict resolution rules shall be configurable.

10. External Service Connectors

The platform shall provide configurable connectors for:

- Payment Providers
- SMS Providers
- Email Providers
- GIS Services
- Identity Providers
- Government Platforms
- Cloud Storage Services

Connectors shall support configuration without code changes where feasible.

11. API Documentation

The module shall provide developer documentation including:

- Endpoint Catalog
- Authentication Guide
- Request Examples
- Response Examples
- Error Codes

- SDK References
- Webhook Specifications
- Version History

Interactive API documentation (e.g., OpenAPI/Swagger) shall be supported.

12. Functional Requirements

API Services

FR-API-001

The system shall expose secure REST APIs for all major business entities.

FR-API-002

APIs shall support pagination, filtering, sorting, and searching.

FR-API-003

APIs shall be versioned.

Authentication

FR-API-004

API consumers shall authenticate using supported security mechanisms.

Integration

FR-API-005

The platform shall consume external APIs through configurable connectors.

Monitoring

FR-API-006

API usage, failures, and response times shall be monitored.

Webhooks

FR-API-007

Authorized users shall configure webhook subscriptions.

Synchronization

FR-API-008

Synchronization jobs shall support retry and error recovery.

13. Business Rules

BR-API-001

All API traffic shall be encrypted.

BR-API-002

Deprecated API versions shall remain supported for configurable periods.

BR-API-003

API requests shall be authenticated unless explicitly configured as public.

BR-API-004

Integration failures shall be logged.

BR-API-005

Webhook delivery shall follow configured retry policies.

14. Monitoring & Health

The module shall monitor:

- API Availability
- Response Times
- Error Rates
- Rate Limit Violations

- Synchronization Status
- Webhook Deliveries
- Connector Health
- Authentication Failures

Alerts shall be generated when thresholds are exceeded.

15. Reports

The Integration Module shall provide:

API Reports

- API Usage
- Top Consumers
- Response Time Analysis
- Failed Requests

Integration Reports

- Synchronization Status
- Connector Health
- Failed Integrations
- Retry Statistics

Webhook Reports

- Published Events
 - Delivery Success Rate
 - Failed Deliveries
 - Subscriber Activity
-

16. Configuration

Administrators shall configure:

- API Versions
- Authentication Methods
- Rate Limits
- Connector Settings
- Synchronization Schedules
- Retry Policies
- Webhook Endpoints
- Logging Levels

- Timeout Values
-

17. Permissions Matrix

The module shall expose permissions including:

- View APIs
 - Manage API Keys
 - Configure Integrations
 - Configure Connectors
 - Manage Webhooks
 - View API Logs
 - Export Integration Reports
 - Configure Integration Module
-

18. Acceptance Criteria

The module shall be accepted when:

- APIs expose all supported business resources.
 - Authentication mechanisms operate correctly.
 - Webhooks publish events reliably.
 - External integrations synchronize data successfully.
 - API monitoring captures performance metrics.
 - Connector failures are logged and recoverable.
 - Documentation is complete and accessible.
 - Permissions are enforced consistently.
-

19. Future Enhancements

The architecture shall support future capabilities including:

- GraphQL API support.
- gRPC services for high-performance integrations.
- Event streaming using Apache Kafka or RabbitMQ.
- Marketplace for third-party plugins.
- AI-assisted API documentation generation.
- Low-code integration designer.
- API monetization.
- Multi-region API gateways.
- Integration with IoT devices for smart properties.

20. Module Summary

The Integration, APIs & External Services Management Module establishes RREP as an API-first enterprise platform. It provides standardized, secure, and scalable mechanisms for exposing business services, integrating with external systems, synchronizing data, and enabling digital ecosystems. By centralizing integration capabilities, it ensures interoperability, extensibility, and long-term maintainability across diverse deployment environments.

BUSINESS MODULE 17

BM-017 — SECURITY, IDENTITY, ACCESS & TRUST MANAGEMENT MODULE (SIATM)

1. Purpose

The Security, Identity, Access & Trust Management Module provides enterprise-grade security services for the RehoSpace Real Estate Platform (RREP). It ensures that all users, devices, applications, APIs, and external systems are authenticated, authorized, monitored, and governed according to organizational security policies.

The module protects the confidentiality, integrity, and availability of business data while supporting secure access for employees, customers, property owners, agents, partners, and integrated systems.

2. Business Value

The module enables organizations to:

- Protect sensitive business data.
- Prevent unauthorized access.
- Meet regulatory and compliance requirements.
- Reduce cybersecurity risks.
- Strengthen customer trust.
- Secure API integrations.
- Enable enterprise authentication.

- Support multi-organization deployments.
-

3. Objectives

The module shall:

- Authenticate users securely.
 - Authorize access based on policies.
 - Protect user identities.
 - Manage user sessions.
 - Secure APIs.
 - Encrypt sensitive information.
 - Detect suspicious activities.
 - Record security events.
 - Support compliance requirements.
-

4. Scope

Included

- Authentication
- Authorization
- Identity Management
- User Lifecycle Management
- Role-Based Access Control (RBAC)
- Attribute-Based Access Control (ABAC)
- Multi-Factor Authentication (MFA)
- Password Management
- Session Management
- API Security
- Encryption
- Secrets Management
- Security Monitoring
- Threat Detection
- Security Audit
- Device Trust
- Login Policies

Excluded

- Network firewall management
- Antivirus administration
- Physical building security

5. Identity Management

The module shall manage identities for:

- Employees
- Administrators
- Customers
- Property Owners
- Agents
- Brokers
- Business Partners
- API Clients
- External Systems

Each identity shall have a globally unique identifier.

6. Authentication

Supported authentication methods shall include:

- Username & Password
- Email & Password
- Mobile Number & OTP
- Passkeys (Future)
- OAuth 2.0
- OpenID Connect
- SAML
- LDAP
- Active Directory
- Social Login (Optional)

Authentication policies shall be configurable.

7. Multi-Factor Authentication (MFA)

The module shall support:

- SMS OTP
- Email OTP
- Authenticator Applications (TOTP)
- Hardware Security Keys (Future)

- Push-Based Authentication
- Backup Recovery Codes

Organizations may enforce MFA based on:

- User Role
 - Risk Level
 - Device
 - Location
 - Authentication Method
-

8. Authorization

The platform shall support:

Role-Based Access Control (RBAC)

Permissions assigned through roles.

Examples:

- Super Administrator
 - Organization Administrator
 - Branch Manager
 - Finance Officer
 - Sales Agent
 - Surveyor
 - Customer
 - Property Owner
-

Attribute-Based Access Control (ABAC)

Access decisions may also consider:

- Organization
- Branch
- Department
- Region
- Property Ownership
- Data Classification
- Time
- Device Trust

Both RBAC and ABAC shall operate together where configured.

9. User Lifecycle Management

The module shall manage:

- User Registration
- Identity Verification
- Account Activation
- Account Suspension
- Account Locking
- Password Reset
- Credential Rotation
- Account Deactivation
- Account Deletion (where legally permitted)

10. Password Policies

Organizations shall configure:

- Minimum Length
- Complexity Requirements
- Password Expiration
- Password History
- Failed Login Limits
- Account Lock Duration
- Password Reuse Restrictions

11. Session Management

The module shall manage:

- Active Sessions
- Concurrent Sessions
- Session Timeout
- Idle Timeout
- Forced Logout
- Device Sessions
- Session Revocation

Users may view and terminate active sessions.

12. Encryption

Sensitive data shall be protected using industry-standard encryption.

Supported areas include:

- Password Hashing
- Database Encryption
- File Encryption
- API Transport Encryption (TLS)
- Backup Encryption
- Token Encryption

Encryption keys shall be managed securely.

13. API Security

The module shall secure APIs through:

- OAuth 2.0
 - JWT
 - API Keys
 - Mutual TLS (Future)
 - Rate Limiting
 - IP Restrictions
 - Request Validation
 - Token Revocation
-

14. Secrets Management

The platform shall securely manage:

- API Keys
- Database Credentials
- Encryption Keys
- Service Credentials
- Webhook Secrets
- OAuth Client Secrets

Secrets shall never be stored in plaintext.

15. Security Monitoring

The module shall monitor:

- Login Attempts
- Failed Logins
- Privilege Escalation
- Unauthorized Access Attempts
- API Abuse
- Suspicious Locations
- Device Changes
- Password Changes

Security events shall be centrally logged.

16. Threat Detection

The module shall detect:

- Brute Force Attacks
- Credential Stuffing
- Session Hijacking
- Account Enumeration
- Suspicious API Usage
- Excessive Failed Requests
- Unusual Login Patterns
- Unauthorized Privilege Changes

Administrators shall receive alerts based on configured thresholds.

17. Functional Requirements

Authentication

FR-SEC-001

The system shall authenticate users using configured authentication methods.

FR-SEC-002

MFA shall be supported where required.

Authorization

FR-SEC-003

Access decisions shall enforce RBAC and, where configured, ABAC policies.

FR-SEC-004

Permission changes shall take effect immediately.

Session Management

FR-SEC-005

Users shall view and terminate active sessions.

Encryption

FR-SEC-006

Sensitive data shall be encrypted in transit and at rest where applicable.

Monitoring

FR-SEC-007

Security events shall be logged and monitored continuously.

APIs

FR-SEC-008

API requests shall require authentication unless explicitly designated as public.

18. Business Rules

BR-SEC-001

Every user identity shall be unique.

BR-SEC-002

Passwords shall never be stored in plaintext.

BR-SEC-003

Access shall be denied by default unless explicitly granted.

BR-SEC-004

Security logs shall be immutable.

BR-SEC-005

Compromised credentials shall require immediate reset.

19. Reports

The Security Module shall provide:

Authentication Reports

- Login History
- Failed Login Attempts
- MFA Usage
- Password Reset Activity

Authorization Reports

- User Roles
- Permission Assignments
- Privileged Users
- Access Reviews

Security Reports

- Security Incidents
- Threat Detection Summary
- API Security Events
- Device Trust Status
- Account Lockouts

Compliance Reports

- User Access Audit
- Privileged Access Review
- Security Policy Compliance

- Identity Lifecycle Report
-

20. Configuration

Administrators shall configure:

- Authentication methods
 - MFA policies
 - Password policies
 - Session policies
 - Role definitions
 - Permission assignments
 - ABAC rules
 - API security policies
 - Encryption settings
 - Security alert thresholds
 - Device trust policies
-

21. Permissions Matrix

The module shall expose permissions including:

- Manage Users
 - Manage Roles
 - Manage Permissions
 - Configure Authentication
 - Configure MFA
 - Manage Sessions
 - View Security Logs
 - View Security Reports
 - Configure Security Policies
 - Manage API Credentials
 - Configure Security Module
-

22. Acceptance Criteria

The module shall be accepted when:

- Users authenticate using configured methods.
- MFA operates correctly for applicable users.
- RBAC and ABAC policies are enforced.

- Password and session policies are applied consistently.
 - Sensitive data is encrypted appropriately.
 - Security events are logged and monitored.
 - API security controls prevent unauthorized access.
 - Reports accurately reflect security activities.
 - Administrative security settings function as configured.
-

23. Future Enhancements

The architecture shall support future capabilities including:

- Passwordless authentication using FIDO2/WebAuthn.
 - Adaptive, risk-based authentication.
 - AI-powered anomaly detection.
 - Behavioral biometrics.
 - Continuous authentication during user sessions.
 - Zero Trust Network Access (ZTNA) integration.
 - Security Information and Event Management (SIEM) integration.
 - Automated compliance assessment.
 - Threat intelligence feed integration.
 - Hardware-backed key management (HSM/KMS).
-

24. Module Summary

The Security, Identity, Access & Trust Management Module establishes the security foundation of the RehoSpace Real Estate Platform. By combining enterprise authentication, fine-grained authorization, identity lifecycle management, encryption, API protection, threat detection, and continuous security monitoring, it safeguards organizational assets while enabling secure collaboration across employees, customers, partners, and integrated systems. Its design aligns with modern security principles and provides a scalable foundation for enterprise deployments.

BUSINESS MODULE 18

BM-018 — SYSTEM ADMINISTRATION & PLATFORM CONFIGURATION MODULE

1. Purpose

The System Administration & Platform Configuration Module provides centralized administration, platform configuration, operational management, feature control, and tenant-independent customization capabilities for the RehoSpace Real Estate Platform (RREP).

The module enables authorized administrators to configure the platform according to organizational policies, operational requirements, branding, and business rules without requiring software development or code modifications.

It serves as the primary control center for system administrators.

2. Business Value

The module enables organizations to:

- Configure the platform without programming.
 - Adapt workflows to organizational needs.
 - Manage users and organizational structures.
 - Configure branding and localization.
 - Control enabled features.
 - Reduce implementation costs.
 - Simplify maintenance.
 - Improve operational flexibility.
 - Support product edition licensing.
-

3. Objectives

The module shall:

- Manage platform settings.
 - Configure business rules.
 - Manage organizational structures.
 - Control feature availability.
 - Configure numbering sequences.
 - Manage localization.
 - Configure branding.
 - Manage backups.
 - Monitor platform health.
-

4. Scope

Included

- System Settings
- Organization Configuration
- Branch Management
- Department Management
- Feature Management
- Module Activation
- Numbering Configuration
- Branding
- Localization
- Time Zone Configuration
- Currency Configuration
- Backup Configuration
- Maintenance Mode
- Platform Monitoring

Excluded

- Software development
 - Source code modification
 - Infrastructure provisioning
-

5. Organization Configuration

Administrators shall configure:

- Organization Name
- Organization Logo
- Organization Address
- Contact Information
- Website
- Tax Information
- Registration Numbers
- Business Licenses

Multiple branches shall be supported.

6. Branch Management

The platform shall support:

- Branch Registration
 - Branch Activation
 - Branch Deactivation
 - Branch Managers
 - Branch Contacts
 - Regional Assignment
 - Branch Branding (Optional)
-

7. Department Management

Administrators shall manage:

- Departments
 - Teams
 - Reporting Structures
 - Department Heads
 - Functional Assignments
-

8. Feature Management

One of the core capabilities of RREP shall be **feature-based licensing**.

Administrators shall enable or disable modules and features according to the purchased edition.

Examples:

Starter Edition

- Property Management
 - Customer Management
 - Basic Reports
-

Professional Edition

Includes Starter plus:

- CRM
- Reservations
- Finance
- Marketing

- Notifications

Enterprise Edition

Includes Professional plus:

- GIS
- APIs
- Workflow Automation
- BI Analytics
- AI Features

Individual feature flags shall also be configurable.

9. Business Rule Configuration

Administrators shall configure:

- Reservation Rules
- Sales Rules
- Rental Rules
- Survey Rules
- Finance Rules
- Approval Policies
- Commission Rules
- Document Rules

Rule changes shall be version-controlled.

10. Numbering Configuration

The platform shall support configurable numbering formats.

Examples:

Properties

PROP-2026-000001

Customers

CUS-2026-000001

Invoices

INV-2026-000001

Receipts

REC-2026-000001

Reservations

RES-2026-000001

Numbering schemes shall support prefixes, suffixes, branch codes, financial years, and sequences.

11. Branding Management

Organizations shall configure:

- Logo
 - Primary Colors
 - Secondary Colors
 - Email Branding
 - Portal Branding
 - Report Branding
 - Invoice Branding
 - PDF Templates
-

12. Localization

The platform shall support:

- Multiple Languages
- Multiple Time Zones
- Date Formats
- Number Formats
- Currency Formats
- Regional Settings

Localization shall be configurable without software changes.

13. Maintenance Management

Administrators shall:

- Enable Maintenance Mode
 - Schedule Maintenance Windows
 - Display Maintenance Messages
 - Restrict Access During Maintenance
 - Notify Users
-

14. Backup & Recovery Configuration

The module shall configure:

- Automatic Backups
- Backup Frequency
- Backup Retention
- Backup Encryption
- Restore Points
- Recovery Verification

Infrastructure implementation may vary by deployment.

15. Platform Monitoring

The platform shall monitor:

- System Health
- Storage Usage
- Database Status
- Background Jobs
- Scheduled Tasks
- Integration Health
- Notification Queue
- API Status

Alerts shall be configurable.

16. Functional Requirements

Configuration

FR-ADM-001

The system shall allow administrators to configure platform settings.

FR-ADM-002

Configuration changes shall take effect according to defined policies.

Feature Management

FR-ADM-003

Administrators shall enable or disable modules based on licensed editions.

FR-ADM-004

Feature activation shall not require source code modification.

Organization

FR-ADM-005

Organizations shall manage branches and departments.

Localization

FR-ADM-006

Administrators shall configure supported languages, currencies, and regional settings.

Monitoring

FR-ADM-007

The platform shall provide operational health dashboards.

Backup

FR-ADM-008

Backup schedules shall be configurable.

17. Business Rules

BR-ADM-001

Only authorized administrators may modify system settings.

BR-ADM-002

Configuration changes shall be audited.

BR-ADM-003

Disabled modules shall not be accessible to users.

BR-ADM-004

Critical configuration changes may require approval, if configured.

BR-ADM-005

Feature licensing shall determine available functionality.

18. Reports

The System Administration Module shall provide:

Configuration Reports

- System Configuration Summary
- Feature Activation Report
- Business Rule Changes
- Branding Configuration

Operational Reports

- Platform Health
- Scheduled Jobs
- Backup Status
- Storage Utilization
- Queue Monitoring
- Integration Health

Administrative Reports

- Branch Register
- Department Register

- Module Usage
 - License Utilization
-

19. Configuration

The module itself shall allow configuration of:

- Global Settings
 - Organization Settings
 - Branding
 - Localization
 - Numbering Schemes
 - Feature Flags
 - Licensing
 - Maintenance Policies
 - Monitoring Thresholds
 - Backup Policies
-

20. Permissions Matrix

The module shall expose permissions including:

- Manage System Settings
 - Manage Organizations
 - Manage Branches
 - Manage Departments
 - Configure Features
 - Configure Licensing
 - Manage Branding
 - Configure Localization
 - Configure Backups
 - View System Health
 - Export Administrative Reports
 - Configure Administration Module
-

21. Acceptance Criteria

The module shall be accepted when:

- Administrators configure the platform without code changes.
- Feature licensing controls module availability.

- Branding updates are reflected across the platform.
 - Numbering schemes generate correctly.
 - Localization settings are applied consistently.
 - Backup schedules execute as configured.
 - Monitoring dashboards reflect system health accurately.
 - Configuration changes are audited.
 - Permissions are enforced consistently.
-

22. Future Enhancements

The architecture shall support future capabilities including:

- Low-code configuration designer.
 - Visual workflow configuration.
 - Dynamic form designer.
 - Dynamic field designer.
 - Business rule scripting engine.
 - Plugin/module marketplace.
 - Configuration export/import between environments.
 - Blue-green deployment support.
 - Multi-environment configuration management.
 - Remote diagnostics and health monitoring.
-

23. Module Summary

The System Administration & Platform Configuration Module is the operational control center of the RehoSpace Real Estate Platform. It empowers administrators to configure organizational settings, activate licensed features, manage branding, define business rules, monitor platform health, and control operational behavior without modifying the application code. This flexibility is fundamental to RREP's vision as a reusable, configurable platform that can be deployed across diverse real estate organizations with varying business requirements.

BUSINESS MODULE 19

BM-019 — AUDIT, COMPLIANCE & GOVERNANCE MANAGEMENT MODULE

1. Purpose

The Audit, Compliance & Governance Management Module provides comprehensive audit logging, regulatory compliance, records management, governance controls, policy enforcement, and accountability across the RehoSpace Real Estate Platform (RREP).

The module captures and preserves a complete history of business activities, security events, configuration changes, and data modifications, ensuring transparency, legal defensibility, and compliance with applicable laws, regulations, and organizational policies.

2. Business Value

The module enables organizations to:

- Maintain complete audit trails.
 - Support legal investigations.
 - Meet regulatory requirements.
 - Improve accountability.
 - Protect against fraud.
 - Ensure data integrity.
 - Simplify internal and external audits.
 - Enforce organizational policies.
 - Strengthen corporate governance.
-

3. Objectives

The module shall:

- Record audit events.
 - Track data changes.
 - Monitor compliance.
 - Manage records retention.
 - Support legal holds.
 - Manage consent records.
 - Detect policy violations.
 - Produce compliance reports.
 - Preserve evidence.
-

4. Scope

Included

- Audit Trails
- Activity Logs
- Data Change History
- Configuration Audit
- Security Audit
- Compliance Monitoring
- Records Retention
- Legal Hold
- Consent Management
- Policy Management
- Governance Controls
- Compliance Reporting

Excluded

- Legal advisory services.
 - External regulatory submissions.
 - Court case management.
-

5. Audit Logging

The platform shall record audit events for all critical operations.

Examples include:

- User Login
- User Logout
- Failed Login
- User Creation
- User Modification
- Permission Changes
- Property Created
- Property Updated
- Property Deleted (logical)
- Customer Created
- Survey Approved
- Reservation Approved
- Sale Completed
- Payment Received
- Refund Processed
- Workflow Approved
- Configuration Changed
- API Access
- Data Export
- Report Generation

Each audit event shall include:

- Audit ID
 - Event Type
 - Date & Time
 - User
 - Organization
 - Branch
 - Source IP Address
 - Device Information
 - Before Value (where applicable)
 - After Value (where applicable)
 - Outcome
 - Correlation ID
-

6. Activity History

The system shall maintain activity histories for:

- Properties
- Customers
- Property Owners
- Agents
- Reservations
- Appointments
- Transactions
- Payments
- Documents
- Surveys
- Workflows
- User Accounts

Users with appropriate permissions shall view complete chronological histories.

7. Data Change Tracking

Every modification shall preserve:

- Previous Value
- New Value
- Changed Fields
- User
- Date
- Reason (optional/configurable)

Historical versions shall remain accessible according to retention policies.

8. Compliance Management

The module shall support configurable compliance frameworks.

Examples include:

- Data Protection Laws
- Financial Reporting Requirements
- Property Transaction Regulations
- Electronic Records Regulations
- Organizational Policies
- Information Security Standards

Frameworks shall be configurable by deployment.

9. Records Retention

Administrators shall configure retention policies for:

- Audit Logs
- Financial Records
- Customer Records
- Property Records
- Survey Records
- Contracts
- Notifications
- Reports
- Attachments

Retention periods shall support legal and organizational requirements.

10. Legal Hold

Authorized users shall place records under legal hold.

While under legal hold:

- Records cannot be deleted.
- Retention periods are suspended.

- Modifications are restricted according to policy.
 - Access remains fully audited.
-

11. Consent Management

The platform shall record customer consent for:

- Marketing Communications
- Data Processing
- Document Sharing
- Third-Party Integrations
- Electronic Signatures

Consent records shall include:

- Consent Type
 - Consent Date
 - Consent Method
 - Consent Status
 - Withdrawal Date (if applicable)
-

12. Policy Management

Organizations shall maintain policies including:

- Password Policy
- Data Retention Policy
- Privacy Policy
- Information Security Policy
- Document Classification Policy
- Access Control Policy

Policy revisions shall be version-controlled.

13. Governance Controls

The platform shall support:

- Segregation of Duties
- Dual Authorization
- Four-Eyes Principle

- Approval Accountability
- Risk-Based Approvals
- Delegated Authority

Governance controls shall be configurable.

14. Functional Requirements

Audit

FR-AUD-001

The system shall automatically record audit events for configurable business activities.

FR-AUD-002

Audit records shall be immutable.

History

FR-AUD-003

Authorized users shall view historical changes to business records.

Compliance

FR-AUD-004

The platform shall monitor configured compliance rules.

Retention

FR-AUD-005

Retention policies shall automatically govern record preservation and disposal.

Consent

FR-AUD-006

The platform shall maintain complete consent histories.

Governance

FR-AUD-007

Approval governance rules shall be enforceable across supported workflows.

15. Business Rules

BR-AUD-001

Audit records shall never be modified or deleted except through approved archival processes.

BR-AUD-002

Every critical business transaction shall generate an audit record.

BR-AUD-003

Retention policies shall supersede user deletion requests where legally required.

BR-AUD-004

Legal holds shall prevent record destruction.

BR-AUD-005

Compliance violations shall generate alerts.

16. Notifications

The module shall publish notifications for:

- Compliance Violation
- Audit Failure
- Policy Update
- Retention Expiry
- Legal Hold Applied
- Legal Hold Released
- Suspicious Activity
- Governance Exception

Supported channels:

- In-App
 - Email
 - SMS (optional)
 - WhatsApp (optional)
-

17. Reports

The Audit, Compliance & Governance Module shall provide:

Audit Reports

- User Activity Log
- Data Modification Report
- Configuration Change Report
- Security Audit Report
- API Audit Report

Compliance Reports

- Compliance Status Summary
- Policy Violations
- Consent Register
- Retention Compliance
- Regulatory Compliance Dashboard

Governance Reports

- Approval Accountability
- Segregation of Duties Review
- High-Risk Transactions
- Privileged User Activities
- Governance Exceptions

Legal Reports

- Legal Hold Register
- Archived Records
- Record Disposal Register

Reports shall support filtering, scheduled delivery, exporting, and digital signatures where required.

18. Configuration

Administrators shall configure:

- Audit scope
 - Audit retention period
 - Compliance frameworks
 - Retention policies
 - Legal hold policies
 - Consent categories
 - Governance rules
 - Policy documents
 - Alert thresholds
 - Compliance schedules
-

19. Permissions Matrix

The module shall expose permissions including:

- View Audit Logs
 - Export Audit Logs
 - Manage Retention Policies
 - Apply Legal Hold
 - Release Legal Hold
 - Manage Compliance Rules
 - Manage Governance Policies
 - View Compliance Reports
 - Manage Consent Records
 - Configure Audit Module
-

20. Acceptance Criteria

The module shall be accepted when:

- Audit logs capture configured business events.
- Audit records remain immutable.
- Historical record versions are accessible.
- Retention policies are enforced.
- Legal holds prevent unauthorized deletion.
- Compliance violations generate alerts.
- Governance rules operate correctly.
- Reports provide accurate compliance and audit information.
- Permissions are enforced consistently.

21. Future Enhancements

The architecture shall support future capabilities including:

- AI-powered fraud detection.
 - Automated compliance assessments.
 - Digital evidence chain-of-custody management.
 - Blockchain-backed audit verification.
 - Electronic discovery (eDiscovery).
 - Risk scoring for business transactions.
 - Automated regulatory reporting.
 - Compliance heat maps.
 - Enterprise GRC (Governance, Risk & Compliance) integration.
-

22. Module Summary

The Audit, Compliance & Governance Management Module establishes accountability and trust across the RehoSpace Real Estate Platform. By providing immutable audit trails, comprehensive compliance monitoring, records retention, legal hold capabilities, consent management, and governance controls, it ensures that business operations remain transparent, secure, and compliant with organizational and regulatory requirements. This module is essential for enterprise deployments and organizations requiring strong operational governance.

BUSINESS MODULE 20

BM-020 — ARTIFICIAL INTELLIGENCE (AI) & SMART AUTOMATION FRAMEWORK

1. Purpose

The Artificial Intelligence (AI) & Smart Automation Framework provides intelligent decision support, predictive analytics, automation, natural language interaction, computer vision, and machine learning capabilities across all modules of the RehoSpace Real Estate Platform (RREP).

Rather than functioning as a separate application, the AI Framework acts as a centralized intelligence engine that continuously analyzes platform data, assists users, automates routine tasks, and generates actionable recommendations to improve productivity, customer experience, operational efficiency, and business performance.

2. Business Value

The AI Framework enables organizations to:

- Reduce manual work.
 - Improve decision-making.
 - Increase sales conversions.
 - Improve customer satisfaction.
 - Detect fraud and anomalies.
 - Automate repetitive processes.
 - Generate business insights.
 - Improve property marketing.
 - Support intelligent customer service.
 - Increase organizational productivity.
-

3. Objectives

The AI Framework shall:

- Provide conversational AI assistants.
 - Generate intelligent recommendations.
 - Automate repetitive business processes.
 - Predict business outcomes.
 - Analyze documents.
 - Analyze property images.
 - Generate reports and summaries.
 - Detect anomalies.
 - Support natural language search.
 - Continuously learn from organizational data where permitted.
-

4. Scope

Included

- AI Assistants
- Natural Language Processing (NLP)

- Intelligent Search
- Predictive Analytics
- Recommendation Engine
- Computer Vision
- OCR & Document Intelligence
- Generative AI
- Workflow Optimization
- Smart Notifications
- Fraud Detection
- AI Reporting
- AI APIs
- AI Model Management

Excluded

- Development of proprietary foundation language models.
 - Autonomous legal decision-making.
 - Autonomous financial approvals without configurable human oversight.
-

5. AI Assistant

The platform shall provide AI assistants for different user groups.

Administrator Assistant

Capabilities include:

- Explain system configuration.
 - Recommend best practices.
 - Diagnose configuration issues.
 - Summarize platform health.
 - Explain audit findings.
-

Sales Assistant

Capabilities include:

- Recommend properties.
 - Prioritize leads.
 - Draft customer communications.
 - Schedule follow-ups.
 - Suggest negotiation strategies.
 - Predict closing probability.
-

Customer Assistant

Capabilities include:

- Search for properties using natural language.
- Recommend properties.
- Explain buying or renting processes.
- Schedule appointments.
- Answer frequently asked questions.
- Track transaction status.

Property Owner Assistant

Capabilities include:

- Explain portfolio performance.
- Recommend rental pricing.
- Forecast occupancy.
- Suggest marketing actions.
- Summarize income trends.

Executive Assistant

Capabilities include:

- Summarize business performance.
- Explain KPI changes.
- Generate executive reports.
- Recommend strategic actions.
- Forecast revenue.

6. Intelligent Property Recommendation Engine

The platform shall recommend properties using factors such as:

- Budget
- Preferred location
- Property type
- Number of bedrooms
- Customer behavior
- Search history
- Saved properties
- Similar customer preferences

- Investment goals

Recommendations shall improve over time using historical interaction data where permitted.

7. Smart Property Valuation

The AI Framework shall estimate property values using:

- Historical transactions.
- Comparable properties.
- Geographic location.
- Property features.
- Market trends.
- Demand indicators.
- Survey information.

Estimated values shall always be identified as advisory rather than official valuations.

8. Lead Intelligence

The AI Framework shall support:

- Lead scoring.
 - Conversion probability.
 - Customer segmentation.
 - Next best action recommendations.
 - Customer lifetime value estimation.
 - Lead prioritization.
-

9. Intelligent Marketing

AI shall assist with:

- Property descriptions.
- Advertisement copy.
- Social media captions.
- Email content.
- Campaign recommendations.
- Audience segmentation.
- Best publishing times.

Generated content shall be editable by users before publication.

10. Document Intelligence

The platform shall support AI-assisted document processing including:

- OCR (Optical Character Recognition).
- Document classification.
- Data extraction.
- Contract summarization.
- Duplicate detection.
- Missing document detection.
- Metadata extraction.

Supported documents include:

- Title deeds
 - Sale agreements
 - Lease agreements
 - Survey plans
 - Identity documents
 - Receipts
 - Invoices
 - Supporting attachments
-

11. Computer Vision

The AI Framework shall analyze property images to:

- Detect image quality issues.
- Identify rooms and property features.
- Detect duplicate images.
- Generate image tags.
- Recommend image ordering.
- Identify missing property photos.

Future capabilities may include automatic floor plan interpretation.

12. Conversational Search

Users shall search using natural language.

Examples:

- "Show available apartments under TZS 250 million in Dar es Salaam."
- "Which properties generated the highest rental income this year?"
- "Show overdue payments from last month."
- "Find customers waiting for survey approval."

The AI engine shall translate these requests into platform queries.

13. Intelligent Reporting

The AI Framework shall:

- Summarize reports.
- Explain trends.
- Identify anomalies.
- Highlight opportunities.
- Recommend actions.
- Answer questions about reports.

Example:

"Revenue increased by 18% this quarter, primarily due to increased apartment sales in the Eastern Region."

14. Predictive Analytics

The platform shall support predictive models including:

- Sales forecasting.
- Revenue forecasting.
- Occupancy prediction.
- Property demand forecasting.
- Customer churn prediction.
- Lead conversion prediction.
- Commission forecasting.
- Cash flow forecasting.

Predictions shall include confidence indicators where available.

15. Fraud & Anomaly Detection

The AI Framework shall identify:

- Suspicious payment patterns.
- Duplicate customers.
- Unusual property pricing.
- Fake property listings.
- Unauthorized access behavior.
- Abnormal API activity.
- Duplicate commissions.
- High-risk transactions.

Detected anomalies shall generate alerts for review.

16. Workflow Optimization

The AI Framework shall recommend:

- Process improvements.
- Bottleneck reduction.
- Task prioritization.
- Approval routing optimization.
- Staffing recommendations.
- Resource allocation improvements.

Recommendations shall not modify workflows automatically unless explicitly configured.

17. Functional Requirements

AI Assistants

FR-AI-001

The platform shall provide role-specific AI assistants.

FR-AI-002

AI responses shall respect user permissions and access controls.

Recommendations

FR-AI-003

The system shall generate personalized recommendations for users where applicable.

Document Intelligence

FR-AI-004

AI shall extract structured information from supported documents.

Predictive Analytics

FR-AI-005

Predictive models shall generate configurable forecasts.

Conversational Search

FR-AI-006

Users shall retrieve platform information using natural language.

Content Generation

FR-AI-007

The system shall generate editable marketing and business content.

18. Business Rules

BR-AI-001

AI-generated recommendations are advisory unless explicitly approved by authorized users.

BR-AI-002

AI shall never expose information beyond the requesting user's permissions.

BR-AI-003

AI-generated content shall be identifiable as AI-generated.

BR-AI-004

Predictions shall include timestamps and model version references where applicable.

BR-AI-005

Organizations may enable or disable individual AI capabilities.

19. Reports

The AI Framework shall provide:

AI Usage Reports

- AI Assistant Usage
- Most Common Queries
- AI Adoption by Department
- Generated Content Statistics

Prediction Reports

- Forecast Accuracy
- Lead Scoring Performance
- Revenue Prediction Accuracy
- Occupancy Prediction Trends

Automation Reports

- Tasks Automated
- Time Saved
- Workflow Optimization Impact
- AI Recommendation Acceptance Rate

AI Governance Reports

- AI Decision Logs
- AI Model Versions
- AI Error Analysis
- AI Policy Compliance

20. Configuration

Administrators shall configure:

- AI providers
 - AI models
 - Prompt templates
 - Language preferences
 - AI feature availability
 - Prediction thresholds
 - Recommendation rules
 - Data sources
 - AI usage limits
 - Human approval requirements
 - Privacy and governance settings
-

21. Permissions Matrix

The module shall expose permissions including:

- Use AI Assistant
 - Manage AI Models
 - Configure AI Providers
 - Manage Prompt Templates
 - View AI Reports
 - Configure AI Features
 - Review AI Decisions
 - Approve AI Recommendations
 - Export AI Analytics
 - Configure AI Framework
-

22. Acceptance Criteria

The module shall be accepted when:

- AI assistants respond accurately within authorized data boundaries.
 - Property recommendations are relevant to user preferences.
 - Document intelligence extracts data with acceptable accuracy.
 - Conversational search retrieves correct platform information.
 - Predictive analytics generate forecasts successfully.
 - AI-generated content is editable and properly identified.
 - Fraud detection identifies configurable anomalies.
 - AI governance controls are enforced.
 - Permissions are consistently respected.
-

23. Future Enhancements

The architecture shall support future capabilities including:

- Autonomous AI agents for repetitive administrative tasks.
- Voice assistants in web and mobile applications.
- Real-time speech translation for multilingual customer interactions.
- AI-powered virtual property tours.
- Integration with drone imagery for land and property analysis.
- Automated legal document review with legal professional oversight.
- Digital twins for large property developments.
- Federated learning across deployments while preserving customer privacy.
- AI copilots for software administrators and developers.
- Continuous model improvement pipelines with human feedback.

24. Module Summary

The Artificial Intelligence & Smart Automation Framework is the intelligence layer of the RehoSpace Real Estate Platform. It enhances every business module with conversational assistance, predictive analytics, intelligent automation, document understanding, computer vision, personalized recommendations, and advanced decision support. Designed with strong governance, security, and human oversight, it enables organizations to improve efficiency, customer experience, and strategic decision-making while remaining adaptable to future advances in artificial intelligence.

Product Editions (Highly Recommended)

Since your vision is to sell this platform to different companies with different budgets, define the editions now.

Example:

Edition	Target Market	Included Modules
Community	Individual agents	Basic Property + CRM + Website
Starter	Small real estate companies	Core operations
Professional	Growing companies	CRM + Finance + Marketing + Reporting
Enterprise	Large companies	All modules
Ultimate	Enterprise + AI	Everything including AI

This will directly influence licensing, pricing, development priorities, and sales.

2. Feature Matrix

Create a large table like:

Module	Community	Starter	Professional	Enterprise	Ultimate
Property Management	✓	✓	✓	✓	✓
Survey Management	X	✓	✓	✓	✓
Finance	X	X	✓	✓	✓
AI Framework	X	X	X	X	✓

This becomes your sales team's reference.

3. User Roles Catalogue

Instead of scattering roles across modules, create one master chapter listing every role.

Examples:

- Super Administrator
- Organization Administrator
- Branch Manager
- Sales Manager
- Sales Agent
- Marketing Officer
- Finance Officer
- Accountant
- Surveyor
- GIS Officer
- Legal Officer
- Property Manager
- Customer
- Property Owner
- Tenant
- Investor
- Business Partner
- API Client

Then define:

- Responsibilities
- Permissions
- Accessible Modules

4. Master Data Catalogue

This is often forgotten but is essential.

Examples:

- Countries
- Regions
- Districts
- Wards
- Streets
- Property Types
- Property Statuses
- Transaction Types
- Payment Methods
- Banks
- Mobile Money Providers
- Currencies
- Tax Rates
- Amenities
- Units of Measure
- Languages

This will simplify development tremendously.

5. Cross-Module Integration Matrix

For example:

Module Sends Data To Receives Data From

Property Marketplace Survey

CRM Marketing Marketplace

Finance Reporting Sales

AI Every Module Every Module

This becomes invaluable during backend development.

6. Product Architecture Overview

One high-level diagram showing:

```
Web Portal
Mobile Apps
Customer Portal
Owner Portal
Agent Portal
|
API Gateway
|
Business Modules
|
Shared Services
|
Database
Storage
Notification Engine
AI Engine
```

This bridges Volume I and Volume II.

7. Non-Functional Requirements

Currently, we've focused primarily on functionality. Add a dedicated chapter covering:

- Performance
- Availability
- Scalability
- Reliability
- Security
- Backup & Recovery
- Accessibility
- Localization
- Browser Support
- Mobile Responsiveness
- Maintainability
- Logging
- Monitoring

This is a critical section.

8. Product Roadmap

Define future releases.

Example:

Version 1.0

- Core real estate operations

Version 2.0

- Mobile applications
- GIS enhancements

Version 3.0

- AI assistants
- Predictive analytics

Version 4.0

- Smart property ecosystem
- IoT integration

9. Glossary

Define all business terms.

Examples:

- Property
- Listing
- Reservation
- Survey
- Parcel
- Agent
- Tenant
- Lease
- Purchase Agreement
- Commission
- Occupancy
- GIS

This avoids ambiguity during development.

10. Appendices

Include:

- Business process diagrams
- State diagrams
- Workflow diagrams
- Approval flowcharts
- Sample forms
- Sample reports
- Notification templates
- Numbering conventions